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Topology Optimization of Flow Configurations Operating in the Turbulent Regime

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Preface

This thesis marks the culmination of my Master of Science in Mechanical Engineering at KU Leuven. The journey of exploring computational fluid dynamics and mathematical optimization has been both intensely challenging and deeply rewarding. Bridging the gap between theoretical mathematics and practical engineering applications required countless hours of coding, debugging, and analyzing flow fields, but seeing the optimizer successfully carve out organic, aerodynamic fluid channels made the entire process worthwhile.

I would like to express my sincere gratitude to my supervisors, Prof. Niels Horsten and Prof. Maarten Blommaert, for providing me with the opportunity to work on this fascinating topic. Their academic guidance, critical insights, and continuous support were instrumental in shaping the direction and rigor of this research. A very special thanks goes to my assistant-supervisor, Esteban Foronda Obando, for his day-to-day patience, technical expertise, and willingness to help me navigate the complex numerical hurdles within FEniCS.

Furthermore, I would like to thank the assessors, Prof. Elke Deckers and Olivier Renders, for their time and valuable feedback in evaluating this work.

Finally, I want to thank my family and friends for their unwavering encouragement, patience, and perspective throughout the course of my studies.

Tiebert Lefebure

Contents

Abstract

As modern electronic, automotive, and energy systems grow increasingly compact and power-dense, their thermal performance increasingly depends on low-loss internal-flow paths that can distribute coolant without excessive pumping power. This thesis focuses on the flow-only part of that broader thermal-management problem. It develops a density-based topology optimization framework for turbulent internal-flow configurations in FEniCS, using Brinkman penalization, wall-resolved Spalart-Allmaras (SA) turbulence modeling, a relaxed Eikonal wall-distance calculation, and a frozen-turbulence weak-form adjoint in which the eddy-viscosity and wall-distance fields are updated in the forward loop but treated as fixed coefficients during sensitivity assembly.

The work first establishes the modeling scope of SA for curved internal flows and then verifies the custom FEniCS-SA implementation against Ansys Fluent on a periodic turbulent channel and a curved U-bend. These verification cases show that the implemented wall-distance, SA transport, and segregated flow-coupling strategy reproduce the relevant Ansys-SA behavior sufficiently for the topology optimization studies, while retaining the known limitations of a two-dimensional, one-equation eddy-viscosity model.

The topology optimization framework is evaluated on pipe-bend and U-bend benchmarks from Bayat et al. and a turbulent diffuser benchmark from Yoon. For the pipe bend, the implemented frozen sensitivity agrees with central finite differences to a maximum coordinate relative discrepancy of 1.47×10^{-4}, and the directional Taylor check shows approximately second-order remainder convergence. The extracted pipe-bend geometry is then validated in Ansys Fluent with both SA and SST $k - \omega$ closures: switching from Ansys-SA to SST $k - \omega$ increases the validated pressure drop by about 4.5% and viscous dissipation by about 8.5%, indicating moderate turbulence-model sensitivity rather than collapse of the optimized design. Under identical SST $k - \omega$ operating conditions, the turbulent pipe-bend design reduces pressure drop by approximately 72% and viscous dissipation by approximately 47% relative to the laminar-optimized reference.

The thesis therefore concludes that a frozen-SA topology optimization framework can produce credible low-loss turbulent internal-flow designs when its modeling assumptions are stated explicitly and the final geometries are independently re-analyzed. The main remaining limitations are the frozen-adjoint approximation, two-dimensional suppression of secondary-flow physics, and sensitivity of final performance to geometry extraction. These limitations define the most important next steps: three-dimensional extension, conjugate heat-transfer coupling, and adjoint hierarchies that reintroduce selected turbulence sensitivities.

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List of Abbreviations and Symbols

Abbreviations

AMG	Algebraic Multigrid
BiCGSTAB	Bi-Conjugate Gradient Stabilized method
CAD	Computer-Aided Design
CEV	Constant Eddy Viscosity
CFD	Computational Fluid Dynamics
CG1	Continuous Galerkin (Linear Basis Functions)
CG2	Continuous Galerkin (Quadratic Basis Functions)
CHT	Conjugate Heat Transfer
DG0	Discontinuous Galerkin (Piecewise Constant Elements)
DNS	Direct Numerical Simulation
DOF	Degree of Freedom
FEM	Finite Element Method
FD	Finite Difference
FVM	Finite Volume Method
GMRES	Generalized Minimal Residual method
IPCS	Incremental Pressure Correction Scheme
ILU	Incomplete LU factorization
MMA	Method of Moving Asymptotes
MUMPS	Multifrontal Massively Parallel Sparse direct Solver
NS-SA	Navier-Stokes-Spalart-Allmaras coupling
PDE	Partial Differential Equation
RAMP	Rational Approximation of Material Properties
RANS	Reynolds-Averaged Navier-Stokes
SA	Spalart-Allmaras
SIMP	Solid Isotropic Material with Penalization
SNES	Scalable Nonlinear Equations Solver
SST $k-\omega$	Shear Stress Transport $k-\omega$ model
TO	Topology Optimization
UFL	Unified Form Language
VMFL	Ansys Fluent verification benchmark case
XFEM	Extended Finite Element Method

Symbols

Fluid Dynamics & Turbulence

D, D_h	Pipe diameter and hydraulic diameter [m]
De	Dean number [-]
H, L	Channel height and characteristic length [m]
k	Turbulent kinetic energy [m^2/s^2]
$p, \bar{p}, p'$	Instantaneous, mean, and fluctuating static pressure [Pa]
ΔP	Macroscopic total pressure drop [Pa]
R_c	Radius of curvature [m]
Re	Bulk Reynolds number [-]
$S, \tilde{S}$	Scalar vorticity magnitude and modified SA vorticity magnitude [1/s]
t	Time / Pseudo-time [s]
τ_w	Wall shear stress [Pa]
$\mathbf{u}, \bar{u}_i$	Velocity vector / Mean velocity component [m/s]
u'_i	Fluctuating velocity component [m/s]
$U_{\mathrm{ref}}, U_{\mathrm{bulk}}, U_{\mathrm{max}}$	Reference, bulk, and maximum velocity [m/s]
u^+	Non-dimensional mean velocity [-]
u_τ	Friction velocity [m/s]
y, d_w	Distance to the nearest wall [m]
y^+	Non-dimensional wall distance [-]
C_f	Skin-friction coefficient [-]
c_{b1}, c_{b2}, c_{v1}	Spalart-Allmaras production and viscosity-ratio constants [-]
c_{w1}, c_{w2}, c_{w3}	Spalart-Allmaras wall-destruction constants [-]
f_{v1}, f_{v2}, f_w	Spalart-Allmaras damping and wall-destruction functions [-]
$r_{\mathrm{SA}}, g_{\mathrm{SA}}$	Spalart-Allmaras wall-destruction auxiliary variables [-]
δ_{ij}	Kronecker delta [-]
μ, μ_t	Dynamic and turbulent eddy viscosity [Pa·s]
ν, ν_t	Kinematic and turbulent eddy kinematic viscosity [m^2/s]
$\tilde{\nu}$	Spalart-Allmaras working variable [m^2/s]
ρ	Fluid density [kg/m^3]
ω	Scalar vorticity in two-dimensional flow [1/s]
Ω_{ij}	Vorticity tensor [1/s]
κ	von Kármán constant [-]

Numerical Implementation (FEniCS)

G, G_0	Relaxed Eikonal field variable and boundary value [1/m]
h, h_{avg}	Finite element cell diameter and average cell size [m]
p_k	Kinematic pressure [$\mathrm{m}^2/\mathrm{s}^2$]
s	Arclength coordinate along a profile line [m]
e_i	Pointwise profile error or design-space basis vector, depending on context [-]
E_∞, E_{L_2}	Maximum and relative L_2 profile discrepancies [-]
Δh	Finite-difference design perturbation [-]
$S_i^{\mathrm{FD}}, S_i^{\mathrm{fr}}$	Finite-difference and frozen-adjoint sensitivity of sampled design variable i
$\varepsilon_i^{\mathrm{fr}}$	Relative frozen-adjoint sensitivity discrepancy [-]
$\lvert\lvert \cdot \rvert\rvert_{L_2}$	L_2 (Euclidean) norm [-]
$[\![\cdot]\!]$	Jump operator across an internal element facet [-]
$\Gamma_{\mathrm{in}}, \Gamma_{\mathrm{out}}, \Gamma_D$	Inlet, outlet, and profile-extraction boundaries
$\mathbf{n}$	Outward-pointing normal vector [-]
v, ξ	Test functions for pressure and turbulence equations [-]
$\alpha_u, \alpha_{\tilde{\nu}}$	Relaxation factors for velocity and turbulence [-]
Δt	Pseudo-time step [s]
ϵ_{tol}	Convergence tolerance [-]
ϕ	General scalar field variable [-]
σ_w	Eikonal relaxation constant [-]

Topology Optimization

$\alpha(\gamma)$	Continuous Brinkman penalization term / inverse permeability
$\alpha_{solid}, \alpha_{fluid}$	Maximum solid and minimum fluid Brinkman penalties
α_{dg}	Artificial diffusion scaling factor for DG0 filter [-]
β	Heaviside projection steepness parameter [-]
γ, γ_{raw}	Continuous and raw design variable (1 for fluid, 0 for solid) [-]
$\tilde{\gamma}, \bar{\gamma}$	Filtered and Heaviside-projected design variables [-]
η	Heaviside projection threshold [-]
Π_β	Heaviside projection operator [-]
$\mathcal{H}$	Helmholtz filter operator [-]
M	Active design mask [-]
J	Generic objective function used in the optimization problem [problem dependent]
J_D	Dissipation objective, evaluated from viscous dissipation [W, or W/m for two-dimensional unit-depth cases]
J_p	Average inlet-pressure objective [Pa]
Φ_D	Sharp-geometry viscous dissipation metric [W, or W/m for two-dimensional unit-depth cases]
q	RAMP penalization tuning parameter [-]
r	Continuous Helmholtz filter radius [m]
$r_V^{(k)}$	Volume residual at optimization iteration k [-]
$v_f^{(k)}$	Realized fluid volume fraction at optimization iteration k [-]
$\mathbf{s}, \mathbf{s}^*$	Monolithic forward state and frozen-flow adjoint state vectors
V_f, V_{domain}	Physical fluid volume and total design domain volume [m^3]
V_{mask}	Volume-mask measure [m^3]
V_{max}	Maximum allowable fluid volume fraction constraint [-]
$\Omega, \partial\Omega, \Gamma_{int}$	Computational domain, domain boundary, and internal element facets
Ω_D, Ω_V	Objective-integration and volume-integration regions

Chapter 1

Introduction

1.1 Background and Motivation

Energy systems increasingly rely on compact flow devices that must move heat, mass, or working fluids through limited space with minimal loss. Examples include liquid-cooled electronics, battery and power-electronics thermal management, fuel-cell and electrolyzer flow fields, compact heat exchangers, and coolant manifolds in automotive and aerospace applications. In these systems, the thermal design is often constrained before any heat-transfer equation is considered: the fluid must first be distributed, collected, redirected, or expanded without creating an excessive pressure drop. A cooling plate or heat exchanger with excellent local heat-transfer potential can still be impractical if the manifold requires too much pumping power or if the flow distribution is poor.

This thesis is positioned within that broader energy and thermal-management context, but its actual framework is deliberately narrower. It does not solve the energy equation, does not perform conjugate heat-transfer optimization, and does not optimize temperature directly. Instead, it studies the flow-only configurations that make those applications possible: manifolds, bends, U-bends, collectors, and diffusers whose performance is governed by pressure loss, viscous dissipation, separation, and turbulence-model behavior. These flow configurations are a necessary stepping stone toward turbulent heat-transfer optimization because reliable thermal design requires a credible and stable prediction of the hydraulic losses first.

Mathematical optimization offers a way to design such internal-flow layouts without prescribing the final channel topology in advance. Among these strategies, topology optimization has emerged as a powerful computational design tool for fluid and thermal-management problems [21]. In a fully coupled thermal problem, the eventual objective may involve heat-transfer rate, maximum temperature, or thermal resistance. In the flow-only problem treated here, the relevant objectives are instead pressure drop and dissipated mechanical energy. This restriction is not merely a simplification for convenience: turbulent density-based topology optimization remains challenging even before the energy equation is added, because the flow solver, turbulence model, wall treatment, geometry representation, and adjoint sensitivity calculation all interact with the evolving topology.

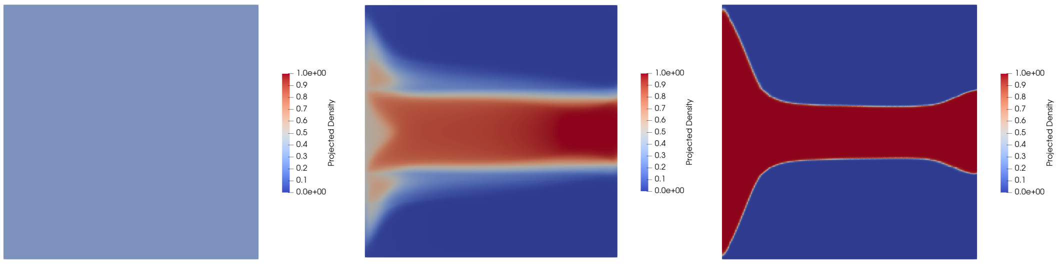
The central gap addressed in this thesis is therefore that density-based fluid topology optimization is far more mature for laminar flows than for turbulent internal flows. Laminar assumptions are attractive because they simplify both the governing equations and the associated adjoint sensitivities, but they are often not representative of compact engineering flow paths operating at industrial Reynolds numbers. In turbulent internal flows, the chosen Reynolds-Averaged Navier-Stokes (RANS) closure, near-wall treatment, and response to curvature or separation can influence the optimized topology itself. Before turbulent conjugate heat-transfer optimization can be treated as a reliable design tool, the underlying turbulent flow-only optimization problem must first be stabilized, verified, validated, and interpreted critically.

1.2 Conceptual Overview of Density-Based Topology Optimization

To understand the challenges of applying optimization to turbulent fluids, it is first necessary to distinguish topology optimization from traditional design methodologies. Historically, computational design relied on size optimization, which changes the thickness of predefined components, or shape optimization, which moves the boundary nodes of a predefined geometry. While effective, these methods are inherently limited by their starting topology. A shape optimization algorithm cannot spontaneously create a new hole or split a single fluid channel into two.

Density-based topology optimization, pioneered by Bendsoe and Kikuchi in 1988 for solid mechanics [13], approaches the problem differently. Instead of moving boundaries, the design space is discretized into a fixed grid of finite elements. The algorithm then determines the optimal distribution of material across this grid. In fluid problems, this is achieved by assigning a continuous design variable, $\gamma \in [0, 1]$, to every element in the domain. In the convention adopted for this thesis, $\gamma = 1$ represents a pure fluid channel, while $\gamma = 0$ represents a solid, impermeable structure.

Because gradient-based optimizers require continuous and differentiable functions, this binary choice is mathematically relaxed. The design variable is allowed to take intermediate "gray" values between 0 and 1. To give physical meaning to these intermediate states, the fluid equations are modified using a fictitious porous medium approach, where intermediate densities correspond to varying levels of flow permeability. The optimization algorithm iteratively updates the design field to minimize a specific objective function, such as pressure loss or dissipated power, while satisfying constraints such as a maximum allowable fluid volume fraction. In the conceptual figures in Figure 1.1, the displayed field should be interpreted as the physical, projected density used by the flow equations. Chapter 5 distinguishes this field from the raw optimizer variable and the filtered density.



(A) Initial projected field (B) Continuous distribution (C) Discrete topology

FIGURE 1.1: Iterative density-based topology optimization process: initialization with a uniform projected density field, intermediate continuous material distribution, and final discrete fluid topology. The full raw-filtered-projected density chain is introduced in Chapter 5.

1.3 The Evolution of Fluid Topology Optimization

While density-based topology optimization became a standard tool in structural engineering in the late 1990s, its application to fluid dynamics is relatively recent. The optimization of fluid manifolds remains an active and highly challenging area of research [21]. The genesis of fluid topology optimization is widely attributed to Borrvall and Petersson (2003), who successfully applied the technique to minimize power dissipation in creeping Stokes flow [14]. This was quickly followed by Gersborg-Hansen et al. (2005), who extended the framework to steady, laminar Navier-Stokes flows [23]. Over the past decade, the literature surrounding laminar flow optimization expanded extensively, adapting the methodology to standard incompressible Navier-Stokes equations [18, 4, 59].

However, the assumption of laminar flow significantly simplifies both the governing equations and the derivation of adjoint sensitivities required for gradient-based optimization. Unfortunately, this assumption prevents the application of these methodologies to many industrial and high-performance applications, where flow velocities are high and turbulence dominates the flow physics. Only recently has the computational mechanics community begun to tackle the severe non-linearities and numerical instabilities associated with optimizing geometries directly in the turbulent regime [56, 19, 57].

Extending topology optimization to the turbulent regime is highly non-trivial. It requires coupling optimization algorithms with Reynolds-Averaged Navier-Stokes (RANS) turbulence models. These models introduce steep velocity gradients, highly non-linear source terms, near-wall resolution requirements, and severe numerical instabilities during the iterative optimization process. To mitigate this, gradient-based turbulent optimizations frequently rely on the “frozen turbulence” assumption, where the turbulent viscosity field is updated in the forward solve but treated as fixed during sensitivity assembly. While this assumption provides necessary solver stability and has driven success in industrial adjoint shape optimization for automotive aerodynamics [36], advancing toward fully coupled turbulent sensitivities in topology optimization remains an ongoing mathematical challenge.

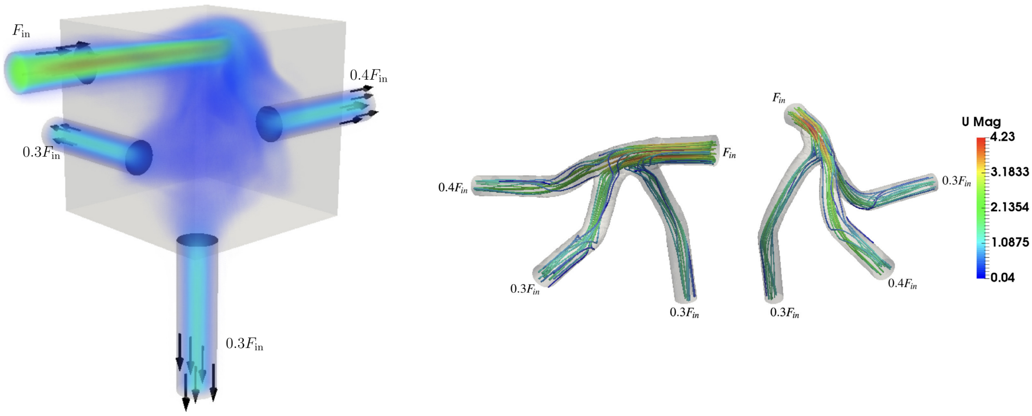


Figure 1.2: Topology optimized manifolds with turbulent flow. Adapted from Dilgen et al. [19].

Pioneering works by Yoon (2016) [56] and Dilgen et al. (2018) [19] have pushed the boundaries of turbulent frameworks. As seen in Figure 1.2, integrating turbulence models allows the optimizer to generate complex fluid manifolds tailored to guide turbulent flow distributions efficiently.

1.4 Objectives and Research Questions

The main objective of this Master's thesis is to extend density-based topology optimization to internal flow configurations operating in the turbulent regime. The phrase "flow configurations" in the title is intentional: the work targets the hydraulic part of energy and thermal-management devices, not the complete heat-transfer device. To achieve this goal, the thesis investigates the coupling of incompressible flow, RANS turbulence modeling, Brinkman topology penalization, and gradient-based optimization within an automated finite-element framework. By isolating the fluid mechanics from heat-transfer calculations, the thesis bounds its scope to establish a stable and verifiable baseline for future turbulent thermal design.

Rather than relying solely on black-box commercial CFD software, the core framework is implemented in **FEniCS**, an open-source computing platform for solving partial differential equations [5]. FEniCS and UFL provide the mathematical flexibility required to write finite-element weak forms directly, verify the one-equation Spalart-Allmaras (SA) turbulence model, and assemble the residual derivatives used by the frozen-turbulence adjoint. For transparency and reproducibility, the complete FEniCS source code developed for this thesis is open-source and publicly available at: `https://github.com/TiebertLefebure/TurbulentTO.git`.

At this stage, it is important to distinguish two different comparisons that appear later in the thesis. A comparison between laminar-optimized and turbulent-optimized layouts is useful as supporting design context, because it shows how inertia and turbulence alter the preferred topology. However, this comparison is not treated as the principal CFD validation step. The decisive assessment instead consists of re-analyzing the final SA-optimized geometries in Ansys Fluent, first with the same SA model and then with the SST $k-\omega$ closure.

Based on this methodology, this thesis seeks to answer the following research questions:

1. **How accurately does the Spalart-Allmaras turbulence model predict curvature-dominated internal turbulent flows, and where do its physical limitations appear?**

   Chapter 2 answers this question by combining turbulence-model theory, Dean-number-based literature evidence, and Ansys Fluent comparisons on curved internal-flow benchmarks. The goal is not to prove a universal Dean-number threshold, but to define the operating range in which SA is credible for the present thesis.

2. **Can a custom FEniCS implementation of the Spalart-Allmaras turbulence model achieve quantitative agreement with Ansys Fluent for turbulent internal-flow benchmarks?**

   Chapters 3 and 4 answer this question by presenting the weak forms, implementing the standalone SA solver, and comparing it against Ansys Fluent on a turbulent channel and a U-bend benchmark.

3. **To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis with SA and SST $k-\omega$, and how do they compare with laminar-optimized designs under identical turbulent operating conditions?**

   Chapter 6 answers this question through frozen-sensitivity verification, FEniCS optimization results, sharp-geometry Ansys-SA validation, SST $k-\omega$ re-analysis, and laminar-versus-turbulent design comparisons for the pipe-bend, U-bend, and diffuser benchmarks.

1.5 Outline of the Thesis

The remainder of this thesis is structured into the following six chapters:

- **Chapter 2 (Turbulence Modeling):** Provides the theoretical background on fluid dynamics and turbulence. It discusses the Navier-Stokes equations, RANS modeling, near-wall treatment, the physical characteristics of the Spalart-Allmaras model, and the curvature limits relevant to this thesis.

- **Chapter 3 (Implementation of the Spalart-Allmaras Model in FEniCS):** Details the numerical implementation of the standalone SA solver in FEniCS, including weak forms, IPCS flow solves, Picard coupling, and wall-distance computation.

- **Chapter 4 (Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows):** Verifies the standalone FEniCS-SA forward solver against Ansys Fluent on turbulent channel-flow and U-bend benchmarks.

- **Chapter 5 (Turbulent Topology Optimization Methodology):** Introduces the density-based optimization framework. It details Brinkman momentum penalization, topology penalties, reciprocal wall-distance penalization, filtering, projection, MMA, and the frozen-turbulence adjoint.

- **Chapter 6 (Turbulent Topology Optimization Results):** Presents the turbulent topology optimization results for the pipe-bend and U-bend benchmarks from Bayat et al. and the diffuser benchmark from Yoon, documents the pipe-bend sensitivity verification, and evaluates extracted designs with both SA and SST $k - \omega$ turbulence models.

- **Chapter 7 (Conclusion and Future Work):** Summarizes the main findings, answers the research questions conservatively, and outlines remaining limitations and future extensions toward three-dimensional and conjugate heat-transfer applications.

Chapter 2

Turbulence Modeling

2.1 Introduction to Fluid Dynamics and Turbulence

Fluid dynamics is governed by the Navier-Stokes equations, which describe the conservation of mass and momentum for a viscous continuum fluid [54]. In this introductory text, these equations are presented using index notation to clearly illustrate the individual tensor components. The indices i and j represent the three spatial dimensions, and a repeated index within a term implies summation over all spatial directions. In subsequent chapters dealing with finite element implementation, this notation shifts to standard vector calculus to align with the FEniCS Unified Form Language syntax.

The fundamental principle of mass conservation is expressed by the continuity equation. For a general compressible fluid, this is defined as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u_i)}{\partial x_i} = 0 \tag{2.1}$$

where ρ is the fluid density. For most liquid cooling applications and low-speed fluid manifolds, the fluid can be assumed incompressible. Under this assumption, ρ is constant in space and time, and the continuity equation reduces to

$$\frac{\partial u_i}{\partial x_i} = 0. \tag{2.2}$$

For an incompressible Newtonian fluid with constant dynamic viscosity μ, the conservation of momentum is expressed as:

$$\rho \left(\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \tag{2.3}$$

where p represents the static pressure. In principle, solving Equation 2.3 directly yields the exact flow field. This approach is known as Direct Numerical Simulation (DNS). However, as the Reynolds number (Re) increases, the flow transitions into a chaotic turbulent state characterized by a broad spectrum of spatial and temporal scales [38]. Resolving the smallest turbulent eddies requires extremely fine spatial grids. For industrial applications, including electronics cooling manifolds, DNS is computationally prohibitive. Engineers therefore rely on turbulence modeling to predict the averaged effects of these fluctuations without resolving the full turbulent spectrum [52, 6].

2.2 Reynolds-Averaged Navier-Stokes (RANS) Equations

The most common approach to turbulence modeling in engineering is the Reynolds-Averaged Navier-Stokes (RANS) framework, based on the theories originally introduced by Osborne Reynolds [38]. The core principle of RANS is the Reynolds decomposition, which separates instantaneous flow variables into a time-averaged mean component ($\bar{u}_i$, $\bar{p}$) and a fluctuating component (u_i', p'):

$$u_i = \bar{u}_i + u_i' \quad \text{and} \quad p = \bar{p} + p'. \tag{2.4}$$

Figure 2.1 illustrates this separation for velocity and pressure signals measured at a fixed point in a turbulent flow. The instantaneous signal contains rapid turbulent fluctuations, while the dashed mean line represents the slowly varying quantity retained by the RANS equations. By construction, the mean of the fluctuating component is zero, but products of fluctuations, such as $\overline{u_i'u_j'}$, remain non-zero and generate the Reynolds-stress terms.

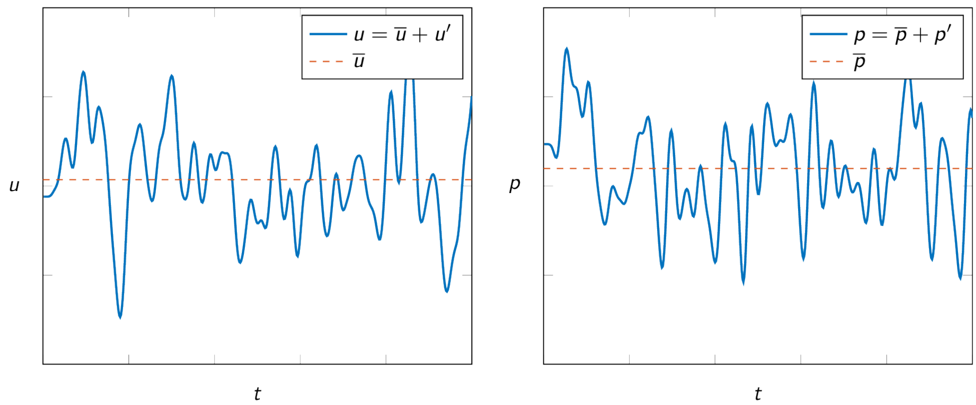


Figure 2.1: Reynolds decomposition of velocity and pressure signals into mean and fluctuating components. Adapted from the Fluid Mechanics on the Web Chapter 6 material based on White's viscous-flow treatment [34].

Substituting these decomposed variables into the standard Navier-Stokes equations and taking the time average yields the RANS momentum equations:

$$\rho\left(\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j}\right) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu \frac{\partial \bar{u}_i}{\partial x_j} - \rho\overline{u_i'u_j'}\right). \tag{2.5}$$

The averaging process introduces a new term, $-\rho\overline{u_i'u_j'}$, known as the Reynolds stress tensor. This tensor represents the momentum transfer caused by turbulent fluctuations. The introduction of this term creates a mathematical closure problem because the RANS equations now contain more unknowns than available physical equations [52].

2.2.1 The Boussinesq Hypothesis and the Isotropic Assumption

To close the system and solve for the mean flow, most RANS models rely on the Boussinesq eddy viscosity hypothesis [38]. This hypothesis assumes that turbulent momentum transfer is analogous to molecular momentum transfer. It relates the Reynolds stresses linearly to the mean velocity gradients using a scalar turbulent eddy viscosity (μ_t):

$$-\rho\overline{u_i'u_j'} = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3}\rho k \delta_{ij} \tag{2.6}$$

where k is the turbulent kinetic energy and δ_{ij} is the Kronecker delta. The primary task of any Boussinesq-based turbulence model is to calculate this local eddy viscosity.

It is crucial to highlight a fundamental physical limitation of Equation 2.6: the Boussinesq hypothesis assumes that μ_t is an isotropic scalar quantity. It maps all turbulence effects into one scalar multiplier aligned with the mean strain rate. While this assumption is often adequate for attached shear flows, it is less reliable when curvature, rotation, or separation create Reynolds-stress anisotropy. Curved ducts and bends are especially relevant here because centrifugal effects redistribute momentum across the cross-section and can create secondary motion that a scalar eddy-viscosity closure cannot represent exactly [41, 40].

2.3 Boundary Layer Theory and Near-Wall Treatment

Before selecting a specific turbulence equation to calculate μ_t, the physical structure of a turbulent boundary layer must be considered. The boundary layer is traditionally divided into three regions based on the non-dimensional wall distance (y^+) and non-dimensional velocity (u^+):

$$y^+ = \frac{y u_\tau}{\nu}, \quad u^+ = \frac{\bar{u}}{u_\tau} \tag{2.7}$$

where y is the distance from the wall, ν is the kinematic viscosity, and $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity derived from the wall shear stress (τ_w) [38].

Figure 2.2 shows the usual wall-unit subdivision used when discussing RANS wall treatment. The viscous sublayer lies closest to the wall and is controlled mainly by molecular viscosity. The buffer layer is a transition region where molecular and turbulent shear are both important. Farther from the wall, the logarithmic layer follows the classical law-of-the-wall behavior, while the outer wake region becomes increasingly sensitive to pressure-gradient and outer-flow effects [7].

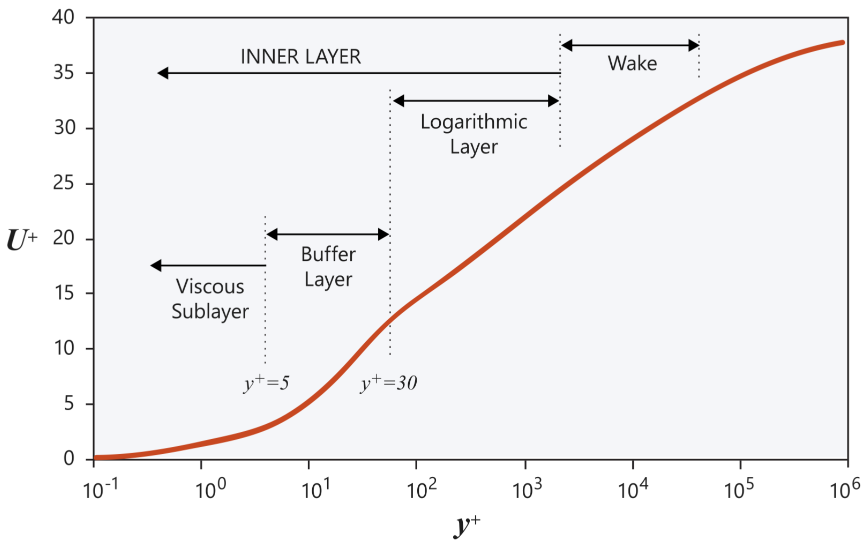


FIGURE 2.2: Near-wall turbulent boundary-layer regions in wall units. Adapted from Altair's AcuSolve near-wall modeling documentation [7].

1. **The viscous sublayer ($y^+ < 5$):** Immediately adjacent to the wall, turbulent fluctuations are damped by kinematic constraints. Viscous shear dominates, and the velocity profile is approximately linear ($u^+ = y^+$).

2. **The buffer layer ($5 < y^+ < 30$):** A transition region where viscous and turbulent shear stresses are both important.

3. **The log-law region ($y^+ > 30$):** Farther from the wall, turbulence dominates and the velocity follows a logarithmic distribution controlled by the von Karman constant [52, 45].

High-Reynolds models, such as the standard $k-\epsilon$ model, use empirical wall functions to bridge the gap between the wall and the log-law region, requiring the first computational mesh element to be placed at $y^+ > 30$. For topology optimization, where solid-fluid boundaries continuously evolve during the optimization loop, empirical wall functions introduce severe mathematical discontinuities and numerical instabilities. Therefore, a wall-resolved model that integrates the governing equations through the viscous sublayer ($y^+ \approx 1$) is required for robust gradient-based optimization.

2.4 The Spalart-Allmaras One-Equation Model

While two-equation models such as $k - \epsilon$ or $k - \omega$ solve separate transport equations for turbulent kinetic energy and a turbulent length scale, they introduce highly non-linear source terms. These terms frequently cause numerical instabilities during the adjoint sensitivity analysis required for topology optimization. To achieve a trade-off between physical accuracy and computational efficiency, this thesis utilizes the Spalart-Allmaras (SA) model, originally developed in 1992 for aerodynamic flows with adverse pressure gradients [43]. The SA model solves a single transport equation for a working variable, $\tilde{\nu}$, which is subsequently linked to the turbulent kinematic eddy viscosity ($\nu_t = \mu_t/\rho$) [44].

The transport equation for the standard SA model is given by:

$$\frac{\partial \tilde{\nu}}{\partial t} + \bar{u}_j \frac{\partial \tilde{\nu}}{\partial x_j} = c_{b1} \tilde{S} \tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2 + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + c_{b2} \left(\frac{\partial \tilde{\nu}}{\partial x_j} \right)^2 \right]. \tag{2.8}$$

The terms on the right-hand side represent production, destruction, and diffusion of the turbulent working variable, respectively. The production term relies on a modified vorticity scalar, $\tilde{S}$, so turbulence generation in the SA model is fundamentally tied to shear. The destruction term is explicitly a function of d, the distance to the nearest wall. This makes the SA model a wall-resolved low-Reynolds model that integrates through the viscous sublayer without requiring empirical wall functions or a second length-scale equation [39].

At this stage of the thesis, the SA model is treated as a turbulence closure rather than an implementation object. Therefore, Chapter 2 only states the main transport equation and its physical role. Detailed constants, damping functions, linearization choices, and weak forms are deferred to Chapter 3, where the FEniCS implementation is discussed.

2.5 Evaluation of Curvature Effects and the Dean Number

Despite its computational advantages for topology optimization, the standard SA model relies on a single length scale tied to the nearest wall distance. It lacks an explicit rotation/curvature correction and remains constrained by the Boussinesq eddy-viscosity assumption. It can therefore struggle to capture the anisotropic physics of streamline curvature.

When fluid passes through a bend, centrifugal forces induce secondary motion perpendicular to the main flow direction. To quantify the strength of these curvature effects, the dimensionless Dean number (De) is utilized:

$$De = Re_d \sqrt{\frac{d}{2R_c}} \tag{2.9}$$

where d is the characteristic length of the cross-section, such as the pipe diameter D or hydraulic diameter D_h, R_c is the radius of curvature, and Re_d is the bulk Reynolds number based on that same characteristic length [17]. A higher Dean number indicates a stronger cross-sectional influence of centrifugal forces.

The Dean number is academically relevant here because it separates curvature severity from Reynolds number alone. However, it should not be interpreted as a universal threshold for SA validity. The breakdown of an eddy-viscosity model depends on geometry, separation, mesh resolution, boundary conditions, and the validation quantity being compared. In this thesis, the Dean number is used as a physically meaningful way to bracket the curvature range of the benchmark cases.

For mildly curved flows, the SA model can perform relatively well. Apalowo [9] evaluated a 90° circular pipe bend with a Dean number of $De \approx 9{,}000$ at $Re_D \approx 34{,}000$. Under these moderate conditions, where $R_c = 7D$, the SA model provided an acceptable approximation of the streamwise velocity profiles compared to higher-order models and experiments. This moderate-curvature agreement is illustrated in Figure 2.3.

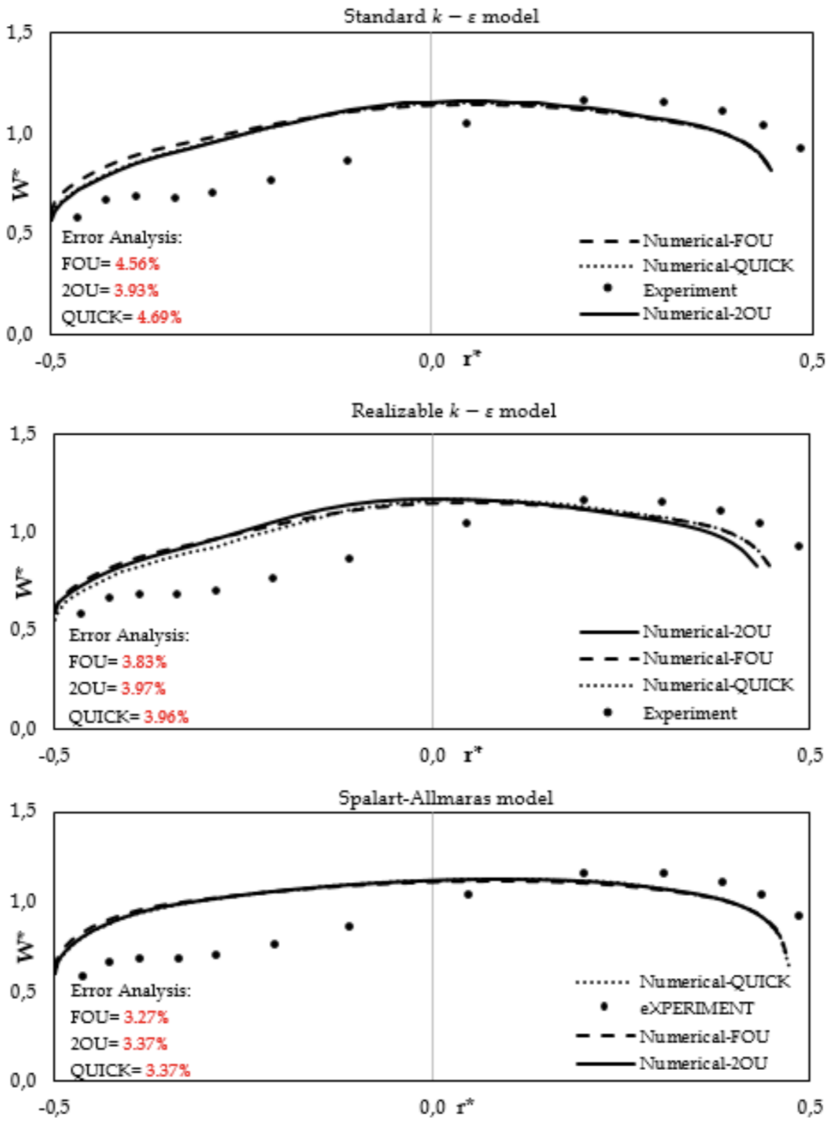


Figure 2.3: Streamwise velocity profiles in a 90° pipe bend ($R_c = 7D$, $Re_D \approx 34{,}000$, $De \approx 9{,}000$). The SA model performs relatively well for mild curvature. Reprinted from Apalowo [9].

As curvature tightens and the Dean number increases, the limitations of the isotropic Boussinesq assumption become more severe. Abuhatira et al. [2] evaluated turbulent flow through a 90° pipe bend with $R_c = 2.0D$ and $Re_D = 60,000$, corresponding to $De \approx 30,000$. In this configuration, the SA model mischaracterized the near-wall velocity profiles at the bend exit, particularly close to the inner wall, while two-equation models resolved the measured trend more accurately. This provides supporting evidence that strong curvature can expose the limitations of the standard SA closure, but it is not treated as proof of a universal failure threshold.

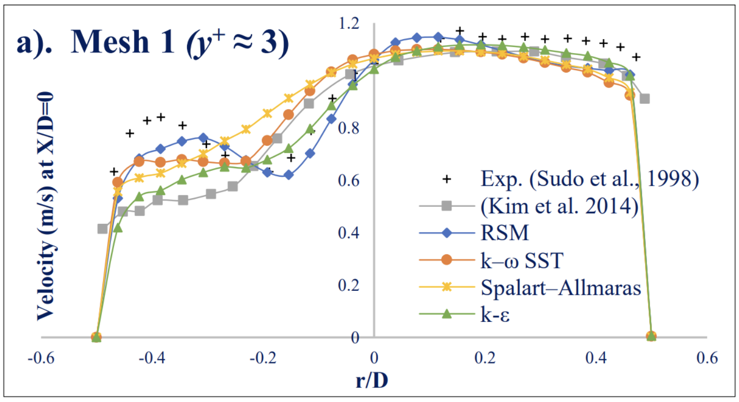


FIGURE 2.4: Turbulence model predictions at the 90° bend exit ($X/D = 0$, $R_c = 2.0D$, $Re_D = 60,000$, $De \approx 30,000$). Due to the strong curvature, the standard Spalart-Allmaras model fails to accurately capture the near-wall velocity profile compared to two-equation closures. Adapted from Abuhatira et al. [2].

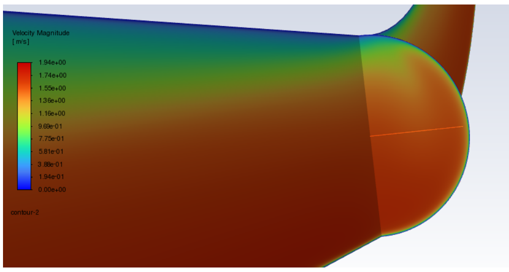
The literature therefore provides useful bracketing evidence: SA can be acceptable for mild curvature and can fail for stronger curvature. The thesis-specific evidence must come from the Ansys comparisons below, which evaluate the geometries and quantities most relevant to the present optimization study.

2.6 Benchmark Studies in Ansys Fluent

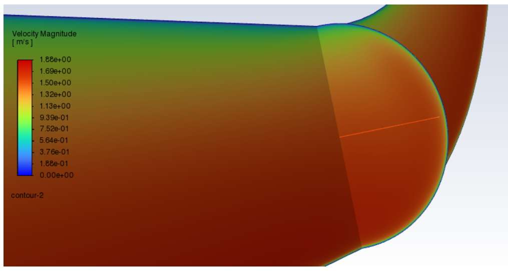
To investigate where the SA model's accuracy deteriorates for the present study, benchmark simulations were performed using **Ansys Fluent**. The study focused on two fully three-dimensional internal flow geometries: a 180° U-bend and a tighter 90° pipe bend. To ensure a standardized numerical baseline, the computational meshes, fluid properties, and boundary conditions were adopted from the official Ansys Fluid Dynamics Verification Manual [8]. Specifically, the 90° bend corresponds to verification case VMFL030, while the 3D U-bend corresponds to case VMFL048. The results were compared directly against the SST $k-\omega$ model [32], an industry standard for adverse pressure gradients and flow separation.

2.6.1 Mild Curvature: The U-Bend Case

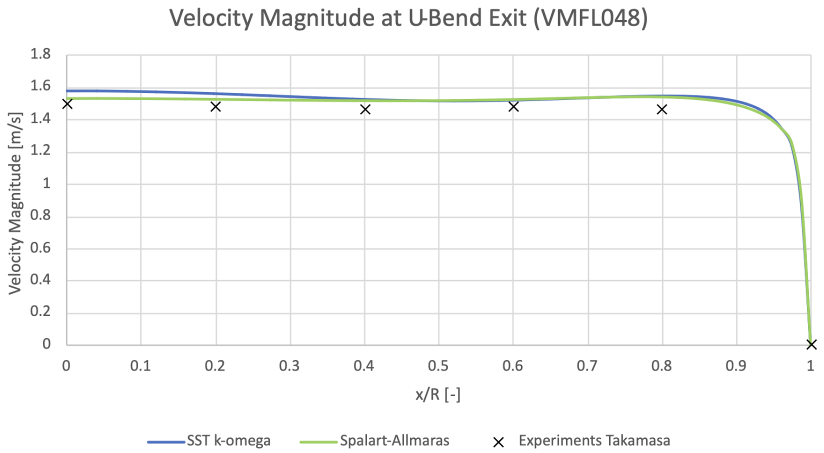
The first benchmark evaluated a 3D U-bend geometry characterized by a moderate curvature ratio of $R_c \approx 4.46D$. Operating at a bulk Reynolds number of $Re_D \approx 44,700$, this geometry resulted in a Dean number of $De \approx 15,000$. Consistent with the moderate-curvature literature evidence, the SA model performed well under these conditions. As shown in Figure 2.5, the qualitative velocity contours of the SA model reproduce the bulk flow distribution of the SST $k - \omega$ model. The velocity profiles in the lower panel are extracted along the pipe-radius line visible on the contour plots above, linking the one-dimensional profile comparison directly to the displayed exit-plane flow field. The quantitative velocity profiles agree well with the experimental data from Takamasa and Kondo [49].



(A) SST $k - \omega$ contour



(B) Spalart-Allmaras contour



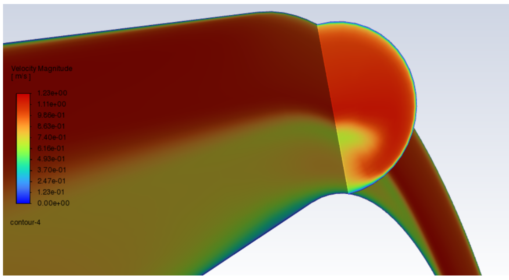
(C) Velocity profiles

Figure 2.5: Flow predictions for a 180° U-bend ($R_c \approx 4.46D$, $Re_D \approx 44,700$, $De \approx 15,000$). The SA model matches the SST $k - \omega$ contours and aligns with the experimental data from Takamasa and Kondo [49].

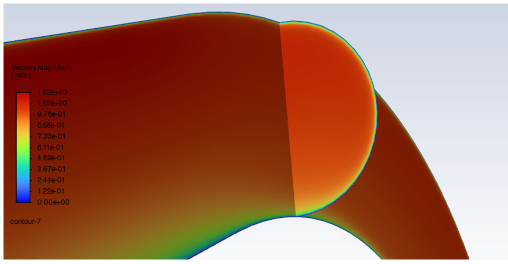
2.6.2 Strong Curvature: The 90° Bend Case

The second benchmark evaluated a tighter 90° pipe bend characterized by $R_c = 2.8D$. Despite operating at a similar bulk Reynolds number of $Re_D \approx 43,000$, this tighter bend radius pushed the Dean number to $De \approx 18,000$. Because the Reynolds numbers are similar between the two benchmark cases, the comparison isolates the geometric effect of streamline curvature more clearly than a Reynolds-number-only comparison.

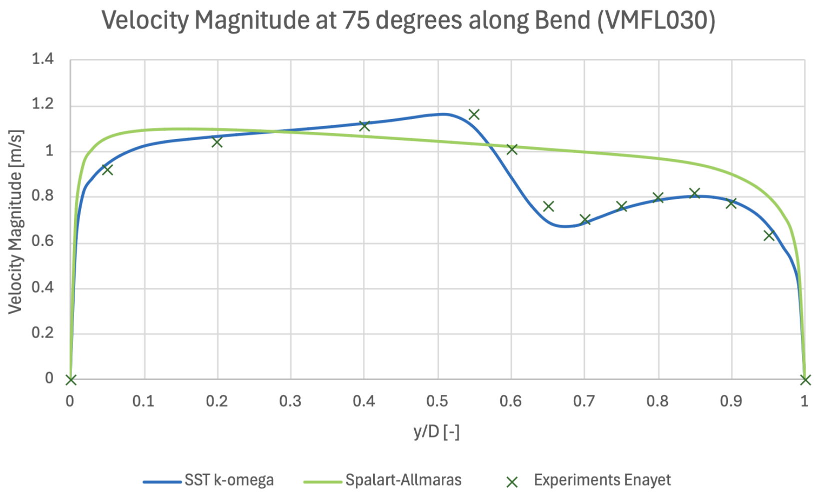
As shown in Figure 2.6, the velocity profile at the bend exit ($\theta = 90°$) demonstrates clear deterioration of the SA prediction. Driven by the tighter bend radius and stronger anisotropic centrifugal effects, the SA model over-predicts the turbulent viscosity. This causes it to over-smooth the velocity profile, failing to capture the sharp velocity gradients near the inner wall ($y/D = 0$) when compared against both the SST $k-\omega$ prediction and the experimental data from Enayet et al. [20].



(A) SST $k-\omega$ contour



(B) Spalart-Allmaras contour



(C) Velocity profiles

FIGURE 2.6: Flow predictions at the 90° pipe bend exit ($R_c = 2.8D$, $Re_D \approx 43,000$, $De \approx 18,000$). The SA model over-smooths the velocity gradients near the inner wall, deviating from both the SST $k-\omega$ model and the experimental data from Enayet et al. [20].

2.7 Conclusion

This chapter answers the first research question regarding the predictive capabilities and physical limitations of the Spalart-Allmaras model for curvature-dominated internal turbulent flows. The SA model is attractive for topology optimization because it is a one-equation, wall-resolved closure that avoids empirical wall functions and remains substantially easier to stabilize than two-equation alternatives. The channel-like and mildly curved evidence supports its use as a practical forward model for wall-resolved internal-flow topology optimization.

The limitation is not simply high Reynolds number. The main issue is curvature- and separation-driven Reynolds-stress anisotropy under a scalar eddy-viscosity closure. Literature cases at $De \approx 9,000$ and $De \approx 30,000$ provide useful supporting evidence, while the thesis-specific Ansys comparisons bracket the present operating range: the U-bend case at $De \approx 15,000$ remains credible, whereas the tighter 90° bend at $De \approx 18,000$ already shows clear SA deterioration for the selected profile comparison. These values are treated as case-specific guidance, not universal thresholds.

Consequently, the SA model is suitable as the baseline turbulence model for the FEniCS topology optimization framework, but final topologies cannot be accepted on the SA result alone. Any complex manifold generated by the SA optimizer must be re-analyzed with an independent turbulence closure, such as SST $k-\omega$, to assess whether the optimized low-loss behavior survives a change in turbulence model.

Chapter 3

Implementation of the Spalart-Allmaras Model in FEniCS

3.1 Introduction

With the theoretical foundation of the Spalart-Allmaras (SA) model established in Chapter 2, this chapter details its numerical translation into a working FEniCS solver. The focus is methodological: the finite-element weak forms, Picard coupling strategy, pressure-correction flow solver, SA transport equation, and relaxed Eikonal wall-distance calculation are described before the solver is verified separately in Chapter 4.

The numerical simulations and eventual topology optimization in this thesis are built upon FEniCS, an open-source computing platform for solving partial differential equations (PDEs). Unlike commercial computational fluid dynamics (CFD) software such as Ansys, which typically relies on the Finite Volume Method (FVM), FEniCS uses the Finite Element Method (FEM) and requires the user to explicitly define the mathematical formulation of the problem [30]. FEniCS is useful in this thesis because the same weak-form language used to assemble the forward equations can also be used to assemble residual derivatives for the topology-optimization adjoint in Chapter 5.

To solve a PDE in FEniCS, the governing equations (the "strong form") must be mathematically transformed into a weak formulation, or variational form. FEniCS utilizes the Unified Form Language (UFL) to express these weak forms in Python code that closely mirrors the mathematical notation [28]. This high-level mathematical scripting is particularly advantageous for topology optimization, as it makes the connection between the implemented residuals and the sensitivity equations explicit.

To ensure complete transparency and reproducibility, the entire custom codebase developed for this thesis is available in the public GitHub repository `TurbulentTO`. To maintain modularity, the repository is divided into two primary directories. The `TurbulenceModels` directory contains the forward-flow Spalart-Allmaras solver and the verification benchmarks reported in Chapter 4. The `FluidTO` directory builds on this verified forward solver by adding the optimization algorithms, Brinkman and turbulence penalties, objective functions, and sensitivity calculations detailed in Chapter 5.

3.1.1 Derivation of a Weak Formulation

Before diving into the complex turbulence equations, it is instructive to demonstrate how a continuous PDE is translated into a discrete FEniCS formulation. To illustrate this methodology, consider the Poisson equation for pressure, which is a fundamental component of incompressible fluid solvers. The strong form of the Poisson equation is given by:

$$-\nabla^2 p = f \quad \text{in}\, \Omega \tag{3.1}$$

where p is the pressure field, f is a source term derived from the velocity divergence, and Ω is the computational domain. To obtain the weak form, the equation is multiplied by a sufficiently smooth test function v and integrated over the entire domain Ω, following standard Galerkin finite element procedures [30, 28]:

$$-\int_\Omega (\nabla^2 p) v \, dx = \int_\Omega f v \, dx. \tag{3.2}$$

Applying integration by parts to the left-hand side yields:

$$\int_\Omega \nabla p \cdot \nabla v \, dx - \int_{\partial\Omega} (\nabla p \cdot \mathbf{n}) v \, ds = \int_\Omega f v \, dx, \tag{3.3}$$

where $\partial\Omega$ is the boundary of the domain and $\mathbf{n}$ is the outward-pointing normal vector. This integral formulation lowers the continuity requirements of the solution, requiring only first-order derivatives to be square-integrable. Neumann boundary conditions can also be substituted directly into the boundary integral. This variational approach forms the foundation of all fluid and turbulence equations implemented in this thesis.

3.2 Numerical Solution Strategy for Fluid Dynamics

Solving the full Reynolds-Averaged Navier-Stokes (RANS) equations is numerically challenging due to the non-linear convective terms, the saddle-point nature of the coupled velocity-pressure system, and the intense cross-coupling with the turbulence model. To achieve a stable and efficient steady-state solution in FEniCS, the verification solver in this chapter uses segregated Picard coupling between the velocity-pressure and SA fields.

3.2.1 Decoupling the Physics: The Picard Iteration Scheme

Because the momentum equation relies on the turbulent eddy viscosity (ν_t), and the turbulence transport equation relies on the convective velocity ($\mathbf{u}$), solving the entire fluid-turbulence system simultaneously in a massive monolithic matrix frequently leads to ill-conditioning and solver divergence.

To resolve this, the forward physics are decoupled using a **Picard iteration scheme** (successive substitution), a robust approach for segregated RANS solvers [22, 52]. During each Picard step, the flow solver advances using the eddy viscosity from the previous iteration. The newly updated velocity field is then passed to the Spalart-Allmaras solver to calculate a new eddy viscosity field. This alternating sequence forms the outer loop of the solver, iteratively updating the decoupled equations until the coupled physical system reaches a steady-state equilibrium.

3.2.2 Incremental Pressure Correction Scheme (IPCS)

Within the Picard loop, the Navier-Stokes equations themselves must be solved. Rather than solving for velocity and pressure simultaneously, the standalone verification solver utilizes an Incremental Pressure Correction Scheme (IPCS). This algorithmic splitting is based on the fractional-step projection methods originally pioneered by Chorin [16] and Temam [50]. To drive the non-linear convective momentum terms toward a stable steady state, a pseudo-time stepping approach (Δt) is employed for the flow variables. The IPCS algorithm splits the fluid equations into three sequential steps per pseudo-time step:

1. **Tentative velocity step:** The momentum equation is solved using the pressure field from the previous iteration (p^n) to compute an intermediate, non-divergence-free velocity field ($\mathbf{u}^*$). Because this is a RANS framework, the diffusion term utilizes the effective kinematic viscosity, $(\nu + \nu_t)$. Treating the convective term explicitly and the diffusive term implicitly, the semi-discrete equation is:
$$\frac{\mathbf{u}^* - \mathbf{u}^n}{\Delta t} + (\mathbf{u}^n \cdot \nabla)\mathbf{u}^* - \nabla \cdot ((\nu + \nu_t^n)\nabla \mathbf{u}^*) = -\nabla p^n. \tag{3.4}$$
2. **Pressure correction step:** A Poisson equation is solved to find the pressure variation that enforces the continuity constraint ($\nabla \cdot \mathbf{u}^{n+1} = 0$):
$$\nabla^2 p^{n+1} = \nabla^2 p^n + \frac{1}{\Delta t}\nabla \cdot \mathbf{u}^*. \tag{3.5}$$
3. **Velocity update step:** The tentative velocity is corrected using the gradient of the pressure difference to yield a mass-conserving velocity field:
$$\mathbf{u}^{n+1} = \mathbf{u}^* - \Delta t \nabla (p^{n+1} - p^n). \tag{3.6}$$

In finite-element form, the three IPCS subproblems are written on a velocity space V_h and pressure space Q_h. For all test functions $\mathbf{v} \in V_h$ and $q \in Q_h$, the tentative velocity step solves

$$\begin{aligned} \int_\Omega \frac{1}{\Delta t}\mathbf{u}^* \cdot \mathbf{v}\, dx + \int_\Omega ((\mathbf{u}^n \cdot \nabla)\mathbf{u}^*) \cdot \mathbf{v}\, dx + \int_\Omega (\nu + \nu_t^n)\nabla \mathbf{u}^* : \nabla \mathbf{v}\, dx \\ = \int_\Omega \frac{1}{\Delta t}\mathbf{u}^n \cdot \mathbf{v}\, dx - \int_\Omega \nabla p^n \cdot \mathbf{v}\, dx. \end{aligned} \tag{3.7}$$

The pressure correction is then obtained from

$$\int_\Omega \nabla p^{n+1} \cdot \nabla q \, dx = \int_\Omega \nabla p^n \cdot \nabla q \, dx - \int_\Omega \frac{1}{\Delta t}(\nabla \cdot \mathbf{u}^*) q \, dx, \tag{3.8}$$

and the corrected velocity is recovered through

$$\int_\Omega \mathbf{u}^{n+1} \cdot \mathbf{v} \, dx = \int_\Omega \mathbf{u}^* \cdot \mathbf{v} \, dx - \int_\Omega \Delta t \, \nabla(p^{n+1} - p^n) \cdot \mathbf{v} \, dx. \tag{3.9}$$

In the forward-flow verification scripts, pressure is treated as a kinematic pressure and the viscous operator uses the non-symmetric diffusion form $(\nu + \nu_t)\nabla\mathbf{u}$. This is the forward turbulent-flow solve that is verified in Chapter 4 and then extended in Chapter 5. The topology-optimization implementation retains the same SA transport model, wall-distance machinery, and segregated forward-flow update, while adding Brinkman resistance, topology-dependent turbulence penalties, and a final frozen-viscosity residual with dynamic viscosity and the symmetric strain tensor for adjoint sensitivity assembly.

The simulation is deemed converged to a steady state when the L_2 norm of the residual difference between successive Picard iterations falls below a defined tolerance limit (ϵ_{tol}) for both the fluid and turbulence fields.

3.3 Detailed Formulation of the Spalart-Allmaras Model

As concluded in Chapter 2, the Spalart-Allmaras (SA) one-equation model was selected for its balance of physical accuracy and numerical stability. Unlike the pseudo-transient flow equations, the SA transport equation is solved in its steady-state form within the Picard loop. The steady transport equation for the continuous nonlinear working variable $\tilde{\nu}$ is given by:

$$\mathbf{u} \cdot \nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \tag{3.10}$$

where ν is the molecular kinematic viscosity, $\sigma = 2/3$ is the turbulent Prandtl number, c_{b1} and c_{w1} are calibrated closure constants, $\tilde{S}$ is a modified vorticity magnitude, f_w is the wall-damping function, and y is the physical distance to the nearest solid boundary [44].

3.3.1 Closure Terms and Calibration Constants

The Spalart-Allmaras model relies on several interconnected closure functions to accurately capture boundary layer physics. The production term relies on a modified vorticity magnitude, $\tilde{S}$, which prevents the destruction of turbulence in regions where the vorticity goes to zero, while maintaining correct scaling near the wall [44]. It is defined as:

$$\tilde{S} = S + \frac{\tilde{\nu}}{\kappa^2 y^2} f_{v2} \tag{3.11}$$

where $S = \sqrt{2\Omega_{ij}\Omega_{ij}}$ is the scalar magnitude of the vorticity tensor, κ is the von Karman constant, and y is the distance to the nearest wall. The function f_{v2} is a near-wall modifier:

$$f_{v2} = 1 - \frac{\chi}{1+\chi f_{v1}} \quad \text{where} \quad \chi = \frac{\tilde{\nu}}{\nu} \quad \text{and} \quad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}. \tag{3.12}$$

Similarly, the wall destruction term is modulated by the non-dimensional function f_w, which accelerates the decay of turbulent viscosity inside the viscous sublayer:

$$f_w = g\left(\frac{1+c_{w3}^6}{g^6+c_{w3}^6}\right)^{1/6} \tag{3.13}$$

where the intermediate variables g and r are evaluated as:

$$g = r + c_{w2}(r^6 - r) \quad \text{and} \quad r = \min\left(\frac{\tilde{\nu}}{\tilde{S}\kappa^2 y^2}, 10\right). \tag{3.14}$$

The upper bound of 10 on r ensures numerical stability during solver initialization when $\tilde{S}$ may momentarily approach zero.

The original formulation by Spalart and Allmaras [44] includes additional geometric trip functions (f_{t1} and f_{t2}) designed to manually trigger the transition from laminar to turbulent flow at specific user-defined coordinates. Following the topology optimization methodology established by Yoon [56], these transition functions are omitted from this thesis. Because the topological boundaries are unknown a priori and evolve during optimization, tracking discrete trip points is mathematically infeasible; therefore, the model assumes a fully turbulent flow regime.

To close the system, the standard empirical calibration constants formulated by Spalart and Allmaras are utilized throughout the FEniCS implementation [39]:

$$\begin{aligned} c_{b1} &= 0.1355 & \sigma &= 2/3 & c_{b2} &= 0.622 \\ \kappa &= 0.41 & c_{w2} &= 0.3 & c_{w3} &= 2.0 \\ c_{v1} &= 7.1 & c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}. \end{aligned}$$

3.3.2 Wall-Distance Calculation via the Relaxed Eikonal Equation

A major computational challenge in the SA model is the calculation of the wall distance y. In a standard static CFD simulation, y is purely geometric and can be pre-computed. However, to ensure this verification solver translates into the topology optimization code, where solid boundaries evolve continuously, the implementation uses a PDE-based approach.

The true geometric wall distance satisfies the standard Eikonal equation, $|\nabla y| = 1$. However, this formulation contains singularities at the medial axes of the domain. To guarantee continuous differentiability, a relaxed reciprocal-distance equation proposed by Yoon [56] is solved:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) = (1 + 2\sigma_w)G^4. \tag{3.15}$$

Here, G is an intermediate reciprocal-distance field variable. The parameter σ_w is a relaxation constant, typically set to 0.1, that controls the smoothness of the field. The PDE is subjected to a Dirichlet boundary condition at the physical walls, $G = G_0$, where G_0 is an arbitrarily large constant. Once G is solved over the domain, the physical wall distance y is recovered algebraically using the inverse relationship:

$$y = \frac{1}{G} - \frac{1}{G_0}. \tag{3.16}$$

The code applies small numerical floors and non-negativity safeguards during this recovery to prevent division-by-zero and roundoff artifacts, but the mathematical equation used in the thesis remains the Yoon formulation above.

This system is implemented as a nonlinear weak residual for G. After multiplying by a scalar test function and integrating the $\sigma_w G \nabla^2 G$ term by parts, the residual contains only first derivatives of G:

$$(1-\sigma_w)\int_\Omega |\nabla G|^2 z\, dx - \sigma_w \int_\Omega G \nabla G \cdot \nabla z\, dx - (1+2\sigma_w)\int_\Omega G^4 z\, dx = 0. \tag{3.17}$$

3.4 Variational Formulation and FEniCS Implementation

The standard Galerkin weak formulation is constructed by multiplying the steady transport Equation 3.10 by a test function ξ and applying integration by parts to the diffusive terms. Because the equation is highly non-linear, robust convergence is dependent on the stability of the external velocity field provided by the Picard outer loop.

To manage this non-linearity, the equation is linearized within the Picard iteration scheme by freezing certain parameters using the most recent value of the working variable, $\tilde{\nu}^0$. Specifically, the turbulence production and cross-diffusion terms are treated as explicit sources evaluated using $\tilde{\nu}^0$. Conversely, the wall destruction term is linearized into an implicit reaction sink by scaling the continuous $\tilde{\nu}$ against the frozen variable, taking the form $c_{w1} f_w (\tilde{\nu}^0/y^2)\tilde{\nu}$.

The finite-element implementation mirrors the analytical weak form using the most recent convective velocity field passed from the flow solver. Let $\tilde{\nu}^k$ denote the previous Picard value of the SA working variable, while $\tilde{\nu}^{k+1}$ is the unknown being solved. The modified vorticity $\tilde{S}^k$, wall function f_w^k, diffusion coefficient, and explicit source terms are evaluated from $\tilde{\nu}^k$ and the current velocity field. The Picard-linearized SA step is: find $\tilde{\nu}^{k+1} \in T_h$ such that, for all scalar test functions $\xi \in T_h$,

$$\begin{aligned}
&\int_\Omega (\mathbf{u}\cdot\nabla\tilde{\nu}^{k+1})\xi\, dx + \frac{1}{\sigma}\int_\Omega (\nu + \tilde{\nu}^k)\nabla\tilde{\nu}^{k+1}\cdot\nabla\xi\, dx \\
&\quad + \int_\Omega c_{w1} f_w^k \frac{\tilde{\nu}^k}{y^2}\tilde{\nu}^{k+1}\xi\, dx \\
&= \int_\Omega \left[c_{b1}\tilde{S}^k\tilde{\nu}^k + \frac{c_{b2}}{\sigma}\nabla\tilde{\nu}^k\cdot\nabla\tilde{\nu}^k\right]\xi\, dx.
\end{aligned} \tag{3.18}$$

The left-hand side contains convection, implicit diffusion, and implicit wall destruction. Production and cross-diffusion remain on the right-hand side as explicit Picard sources. After solving Eq. 3.18, the working variable is under-relaxed and bounded before the turbulent eddy viscosity ν_t is recomputed from the standard SA algebraic relation.

3.5 Global Benchmark Solver Path

For the verification benchmarks in Chapter 4, the forward solver starts from initialized velocity, pressure, SA working-variable, and wall-distance fields. The relaxed Eikonal problem is first solved on the fixed physical walls. Each Picard cycle then recomputes ν_t from the current $\tilde{\nu}$, advances the velocity-pressure state through the IPCS weak problems in Eqs. 3.7–3.9, solves the SA weak problem in Eq. 3.18, applies relaxation and admissible bounds, and checks the flow and turbulence residuals. Once the residual tolerances are satisfied, the solver exports the velocity, pressure, SA working variable, eddy viscosity, and wall-distance fields used for the verification plots.

3.6 Conclusion

This chapter has established the numerical formulation of the FEniCS-SA forward solver used for the verification benchmarks. The key ingredients are the weak-form finite-element implementation, Picard decoupling of the velocity-pressure and turbulence fields, IPCS pseudo-time stepping for the incompressible flow solve, the steady SA transport equation, and the relaxed Eikonal wall-distance formulation.

Separating these methods from their verification keeps the thesis logic clean: this chapter explains how the forward SA solver is constructed, Chapter 4 verifies its turbulent-flow behavior against Ansys Fluent benchmark cases, and Chapter 5 extends the verified forward solver to density-based topology optimization and frozen adjoint sensitivities.

Chapter 4

Verification of the Spalart-Allmaras Solver for Turbulent Internal Flows

4.1 Introduction

Chapter 3 introduced the numerical implementation of the Spalart-Allmaras (SA) model in FEniCS. Before using this solver inside the topology optimization framework, the forward fluid solver is verified against independent Ansys Fluent calculations on two benchmark flows. The first benchmark is a fully developed turbulent channel, which isolates wall damping, turbulence production, and Picard coupling in a simple geometry. The second benchmark is a curved U-bend, which tests the same implementation under strong convection, streamline curvature, and separation.

This chapter keeps the essential benchmark definitions in the main text: geometry type, boundary conditions, Reynolds number, wall resolution, and the comparison metrics used to judge agreement. The appendices retain the exhaustive mesh hierarchies, solver tolerances, and full configuration tables. This separation keeps the verification argument readable while still making the numerical setup reproducible. The purpose is to verify the forward-flow solver developed in Chapter 3, because this is the turbulent-flow core on which the topology optimization framework is built. Chapter 5 later adds Brinkman penalization, topology-dependent SA and wall-distance penalties, and a frozen-adjoint sensitivity residual, but those additions build on the same SA transport equation, wall-distance computation, and segregated flow-coupling strategy verified here.

All mesh-independence relative differences referenced from Appendix A follow the appendix convention: coarse–medium differences are normalized by the coarse value, while medium–fine differences are normalized by the fine value.

4.2 Channel Verification Benchmark

To verify the fundamental correctness of the custom fluid and turbulence solvers, the code was first tested on a standard 2D turbulent channel flow. To cleanly isolate and verify turbulence production, wall damping, and Picard coupling without the artifacts of a developing boundary layer, this benchmark was formulated as a pressure-driven periodic channel rather than a conventional inlet-outlet duct.

The velocity field ($\mathbf{u}$) and Spalart-Allmaras working variable ($\tilde{\nu}$) were constrained periodically between the streamwise boundaries, while the top and bottom boundaries were treated as standard no-slip walls ($\mathbf{u} = 0, \tilde{\nu} = 0$). Flow was driven entirely by a fixed pressure drop from the upstream plane ($p = 2$ Pa) to the downstream plane ($p = 0$ Pa). Based on a reference bulk velocity of $U_{\text{ref}} = 18.5$ m/s and using the full channel height ($H = 2.0$ m) as the characteristic length scale, the solver was targeted to resolve a fully turbulent flow regime at a channel Reynolds number of $Re_H \approx 20{,}300$. The complete definition of the geometric, physical, and boundary-condition parameters is provided in Table A.1 of Section A.1.1 in Appendix A.

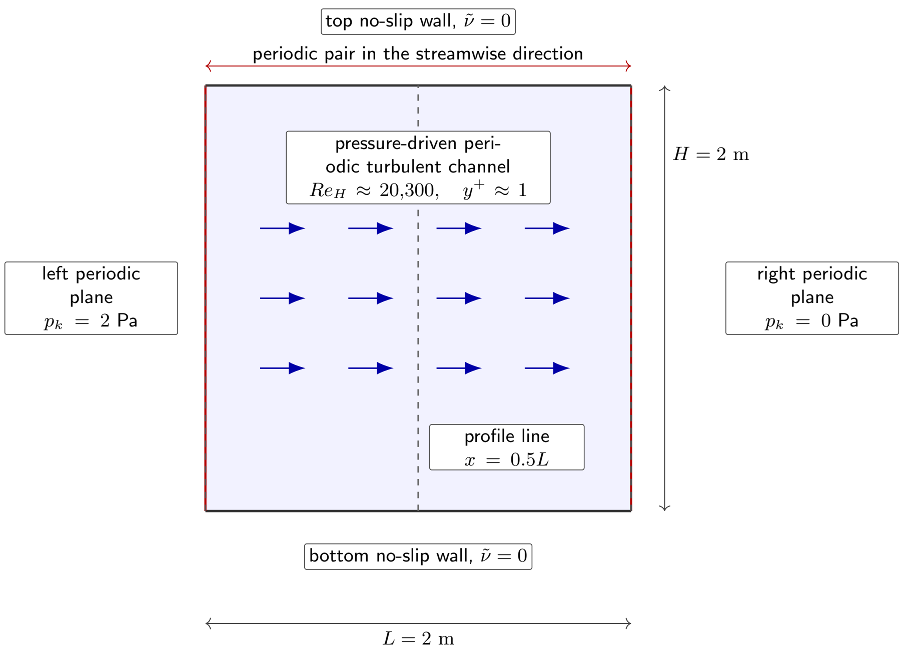


FIGURE 4.1: Schematic of the pressure-driven periodic channel verification setup. The vertical dashed line marks the mid-plane profile extraction location used for the FEniCS–Ansys comparison.

4.2.1 Mesh Resolution Requirements

Because the SA formulation is a low-Reynolds model that does not use empirical wall functions, it requires mathematical integration directly through the viscous sublayer. To satisfy this strict boundary-layer requirement, a custom structured inflation mesh was generated for the channel flow. Because the domain is a simple periodic rectangle, this anisotropic grid was constructed programmatically using NumPy arrays and converted into a FEniCS-compatible format via the `meshio` library. Based on the target bulk Reynolds number, this approach enforced a physical first-cell height of approximately 1.7×10^{-3} m, aiming for a non-dimensional wall distance of $y^+ \approx 1$ across the entire domain.

To verify that the numerical results were not dominated by spatial discretization, a grid-independence study was conducted. The full spatial verification is documented in Section A.1.3 of Appendix A. The mesh hierarchy is detailed in Table A.3, and visual confirmation of convergence is provided by the velocity profiles in Figure A.1.

4.2.2 FEniCS vs. Ansys Comparison

Prior to this comparative analysis, the Ansys baseline itself was verified for grid independence to ensure a robust reference (see Table A.4 and Figure A.2 in Section A.1.3 of Appendix A). The channel flow was then simulated in FEniCS using the decoupled Picard iteration method and compared against the Ansys Fluent baseline.

As shown in Figure 4.2, the qualitative velocity contours of the custom FEniCS implementation show good agreement with the Ansys Fluent baseline. For the quantitative comparison, velocity profiles are extracted from both solvers across the middle of the channel, i.e. at $x = 0.5L$, with y varied from the lower wall to the upper wall. The custom solver resolves the logarithmic region of the turbulent boundary layer and predicts the turbulent diffusion toward the channel center.

Because the channel is driven by a prescribed pressure difference, $\Delta p = 2.0$ Pa is identical by construction and is not a useful comparison metric. The retained global metric is therefore the bulk velocity,

$$U_{\text{bulk}} = \frac{1}{H} \int_0^H u_x(0.5L, y)\,\mathrm{d}y, \tag{4.1}$$

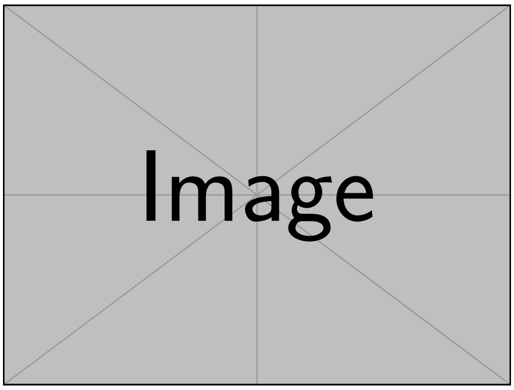
and the corresponding plane-channel skin-friction coefficient,

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho U_{\text{bulk}}^2} = \frac{\Delta p\, H}{\rho L U_{\text{bulk}}^2}. \tag{4.2}$$

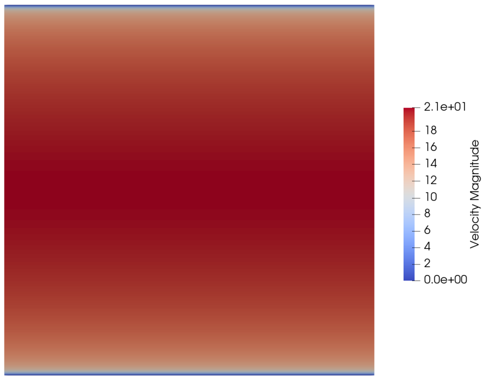
Here u_x is the streamwise velocity component, H is the channel height, L is the streamwise periodic length, τ_w is the mean wall shear stress, ρ is the fluid density, and Δp is the imposed pressure difference. When the solver pressure is treated as kinematic pressure, the same expression is evaluated consistently with $\rho = 1$. Since Δp is fixed, C_f is not independent of U_{bulk}, but it is still useful because it converts the imposed forcing into the wall friction implied by the predicted flow rate.

The two error metrics are cross-code discrepancies, not separate errors for each solver. After interpolating the FEniCS and Ansys profiles onto the same line coordinates y_i, the pointwise difference is

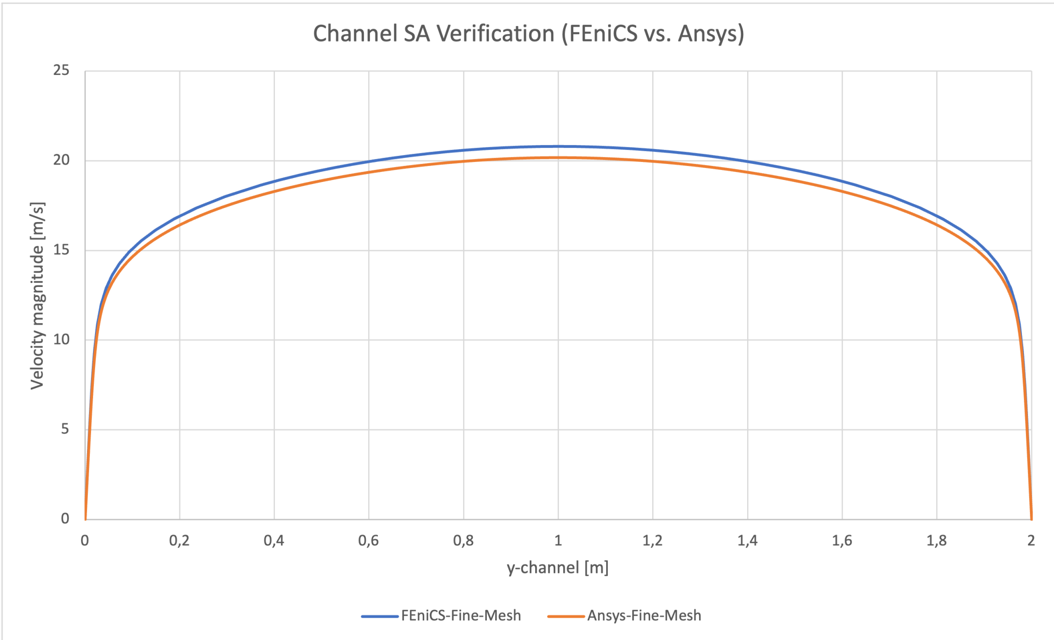
$$e_i = u_{x,\text{FEniCS}}(y_i) - u_{x,\text{Ansys}}(y_i). \tag{4.3}$$



(A) Ansys Fluent velocity magnitude contour



(B) FEniCS velocity magnitude contour



(C) Velocity profiles

FIGURE 4.2: Verification of the 2D channel flow ($Re_H \approx 20{,}300$). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, plus quantitative velocity profiles extracted at the streamwise mid-plane, $x = 0.5L$, while varying the wall-normal coordinate y.

The maximum pointwise velocity discrepancy is normalized by the maximum Ansys profile velocity to avoid singular relative errors at the no-slip walls,

$$E_\infty = \frac{\max_i |e_i|}{\max_i |u_{x,\text{Ansys}}(y_i)|} \times 100\%, \tag{4.4}$$

and the relative profile error is evaluated as a line-integrated norm,

$$E_{L_2} = \left(\frac{\int_0^H |u_{x,\text{FEniCS}} - u_{x,\text{Ansys}}|^2 \,\mathrm{d}y}{\int_0^H |u_{x,\text{Ansys}}|^2 \,\mathrm{d}y} \right)^{1/2} \times 100\%. \tag{4.5}$$

The reported error metrics are therefore cross-code verification discrepancies between the FEniCS and Ansys profiles. They are assessed relative to the grid-independence evidence documented in Appendix A and to the accuracy required before using the FEniCS solver in the topology optimization loop. The quantitative errors and macroscopic metrics are summarized in Table 4.1.

Table 4.1: Quantitative comparison of flow metrics and extracted profile discrepancies for the 2D channel benchmark.

Metric	**Ansys**	**FEniCS**	**Rel. diff.**
Bulk velocity (U_{bulk}) [m/s]	[TBD]	18.472	[TBD]%
Friction coefficient (C_f)	[TBD]	5.861×10^{-3}	[TBD]%
Max pointwise velocity discrepancy (E_∞)	–	–	[TBD]%
Relative L_2 velocity discrepancy (E_{L_2})	–	–	[TBD]%

This agreement supports the verification that the core Navier-Stokes solver, relaxed Eikonal wall-distance field, and SA production/destruction terms behave correctly in a canonical wall-bounded turbulent flow. The main-text parameters are sufficient to interpret the result, while Appendix A provides the detailed mesh and solver settings needed to reproduce it.

4.3 U-Bend Verification Benchmark

Following the channel flow verification, the solver was tested on the highly convective separating U-bend flow evaluated in Chapter 2. This case is closer to the curved internal-flow topologies studied later in the thesis because the flow experiences strong streamline curvature and pressure-gradient-driven redistribution.

The verification used a 2D geometry corresponding to the symmetry mid-plane of the full 3D VMFL048 case. Since the FEniCS domain is mathematically a 2D planar approximation, it cannot resolve true 3D Dean secondary vortices. Instead, this benchmark validates the 2D symmetry-plane approximation and the solver's ability to capture curvature-induced separation and momentum redistribution. The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys verification runs use this same short-leg U-bend geometry.

To drive the flow, a uniform prescribed velocity ($\mathbf{u}_{in} = 1.42$ m/s) was imposed at the inlet alongside a fully turbulent boundary value ($\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s), while a zero relative static pressure ($p = 0$ Pa) was assigned to the outlet. While the domain is 2D, the 3D physical pipe diameter ($D = 0.028$ m) and radius of curvature ($R_c = 0.125$ m) were retained as the characteristic length scales. This preserves the target Reynolds number ($Re_D \approx 44{,}700$) and Dean number ($De \approx 15{,}000$) of the 3D Ansys reference case. The physical setup and boundary conditions are listed in Table A.5 of Section A.2.1 in Appendix A.

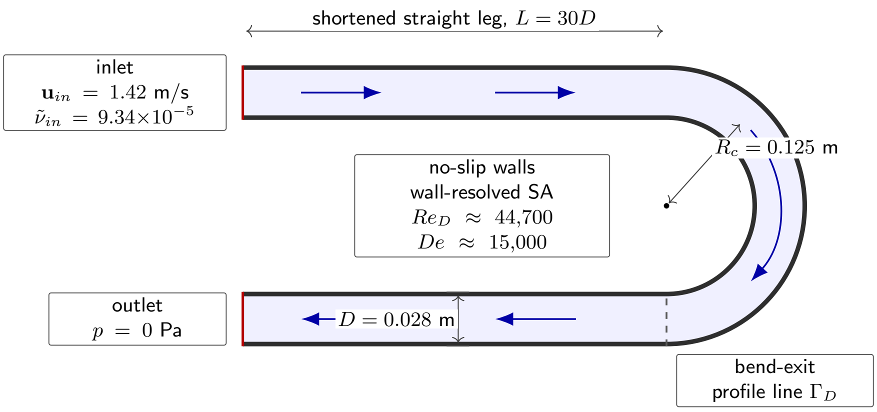


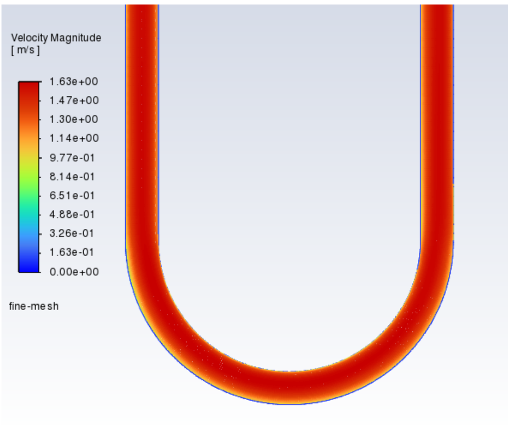
FIGURE 4.3: Schematic of the shortened 2D U-bend verification setup based on the VMFL048 operating point from the Ansys verification manual [8]. The profile line Γ_D is used for the bend-exit velocity comparison.

4.3.1 Mesh Resolution Requirements

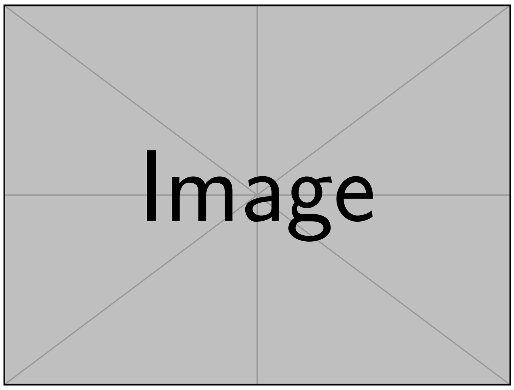
The U-bend geometry features curved inner and outer arcs, complex corner transitions, and anisotropic boundary-layer inflation around non-orthogonal walls. To manage this geometric complexity while satisfying the $y^+ \approx 1$ resolution requirement, a custom mesh was generated using Gmsh [24]. The first element height adjacent to the wall was controlled to 1.20×10^{-5} m across the entire domain. The independence studies are documented in Section A.2.3 of Appendix A: the inflation parameters and element densities of the Gmsh hierarchy are listed in Table A.7, and the extracted velocity profiles confirming grid convergence at the bend exit are illustrated in Figure A.3.

4.3.2 FEniCS vs. Ansys Comparison

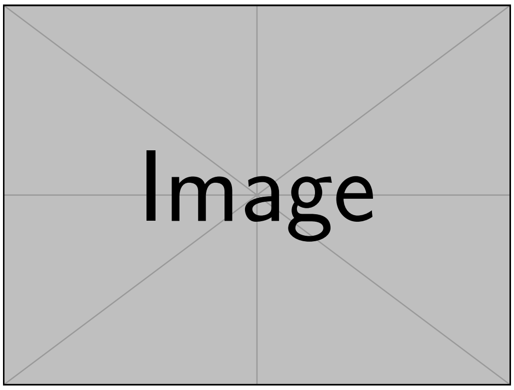
Having validated the spatial resolution of the FEniCS mesh, an equivalent grid-independence study was executed for the commercial Ansys Fluent baseline (detailed in Table A.8 and Figure A.4 of Section A.2.3 in Appendix A). With both solvers operating on spatially converged grids, an independent Ansys Fluent simulation was executed on the shortened 2D domain.



(A) Ansys Fluent SA contour



(B) FEniCS SA contour



(C) Velocity profiles

Figure 4.4: Verification of the 2D U-bend flow ($Re_D \approx 44{,}700$, $De \approx 15{,}000$). The comparison includes qualitative velocity contours from Ansys Fluent and FEniCS, showing separation characteristics, plus quantitative velocity profiles extracted at the bend exit across the pipe diameter $D = 0.028$ m.

As shown in Figure 4.4, the global flow fields and separation regions observed in the qualitative velocity contours demonstrate visual agreement. To provide a more robust assessment, velocity data are extracted along a cross-sectional line probe positioned at the bend exit, spanning the full pipe diameter $D = 0.028$ m from the inner wall to the outer wall. The quantitative velocity profile computed by the custom FEniCS implementation traces the commercial Ansys results closely, adequately capturing the boundary-layer growth and the momentum shift toward the outer wall.

For the U-bend, the inlet velocity is prescribed rather than produced by a fixed pressure gradient, so pressure drop remains a meaningful global comparison metric. A channel-style friction coefficient is not retained because the curved geometry does not have a single fully developed wall-shear relation equivalent to the periodic channel. In FEniCS, the pressure-drop diagnostic is evaluated from boundary-measure-weighted mean pressures on the inlet and outlet facets,

$$\Delta p_{\text{FEniCS}} = \rho \left[\frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p_k \, \mathrm{d}\Gamma - \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} p_k \, \mathrm{d}\Gamma \right], \tag{4.6}$$

where p_k is the kinematic pressure solved by the IPCS formulation, $\rho = 997$ kg/m^3, and $|\Gamma| = \int_\Gamma 1 \, \mathrm{d}\Gamma$. The inlet and outlet integrals correspond to the marked FEniCS facets `ds(2)` and `ds(3)`, respectively, so the reported value is the area-averaged inlet static pressure minus the area-averaged outlet static pressure, converted to pascals. The retained local comparison metrics are the same bend-exit profile discrepancies used for the channel,

$$E_\infty = \frac{\max_i \big| |\mathbf{u}_{\text{FEniCS}}(s_i)| - |\mathbf{u}_{\text{Ansys}}(s_i)| \big|}{\max_i |\mathbf{u}_{\text{Ansys}}(s_i)|} \times 100\%, \tag{4.7}$$

and

$$E_{L_2} = \left(\frac{\int_{\Gamma_D} \left(|\mathbf{u}_{\text{FEniCS}}| - |\mathbf{u}_{\text{Ansys}}| \right)^2 \mathrm{d}s}{\int_{\Gamma_D} |\mathbf{u}_{\text{Ansys}}|^2 \, \mathrm{d}s} \right)^{1/2} \times 100\%, \tag{4.8}$$

where Γ_D denotes the line segment across the bend-exit diameter and s is the corresponding arclength coordinate. Table 4.2 summarizes the quantitative error margins and the macroscopic pressure drop (Δp) across the 2D U-bend domain for both solvers. Since the FEniCS formulation solves for kinematic pressure, its pressure drop is converted to physical static pressure using $\rho = 997$ kg/m^3 before comparison; equivalently, the fine Ansys pressure drop of 792.995 Pa corresponds to $\Delta p/\rho = 0.7954$ m^2/s^2.

TABLE 4.2: Quantitative comparison of flow metrics and extracted profile discrepancies for the 2D U-bend benchmark.

Metric	**Ansys**	**FEniCS**	**Rel. diff.**
Pressure drop (Δp) [Pa]	792.995	[TBD]	TBD%
Max pointwise velocity discrepancy (E_∞)	–	–	[TBD]%
Relative L_2 velocity discrepancy (E_{L_2})	–	–	[TBD]%

4.4 Conclusion

The two verification cases answer the second research question by establishing the practical operating range of the custom FEniCS-SA solver before it is embedded in the topology optimization loop. The periodic channel confirms that the solver reproduces canonical wall-bounded turbulent behavior, including the near-wall damping and production balance that are central to the SA model. The U-bend checks the same implementation in a curved, separation-prone internal flow, where streamline curvature and adverse pressure gradients make the forward-flow prediction more demanding than in the channel. The localized velocity-profile agreement and pressure-drop metric indicate that the framework provides a credible forward-flow basis for the pressure-drop and dissipation objectives in the subsequent topology optimization campaigns.

These results do not remove the known limitations of a 2D SA model: secondary Dean vortices, anisotropic Reynolds stresses, and full three-dimensional separation physics remain outside the present solver scope. They do, however, show that the implemented SA equations, wall-distance calculation, and segregated coupling strategy are sufficiently consistent with Ansys Fluent to support the topology optimization methodology and results presented in Chapters 5 and 6. The verification should therefore be interpreted precisely: it validates the forward turbulent-flow solver that is extended by the optimizer. The separate sensitivity checks in Chapter 6 then verify the additional frozen-adjoint derivative used by MMA.

Chapter 5

Turbulent Topology Optimization Methodology

5.1 Introduction

Building upon the Spalart-Allmaras (SA) fluid dynamics solver [43, 44] developed in Chapter 3 and verified in Chapter 4, this chapter introduces the mathematical framework required to perform Topology Optimization (TO) in the turbulent regime. The present chapter extends the verified forward solver by embedding the same SA transport model and reciprocal wall-distance machinery in a density-based design loop and differentiating the resulting residuals for gradient-based optimization.

The core objective of this methodology is to algorithmically determine the optimal distribution of fluid channels within a given design domain to reduce hydraulic or mechanical energy loss under a prescribed flow condition. This chapter details the density-based fictitious-domain formulation used to represent solid and fluid regions on a fixed computational mesh [14, 23]. It then explains the turbulent-flow ingredients retained for the remainder of the thesis: Brinkman momentum resistance, topology penalties for the SA and wall-distance equations, the reciprocal penalized wall-distance formulation, frozen-turbulence adjoint sensitivities, active-domain filtering, Heaviside projection, and MMA updates.

The topology-optimization code is therefore not a separate turbulence solver. It uses the verified Chapter 3 forward-flow machinery as its starting point and augments it with design-dependent physics. During the optimization loop, IPCS solves are used as robust warm starts and Picard updates. For the adjoint step, the final frozen-viscosity state is assembled as a steady weak residual with Brinkman resistance, dynamic viscosity, and the symmetric strain tensor. This residual is the differentiable state equation used for design sensitivities, while the underlying SA transport and wall-distance updates remain the same forward-model components verified in Chapter 4.

5.2 The Fictitious Domain Method and Brinkman Penalization

In standard shape optimization, the solid-fluid boundary is explicitly defined by the mesh, and the nodes are physically moved to improve the design. In density-based topology optimization, however, the computational domain (Ω) remains strictly fixed. To create solid walls and fluid channels within this fixed grid, the Fictitious Domain Method is employed.

Originally applied to Stokes flow by Borrvall and Petersson [14] and extended to the Navier-Stokes equations by Gersborg-Hansen et al. [23], this method assigns a continuous design variable to the spatial coordinates in the domain. In the convention used here, a physical topology value of $\bar{\gamma} = 1$ represents a pure **fluid** channel, while $\bar{\gamma} = 0$ represents a **solid**, impermeable structure. To force the fluid velocity to zero inside solid regions without mathematically removing the finite elements, the fluid is treated as a fictitious porous medium.

This flow resistance is quantified by an inverse permeability term, or Brinkman penalty function, denoted as $\alpha(\bar{\gamma})$. To ensure the optimization algorithm can transition smoothly between fluid and solid states, α must be continuous and differentiable. The formulation uses a convex Rational Approximation of Material Properties (RAMP) interpolation [46] scaled between a maximum solid penalty (α_{solid}) and a minimum fluid penalty (α_{fluid}):

$$\alpha(\bar{\gamma}) = \alpha_{solid} + (\alpha_{fluid} - \alpha_{solid}) \frac{\bar{\gamma}(1+q)}{\bar{\gamma} + q} \tag{5.1}$$

where α_{solid} is a sufficiently large penalty factor that stagnates the flow when $\bar{\gamma} = 0$, and q is a tuning parameter controlling the convexity of the penalization curve. Equation 5.1 is evaluated using the projected effective density $\bar{\gamma}$, rather than the raw optimizer variable γ, because the physical flow equations are formulated on the filtered, projected, and domain-bounded topology. For visualization, the Brinkman resistance is normalized as

$$\hat{\alpha}(\bar{\gamma}) = \frac{\alpha(\bar{\gamma}) - \alpha_{fluid}}{\alpha_{solid} - \alpha_{fluid}},$$

so that $\hat{\alpha} = 1$ in a fully solid cell and $\hat{\alpha} = 0$ in a fully fluid cell. If $\alpha_{fluid} = 0$, this reduces to $\alpha(\bar{\gamma})/\alpha_{solid}$.

Figure 5.1 shows the normalized resistance curve for increasing values of q. Small values of q create a very steep drop in resistance immediately away from the solid end ($\bar{\gamma} = 0$), which makes the curve almost flat near the fluid end ($\bar{\gamma} = 1$). Larger values of q spread the transition more evenly across intermediate densities. During continuation, the RAMP parameter is adjusted together with the projection steepness to control how aggressively intermediate densities are penalized while keeping the interpolation differentiable. Equivalent Brinkman interpolation curves are sometimes written with a reciprocal penalty parameter, e.g. $\hat{\alpha} = (1 - \bar{\gamma})/(1 + Q\bar{\gamma})$ for $\alpha_{fluid} = 0$, which corresponds to $Q = 1/q$ in the present convention. Unlike traditional structural optimization, which often relies on the SIMP power law [13], fluid optimization generally avoids SIMP for momentum because its derivative can become ineffective near the solid limit. The RAMP formulation retains a useful gradient across the active design domain [46, 48].

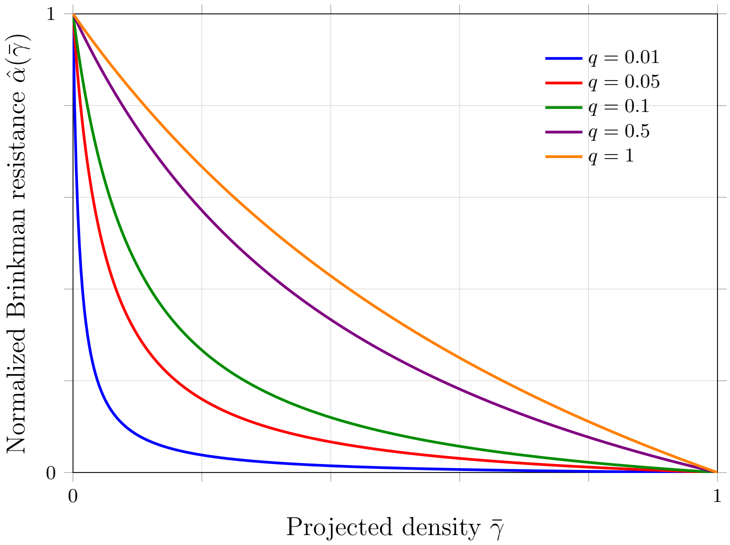


FIGURE 5.1: Normalized RAMP Brinkman resistance from Eq. 5.1 for $q = \{0.01, 0.05, 0.1, 0.5, 1.0\}$, adapted from Stolpe and Svanberg [46].

5.2.1 The Density Chain and Domain Bounding

While conceptually represented by the physical topology field in the governing equations, the layout is managed through a distinct chain of density representations to handle design and non-design domains. To explicitly avoid mathematical confusion with the fluid mass density (ρ) utilized in the Reynolds-Averaged Navier-Stokes equations, this thesis exclusively utilizes the γ notation for all topological variables:

- **Raw design density** (γ): the baseline parameter manipulated by the optimizer.
- **Filtered density** ($\tilde{\gamma}$): the spatially smoothed design field.
- **Projected effective density** ($\bar{\gamma}$): the bounded physical topology field evaluated by the governing equations.

The raw design density, γ, is defined as a Discontinuous Galerkin (DG0) [26] function over the entire mesh. The optimization algorithm strictly manipulates the degrees of freedom corresponding to the actively optimized design space, while passive non-design cells are subsequently restored to their prescribed bounding values.

This raw variable is subjected to a spatial PDE filter [29, 15] to yield a smoothed field $\tilde{\gamma}$. To recover sharp boundaries and strictly preserve required non-design regions, the filtered density is then projected and mapped into the bounded effective density ($\bar{\gamma}$) [25, 53]:

$$\bar{\gamma} = \gamma_{\text{lower}} + (\gamma_{\text{upper}} - \gamma_{\text{lower}})\Pi_\beta(\tilde{\gamma}) \tag{5.2}$$

where Π_β is the projection operator, and γ_{lower} and γ_{upper} define the absolute permissible bounds for each cell. This bounding guarantees that predefined passive-fluid features, such as non-design inlet and outlet extensions ($\gamma_{\text{lower}} = \gamma_{\text{upper}} = 1$), remain fixed without interfering with the mathematical behavior of the active optimization space.

5.2.2 Design and Non-Design Domains

The computational mesh is partitioned into active design cells and passive fluid non-design cells. The active design domain contains the cells whose raw densities are updated by MMA. Fluid non-design domains are fixed as permanent fluid, typically to preserve inlet and outlet extensions, buffer channels, or measurement regions.

These passive fluid cells still participate in the forward flow, turbulence, and wall-distance solves. They are excluded only from the optimization masks: the volume constraint is evaluated over the prescribed volume mask Ω_V, and any domain-integrated post-processing metric is evaluated over its prescribed integration mask. Consequently, prescribed inlet/outlet buffers do not count toward the optimized fluid fraction and do not directly bias reported domain-integrated metrics.

This active/passive partition is used by the active-mask-normalized filtering step introduced in Section 5.7, where the masking convention is shown together with the Helmholtz filtering chain.

5.3 Penalization of the Governing Equations

To optimize turbulent manifolds, the respective penalties must be integrated into the flow, turbulence, and wall-distance equations so that the physics remain consistent as solid boundaries evolve. All physical penalties use the projected effective density, $\bar{\gamma}$.

5.3.1 Modified Reynolds-Averaged Navier-Stokes Equations

The primary application of the Brinkman penalty is the modification of the steady-state, incompressible RANS momentum equation. The RAMP-interpolated inverse permeability $\alpha(\bar{\gamma})$ is added as a linear sink term to the momentum equation:

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \left[(\mu + \mu_t)\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right)\right] - \alpha(\bar{\gamma})\mathbf{u}. \tag{5.3}$$

The topology-optimization residual retains the full symmetric rate-of-strain tensor. Since the turbulent eddy viscosity (μ_t) varies across the spatial domain [38], the effective viscosity is not constant and the transposed gradient term contributes to the momentum balance. In regions where the optimizer places solid material ($\bar{\gamma} \to 0$), the large magnitude of $\alpha_{solid}\mathbf{u}$ dominates the local balance and drives the velocity toward zero.

The final steady flow state used by the frozen adjoint is assembled as a weak residual. With $\mu_{\text{eff}} = \mu + \mu_t^{fr}$ and test functions $(\mathbf{v}, q)$, the implemented residual is: find $(\mathbf{u}, p)$ such that

$$R_{\text{NS}}((\mathbf{u}, p), (\mathbf{v}, q)) = \int_\Omega \rho(\mathbf{u} \cdot \nabla)\mathbf{u} \cdot \mathbf{v}\, dx + \frac{1}{2} \int_\Omega \mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) : \left(\nabla \mathbf{v} + \nabla \mathbf{v}^T\right)\, dx + \int_\Omega \nabla p \cdot \mathbf{v}\, dx + \int_\Omega q\, \nabla \cdot \mathbf{u}\, dx + \int_\Omega \alpha(\bar{\gamma})\mathbf{u} \cdot \mathbf{v}\, dx = 0. \tag{5.4}$$

Equation 5.4 is the differentiable state equation used for the velocity-pressure adjoint. The forward Picard updates may use IPCS warm starts for robustness, but the final sensitivity solve differentiates this steady weak residual.

5.3.2 Modified Spalart-Allmaras Transport Equation

As the Fictitious Domain Method forces velocity to zero in solid regions, turbulence generation inside those regions should theoretically cease. However, numerical diffusion can allow the SA working variable $(\tilde{\nu})$ to bleed into solid regions. This unphysical behavior destabilizes the solver and distorts the adjoint sensitivity analysis.

To enforce a zero-turbulence state inside solid structures, a corresponding penalty sink term is added to the Spalart-Allmaras transport equation [56, 19]:

$$\begin{aligned}\mathbf{u} \cdot \nabla\tilde{\nu} = c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w \left(\frac{\tilde{\nu}}{y}\right)^2 + \frac{1}{\sigma}\left[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) + c_{b2}(\nabla\tilde{\nu})^2\right] \\ - \alpha_{\tilde{\nu}}(1 - \bar{\gamma})^{N_{\tilde{\nu}}}\tilde{\nu}.\end{aligned} \tag{5.5}$$

The last term is the topology penalty used to suppress the SA working variable in solid material. Because $\bar{\gamma} = 1$ denotes fluid and $\bar{\gamma} = 0$ denotes solid, the penalty is applied to the local solid fraction $(1 - \bar{\gamma})$. The coefficient $\alpha_{\tilde{\nu}}$ sets the damping strength and the exponent $N_{\tilde{\nu}}$ controls how sharply the damping grows as a cell becomes solid. In the final benchmark computations these two parameters are held constant throughout the optimization.

In the Picard-linearized finite-element solve, the topology penalty enters as an additional reaction term in the weak SA equation. Using the previous Picard field $\tilde{\nu}^k$ to evaluate $\tilde{S}^k$, f_w^k, and the explicit source terms, the topology-aware SA step solves for $\tilde{\nu}^{k+1}$ from

$$\begin{aligned}&\int_\Omega (\mathbf{u} \cdot \nabla\tilde{\nu}^{k+1})\xi\, dx + \frac{1}{\sigma}\int_\Omega (\nu + \tilde{\nu}^k)\nabla\tilde{\nu}^{k+1} \cdot \nabla\xi\, dx \\ &+ \int_\Omega c_{w1} f_w^k \frac{\tilde{\nu}^k}{y^2}\tilde{\nu}^{k+1}\xi\, dx + \int_\Omega \alpha_{\tilde{\nu}}(1 - \bar{\gamma})^{N_{\tilde{\nu}}}\tilde{\nu}^{k+1}\xi\, dx \\ &= \int_\Omega \left[c_{b1}\tilde{S}^k\tilde{\nu}^k + \frac{c_{b2}}{\sigma}\nabla\tilde{\nu}^k \cdot \nabla\tilde{\nu}^k\right]\xi\, dx.\end{aligned} \tag{5.6}$$

This weak form makes the artificial solid damping explicit: as $\bar{\gamma}$ approaches zero, the reaction term forces the SA working variable toward zero in the same cells where the Brinkman term suppresses velocity.

5.3.3 Reciprocal Penalized Wall-Distance Equation

The Spalart-Allmaras model is highly sensitive to the wall distance, y. While differential equations for wall distance computation are well-established [51], topology optimization introduces a unique challenge. The location of the wall is not only a static mesh boundary, but also the dynamically evolving $\bar{\gamma} = 0$ contour. Therefore, the relaxed Eikonal equation must be penalized to dynamically recognize fictitious solid as a physical boundary [56]:

$$\nabla G \cdot \nabla G + \sigma_w G(\nabla \cdot \nabla G) - (1 + 2\sigma_w)G^4 - \alpha_G(1 - \bar{\gamma})^{N_G}(G - G_0) = 0. \tag{5.7}$$

The wall-distance equation uses the same power-law topology concept. The coefficient α_G sets the strength of the artificial wall-distance forcing and the exponent N_G controls its growth through gray material. In fluid regions the term vanishes, while in topology-created solid regions it pulls G toward G_0. The algebraic recovery function

$$y = \frac{1}{G} - \frac{1}{G_0} \tag{5.8}$$

then collapses the SA wall distance toward zero inside topology-created solid regions while retaining the physical wall distance in fluid regions. As in the standalone solver, small numerical floors and non-negativity safeguards are used in the implementation to avoid division-by-zero artifacts, but they are not part of the mathematical model statement.

The corresponding nonlinear weak residual is obtained by multiplying Eq. 5.7 by a scalar test function z and integrating the $\sigma_w G \nabla^2 G$ contribution by parts:

$$\begin{aligned} R_G(G, z) =& (1 - \sigma_w) \int_\Omega |\nabla G|^2 z \, dx - \sigma_w \int_\Omega G \nabla G \cdot \nabla z \, dx - (1 + 2\sigma_w) \int_\Omega G^4 z \, dx \\ & - \int_\Omega \alpha_G (1 - \bar{\gamma})^{N_G} (G - G_0) z \, dx = 0. \end{aligned} \tag{5.9}$$

The last term is what turns topology-created solid into an artificial wall-distance boundary in the continuous-density setting.

5.4 Boundary Conditions and Turbulence Initialization

To stabilize the topology optimization and preserve physical interpretation, boundary constraints are managed across both fluid and turbulence variables. Non-design fluid buffer regions are mandated at inlets and outlets to isolate the active optimization domain from artificial numerical shear. The framework handles pressure outlets, optional velocity-component constraints, and pressure pin controls to fully constrain the discrete PDE problem.

At the domain inlets, the prescribed turbulence intensity and turbulence length scale are converted into an intermediate eddy-viscosity ratio. This ratio defines the inlet value for the Spalart-Allmaras working variable $\tilde{\nu}$. Throughout the computation, numerical floors, ceilings, and clipping functions are imposed to keep the working variable within stable physical limits during intermediate design cycles.

5.5 Sensitivity Analysis and the Frozen Turbulence Assumption

With the forward state equations fully defined and penalized, the topology optimization algorithm requires the gradient of the objective function with respect to the design variable. Because evaluating finite differences for every design cell is computationally impossible, an adjoint method is required [36, 37].

The implementation used in this thesis is best described as a **frozen-turbulence weak-form residual adjoint of the final steady flow problem**. It differentiates the finite-element residual assembled in UFL for the final frozen-viscosity state. It is not a tape-based differentiation of the entire IPCS/Picard forward algorithm, and it is not a fully coupled turbulence adjoint.

5.5.1 Reduced Adjoint Formulation

Let $\mathbf{s} = (\mathbf{u}, p)$ denote the final velocity-pressure state used for the design derivative. Under the frozen-turbulence approximation, the eddy viscosity ν_t^{fr} and reciprocal wall-distance field G^{fr} are treated as fixed coefficients during the sensitivity calculation. The final flow residual is written as

$$R(\mathbf{s}, \bar{\gamma}; \nu_t^{fr}, G^{fr}) = 0. \tag{5.10}$$

In the implementation, this abstract residual is the weak RANS-Brinkman form in Eq. 5.4. The dependence on $\bar{\gamma}$ therefore enters directly through the Brinkman interpolation $\alpha(\bar{\gamma})$ and, for the dissipation objective, through the objective's Brinkman drag contribution. The dependence of ν_t^{fr} and G^{fr} on the design is not differentiated in the adjoint step. For an objective $J(\mathbf{s}, \bar{\gamma})$, the reduced Lagrangian is

$$\mathcal{L}(\mathbf{s}, \bar{\gamma}, \mathbf{s}^*) = J(\mathbf{s}, \bar{\gamma}) + \mathbf{s}^* R(\mathbf{s}, \bar{\gamma}; \nu_t^{fr}, G^{fr}), \tag{5.11}$$

where $\mathbf{s}^*$ is the adjoint state. In finite-element notation, the product $\mathbf{s}^* R$ represents the assembled residual weighted by the adjoint velocity and pressure test functions, not a pointwise multiplication. The adjoint equation is obtained by eliminating the dependence of the total derivative on the forward-state variation:

$$\left(\frac{\partial R}{\partial \mathbf{s}}\right)^T \mathbf{s}^* = -\left(\frac{\partial J}{\partial \mathbf{s}}\right)^T. \tag{5.12}$$

The design derivative with respect to the projected physical density then becomes

$$\frac{dJ}{d\bar{\gamma}} = \frac{\partial J}{\partial \bar{\gamma}} + \mathbf{s}^* \frac{\partial R}{\partial \bar{\gamma}}. \tag{5.13}$$

This derivative is subsequently chained through the Heaviside projection and the transpose of the active-mask filter before being passed to MMA. The same transpose-filter machinery is used for the volume-constraint sensitivity.

5.5.2 Mechanism of Frozen Turbulence

The frozen turbulence assumption simplifies the sensitivity analysis by decoupling the turbulence transport from the design derivative. During the adjoint formulation, the turbulent eddy viscosity field (ν_t) generated by the forward SA solver is assumed to be locally independent of infinitesimal changes in the physical topology:

$$\frac{\partial \nu_t}{\partial \bar{\gamma}} \approx 0. \tag{5.14}$$

The reciprocal wall-distance field used by the SA model is treated analogously during sensitivity assembly. By treating ν_t and G as frozen spatially varying coefficients, the adjoints of the SA transport equation and the penalized wall-distance equation do not need to be solved.

This does *not* mean that turbulence is frozen in the forward physics. The forward loop still updates the wall distance, SA working variable, and eddy viscosity as the design evolves. The approximation is only in the derivative: the optimizer ignores the infinitesimal derivative of future turbulence and wall-distance fields with respect to the current topology change.

5.5.3 Justification and Limitations

A fully coupled turbulent adjoint would require adjoint equations for the SA working variable, the eddy-viscosity relation, the topology-dependent SA penalty, the reciprocal wall-distance equation, the wall-distance penalty, and the clipping or continuation devices used for numerical robustness. For evolving-wall topology optimization, this creates a very stiff coupled derivative problem. Exact SA adjoints are possible and have been developed for shape optimization and related applications [60, 31], but they are substantially more fragile than frozen-viscosity derivatives in the present topology setting.

The primary advantage of the frozen turbulence approach is therefore practical robustness. It reduces implementation complexity, lowers the cost per design iteration, and avoids differentiating the most nonlinear turbulence and wall-distance source terms. This is why frozen turbulence, or Constant Eddy Viscosity (CEV), remains common in industrial adjoint workflows [36, 42]. Wu et al. [55] show that the CEV assumption can still achieve useful optimization goals in benign flows, while also documenting that sensitivity accuracy and adjoint convergence can deteriorate in more challenging regimes.

The price of this robustness is that the gradients are approximate reduced derivatives. The finite-difference verification in Chapter 6 therefore validates only the implemented frozen derivative: the perturbed checks hold $\tilde{\nu}$ and G fixed in the same way as the adjoint. It should not be interpreted as validation of a fully coupled turbulence sensitivity.

5.6 Optimization Problem Formulation

Using the frozen-turbulence sensitivities described above, the topology optimization task is formulated as a mathematical programming problem: find a spatial density distribution that minimizes a performance metric subject to physical and geometric constraints.

5.6.1 Objective Formulation

The distinction between active and passive domains introduced in Section 5.2 is essential for the objective definition. Dissipation-type objectives are common in density-based flow topology optimization because they penalize mechanical energy loss directly. In Brinkman-penalized formulations, the corresponding power-loss objective is naturally written as the sum of physical viscous dissipation and Darcy/Brinkman drag over the optimized design domain [56, 19]:

$$J_D(\mathbf{u}, \bar{\gamma}) = \int_{\Omega_D} \left[\frac{1}{2} \mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right) : \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T \right) + \alpha(\bar{\gamma})\, \mathbf{u} \cdot \mathbf{u} \right] d\Omega. \tag{5.15}$$

Here Ω_D denotes the optimized design domain used for objective integration, excluding fixed non-design inlet and outlet extensions, and $\mu_{\text{eff}} = \mu + \mu_t$ under the frozen-turbulence flow solve.

For completeness, the implementation also supports an average inlet-pressure objective. Because the outlet pressure is fixed to zero in the pressure-outlet cases, this pressure can be used as a proxy for the pressure loss required to drive the imposed flow rate through the topology:

$$J_p(p) = \frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p \, d\Gamma. \tag{5.16}$$

Here Γ_{in} is the inlet boundary of the fixed non-design extension, not an internal design-domain pressure plane; the division by $|\Gamma_{\text{in}}|$ is required so that the objective is an average pressure. Preliminary pressure-objective runs used this functional, but the final turbulent optimization framework uses J_D from Eq. 5.15 for the pipe-bend, U-bend, and diffuser campaigns. This choice follows the physical interpretation that pressure loss is proportional to mechanical energy dissipation in steady internal flow, while the domain-integrated J_D provides a less localized design signal by summing the objective contribution over all design cells. The code keeps J_p as a diagnostic and fallback objective, but pressure-drop results are interpreted alongside the dissipation objective rather than replacing it.

5.6.2 Volume Constraint

Because the goal is to design a targeted manifold or pipe system, a strict volume constraint must be imposed. The volume fraction constraint is evaluated only over the prescribed volume mask Ω_V, rather than over the entire computational mesh:

$$g(\bar{\gamma}) = \frac{\int_{\Omega_V} \bar{\gamma} \, d\Omega}{V_{\text{mask}}} - V_{\max} \leq 0 \tag{5.17}$$

where V_{mask} is the total volume of the masked region and $V_{\max}$ is the maximum allowable fluid volume fraction inside that region.

5.6.3 The Complete Continuous Optimization Problem

Combining the selected objective function, the governing equations, and the constraints, the continuous topology optimization problem for turbulent flow is formally stated as:

$$\begin{aligned} \underset{\gamma(\mathbf{x})\in[0,1]}{\text{minimize}} \quad & J(\mathbf{u}, p, \bar{\gamma}) \\ \text{subject to} \quad & \text{Modified RANS weak residual (Eq. 5.4)} \\ & \text{Modified Spalart-Allmaras weak form (Eq. 5.6)} \\ & \text{Penalized wall-distance weak residual (Eq. 5.9)} \\ & g(\bar{\gamma}) \leq 0. \end{aligned} \tag{5.18}$$

The Method of Moving Asymptotes (MMA) [48] is used to solve this constrained minimization problem. MMA builds a sequence of local convex approximations from the current objective value, constraint value, and their gradients. The moving asymptotes and move limits restrict how far each active design variable may change in one iteration, which prevents unstable density jumps in a highly nonlinear flow problem. In this implementation, MMA updates only the active raw design degrees of freedom; passive cells are restored to their prescribed bounds after filtering and projection.

5.7 Density Filtering and Projection

A well-known mathematical complication in density-based topology optimization is the lack of existence of a classical solution. If the design space is left unconstrained, the optimizer tends to create infinitely thin structures or checkerboard patterns. To guarantee a mesh-independent and manufacturable design, a minimum length scale must be enforced [15, 29].

5.7.1 The Discontinuous Galerkin Helmholtz Filter

To prevent checkerboarding, a density filter is applied to the raw design variable generated by the optimizer. Because a continuous nodal definition causes the Brinkman penalty to interpolate smoothly across boundary elements, the density field is discretized using piecewise constant elements (DG0) [10, 26]. By assigning a single density value to each element, the structural boundaries remain sharp at element edges.

Unlike an explicit cone-kernel density filter, the Helmholtz filter does not prescribe a direct weighted average over neighboring elements. Instead, it defines the filtered density as the solution of a smoothing problem: the filtered field must remain close to the raw optimizer field while jumps and rapid spatial oscillations are penalized. In a continuous finite-element discretization this smoothing is usually produced by a gradient term. In the present DG0 discretization, however, the density is constant inside each element and therefore has no element-interior gradient. Neighboring cells are coupled only through their shared facets, so the continuous Helmholtz filter is replaced by an interior-penalty form on the internal facets (Γ_{int}):

$$\int_{\Gamma_{int}} r^2 \frac{\alpha_{dg}}{h_{avg}} [\![\tilde{\gamma}]\!] \cdot [\![v]\!] \, dS + \int_{\Omega} \tilde{\gamma} v \, dx = \int_{\Omega} \gamma v \, dx \tag{5.19}$$

where v is the test function, $[\![\cdot]\!]$ denotes the jump operator across a facet, h_{avg} is the average cell size, and $r = R/(2\sqrt{3})$ is the scaled filter radius. The first term penalizes differences between adjacent element densities, while the second term keeps the filtered field anchored to the raw design. Increasing r strengthens the facet coupling and therefore spreads the influence of each raw cell over a wider neighborhood. The filter can thus be interpreted as an implicit, mesh-based averaging operation whose effective kernel is given by the response of Eq. 5.19 to a localized density perturbation.

Active-Mask Normalization

To handle boundary interactions near designated inlet and outlet buffers, this filtering is active-mask-normalized. If the filter were applied blindly across the entire mesh, fixed passive cells could dilute the active design field near non-design regions. The implementation therefore filters only the active contribution and divides by the filtered active mask:

$$\tilde{\gamma} = \frac{\mathcal{H}(M\gamma)}{\mathcal{H}(M)} \tag{5.20}$$

where $\mathcal{H}$ represents the Helmholtz PDE solution operator and M represents the isolated active design indicator mask. The numerator smooths the active raw design field, while the denominator renormalizes the local averaging footprint. As a result, inlet extensions, outlet buffers, and other passive regions do not artificially pull the nearby active design toward their prescribed density values. During sensitivity analysis, the transpose of this normalized filter maps objective and volume gradients back into the active DG0 design space.

Figure 5.2 illustrates this masking convention. The Helmholtz filter is applied to the active design contribution and normalized by the filtered active mask, so fixed passive-fluid cells do not dilute nearby design densities. After projection, the passive-fluid bounds are restored before the physical flow equations are assembled. The same chain is traversed in reverse when objective and volume sensitivities are mapped back to the active raw density variables.

The same Helmholtz operator is used in the reverse sensitivity chain. If $s_{\bar{\gamma}}$ denotes a cellwise derivative with respect to the bounded physical density, then the projection and bound mapping give

$$s_{\tilde{\gamma}} = s_{\bar{\gamma}}(\gamma_{\text{upper}} - \gamma_{\text{lower}}) \frac{\partial \Pi_{\beta}}{\partial \tilde{\gamma}}. \tag{5.21}$$

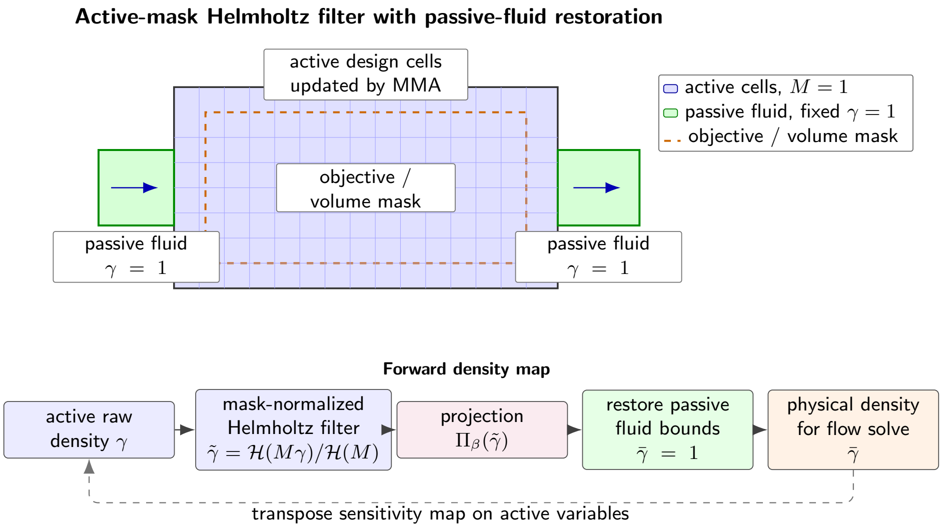


Figure 5.2: Schematic of active design cells, passive fluid cells, objective and volume masks, and the active-mask-normalized Helmholtz filtering chain. The filter concept follows the PDE-filter formulation of Lazarov and Sigmund and the regularization motivation of Bourdin [29, 15].

The transpose of the active-mask-normalized filter then maps this quantity back to the raw active design space:

$$s_\gamma = M\,\mathcal{H}^{-T}\left(\frac{s_{\bar{\gamma}}}{\mathcal{H}(M)}\right). \tag{5.22}$$

Because the Helmholtz bilinear form in Eq. 5.19 is symmetric, $\mathcal{H}^{-T}$ is evaluated by another Helmholtz solve with the transpose-filter right-hand side. Passive cells are set to zero in s_γ because MMA updates only the active design variables.

5.7.2 Heaviside Projection

While the Helmholtz filter imposes a minimum length scale, it smooths the design field and creates a gray transition zone between solid and fluid. To recover a sharper topology, a continuous Heaviside projection is applied to the filtered variable [25, 53]:

$$\Pi_\beta(\tilde{\gamma}) = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\gamma} - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))} \tag{5.23}$$

where $\eta \in [0, 1]$ is the projection threshold, and β is a steepness parameter. The threshold η defines the filtered density around which the projection switches between topology states. In the convention used in this thesis, filtered values below the threshold are projected toward $\bar{\gamma} = 0$ and therefore become solid-like, whereas values above the threshold are projected toward $\bar{\gamma} = 1$ and become fluid-like. Increasing β sharpens this transition; in the limit $\beta \to \infty$, the function approaches a strict Heaviside step.

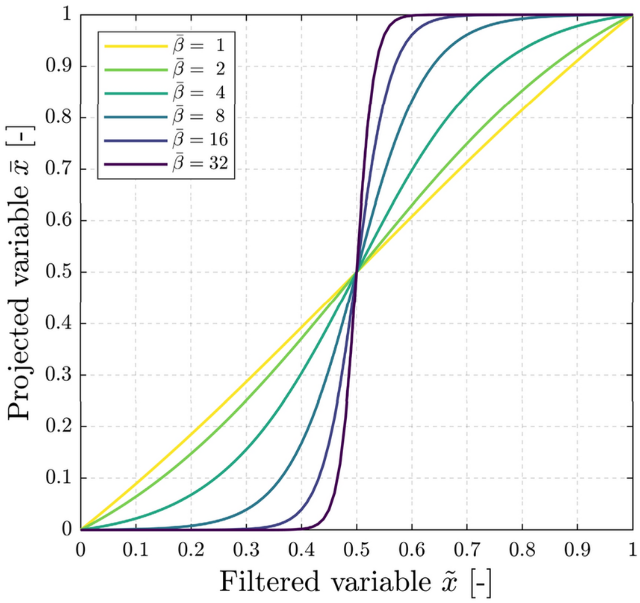


FIGURE 5.3: Heaviside projection from Eq. 5.23 at $\eta = 0.5$ for $\beta = \{1, 2, 4, 8, 16, 32\}$, reproduced from Ooms et al. [35].

5.8 Algorithmic Workflow and Solver Implementation

Executing the optimization requires a robust iterative algorithm. The solver combines density processing, wall-distance updates, Picard-decoupled SA turbulence updates, a final frozen-viscosity state solve, adjoint sensitivity assembly, transpose-filter mapping, volume-gradient assembly, and MMA updates.

Figure 5.4 summarizes the full optimization workflow. The raw active density is first filtered, projected, and restored into the bounded physical density field used by the governing equations. The forward loop then updates the wall-distance, velocity-pressure, and SA turbulence fields until an accepted turbulent state is obtained. Only after this forward update is complete is the frozen-viscosity residual assembled for the final state and adjoint solve. The resulting objective and volume sensitivities are mapped back through the projection slope and transpose filter before MMA updates the active raw design variables.

5.8.1 SA Numerical Controls and Hybrid Forward Flow Recovery

During the optimization loop, the flow solver advances the momentum field using the eddy viscosity from the previous iteration, and the updated velocity is then passed back to the SA solver. During volatile intermediate design steps, the framework uses Stokes-Brinkman warm starts and convection continuation to reintroduce inertial physics gradually. If convergence stalls, the SNES-based final solve can step through recovery schedules with altered tolerances and line-search methods.

The optimization architecture supports non-converged acceptance during difficult intermediate steps, allowing the algorithm to maintain topological momentum.

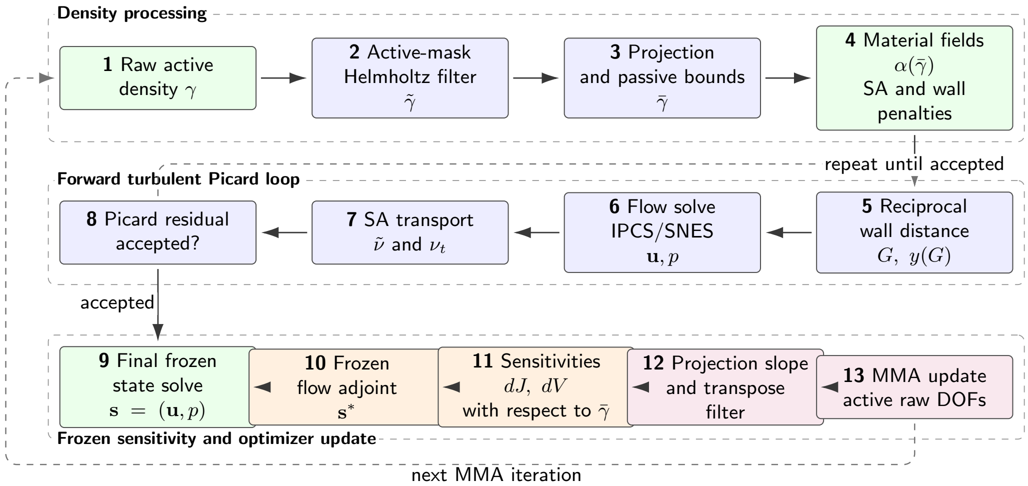


Figure 5.4: Workflow of the density-based frozen-turbulence topology optimization algorithm. The forward turbulence fields are updated in the flow loop, while the adjoint differentiates only the final frozen-viscosity residual.

Prior to adjoint assembly, stricter flow solves can be executed. When used, these strict solves confirm that the sensitivities and frozen turbulent fields are derived from a more equilibrated state. Otherwise, relying on accepted residual criteria is a pragmatic compromise required for highly nonlinear intermediate designs.

5.8.2 Algorithm and Continuation Strategy

To encourage convergence to a physically meaningful binary design, the algorithm iterates over a continuation schedule. The projection steepness (β) is increased as the optimization progresses, while the Brinkman RAMP parameter (q) is advanced according to the prescribed penalty schedule and MMA move limits are tightened. The SA and wall-distance power penalties do not use this continuation in the final benchmark runs; their amplitudes and exponents remain fixed while the local penalty magnitude changes through the evolving topology field.

Within each continuation stage, one MMA iteration consists of the following mathematical sequence. First, the raw active design field is filtered with Eq. 5.19, normalized by the active-mask denominator $\mathcal{H}(M)$, projected with Eq. 5.23, and restored to the prescribed passive-cell bounds. The resulting $\bar{\gamma}$ defines the Brinkman resistance $\alpha(\bar{\gamma})$, the SA damping coefficient in Eq. 5.6, and the wall-distance forcing in Eq. 5.9.

Second, the forward turbulence state is updated. The reciprocal wall-distance residual in Eq. 5.9 is solved for the current topology, the velocity-pressure equations are advanced with the current eddy viscosity, and the penalized SA weak form in Eq. 5.6 updates $\tilde{\nu}$ and therefore μ_t. These Picard updates continue until the configured residual criteria are accepted for the current design step.

Third, the final frozen-viscosity flow state is solved with Eq. 5.4. The adjoint equation from Section 5.5 is then assembled from the same weak residual and the selected objective; in the final turbulent benchmark campaigns this is J_D from Eq. 5.15. The design derivative is obtained from Eq. 5.13, chained through the projection derivative in Eq. 5.21, and mapped back to the active raw variables through the transpose filter in Eq. 5.22. MMA finally updates only those active raw density degrees of freedom, subject to the volume constraint in Eq. 5.17 and the current move limit.

5.9 Conclusion

This chapter has established the mathematical and algorithmic framework used for the turbulent topology optimization results in this thesis. By employing the Fictitious Domain Method with Brinkman momentum resistance [14, 23], the evolving structural topology is integrated into a fixed computational grid. The SA working-variable and reciprocal wall-distance equations are coupled to the topology through power-law penalties, while the DG0 Helmholtz interior penalty filter [29, 15] and continuous Heaviside projection [25, 53] enforce a minimum length scale and reduce mesh-dependent numerical artifacts.

The optimization is driven by a frozen-turbulence weak-form residual adjoint [55, 42]. The forward turbulence fields are updated during the optimization loop; only their derivatives are frozen during sensitivity assembly. This assumption is defensible as a stable engineering approximation, but the thesis treats it as an approximation whose limitations must be checked through finite-difference sensitivity verification and independent CFD validation. Embedded within a Picard-decoupled flow architecture using hybrid IPCS/SNES recovery schemes, this methodology transforms the verified fluid dynamics engine from Chapter 3 into the automated design tool evaluated in Chapter 6.

Chapter 6

Turbulent Topology Optimization Results

6.1 Introduction

Building upon the FEniCS-SA solver verified in Chapter 4 and the topology optimization methodology developed in Chapter 5, this chapter evaluates the turbulent topology optimization solver on the pipe-bend and U-bend internal-flow benchmarks from Bayat et al. [12] and the diffuser benchmark from Yoon [56]. The Bayat cases use reciprocal wall-distance penalization, topology-dependent SA and wall-distance reaction terms, an average inlet-pressure objective, and a frozen-turbulence adjoint. The diffuser benchmark uses the same frozen-SA topology optimization machinery, but it minimizes the dissipation objective J_D.

The chapter first verifies the implemented frozen sensitivity on the same pipe-bend configuration used for optimization with a Dilgen-style coordinate finite-difference check on selected design cells and a directional Taylor check. The optimized pipe-bend, U-bend, and diffuser layouts are then interpreted through their projected density fields, velocity fields, convergence histories, and independent sharp-geometry CFD validation. This ordering keeps the derivative evidence close to the optimization result that uses it, while leaving the standalone forward-flow verification in Chapter 4.

The benchmark sections below introduce the geometry, boundary conditions, objective, and validation metrics needed to read the results without consulting the appendices first. Appendix B retains the full numerical settings and Ansys grid-independence studies for the extracted turbulent designs, while Appendix C retains the corresponding laminar-design validation details.

All mesh-independence relative differences referenced from Appendices B and C follow the appendix convention: coarse–medium differences are normalized by the coarse value, while medium–fine differences are normalized by the fine value.

6.2 Benchmark Cases and Evaluation Metrics

The three benchmarks cover curved internal flows and a diffuser with a prescribed outlet contraction. The pipe bend targets the $V_f = 0.25$, $Re = 5{,}000$ case from Bayat et al. [12], while the U-bend targets their $V_f = 0.27$, $Re = 5{,}000$ case with a 180° flow reversal. The diffuser targets the $Re = 3{,}000$ turbulent diffuser from Yoon [56], where the original solid mass usage of 70% corresponds to a maximum fluid fraction of $V_f = 0.30$ in the present density convention. Table 6.1 summarizes the operating point and analyses reported for each case.

TABLE 6.1: Benchmark cases evaluated in Chapter 6. Detailed physical and numerical parameters are listed in Appendix B.

Case	V_f	Re	**Analyses reported**
Pipe bend	0.25	5,000	FEniCS optimization; finite-difference and Taylor sensitivity verification; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison
U-bend	0.27	5,000	FEniCS optimization; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison
Diffuser	0.30	3,000	FEniCS optimization with J_D; Ansys-SA and Ansys-SST $k-\omega$ CFD validation; laminar-design comparison

6.2.1 Optimization and Validation Metrics

Throughout the optimization process, the pipe-bend and U-bend benchmarks minimize the average inlet-pressure objective J_p defined in Eq. 5.16. This pressure is averaged over the physical inlet boundary of the fixed non-design extension, where the inlet velocity is prescribed. The outlet pressure is fixed to zero, so reducing J_p directly reduces the pressure level required to drive the prescribed inlet flow through the evolving topology. The optimization problem is still solved on a diffuse Brinkman domain, but once a design is extracted for Ansys Fluent, the Brinkman penalty is removed and the solid-fluid interface becomes a sharp wall.

The diffuser benchmark instead minimizes the dissipation objective J_D defined in Eq. 5.15. This matches the objective used by Yoon [56]: physical viscous dissipation plus Brinkman drag is integrated over the design domain and minimized subject to the volume constraint. This distinction matters because the diffuser uses prescribed parabolic velocity profiles at both inlet and outlet, so the most direct optimization loss is the domain-integrated energy loss rather than an inlet-pressure boundary average.

All three benchmark cases use the power-law SA sink, the power-law reciprocal wall-distance penalty, and the frozen-turbulence adjoint introduced in Chapter 5.

Bayat et al. [12] report corresponding final average inlet-pressure objectives for their standard k-ε computations in Table 7. Their "conventional" method is the density-based turbulent topology optimization formulation without the proposed implicit wall-functions: solid regions are imposed through Brinkman/body-force penalization, including damping of k and ε toward zero in solid material, on the same diffuse optimization mesh. It is therefore distinct from the body-fitted re-simulations with explicit wall-functions that they use for validation. Since the present thesis uses a wall-resolved SA model instead of k-ε, the conventional values from Bayat et al. are used only as scale comparisons in the individual Bayat benchmark discussions below. For the diffuser benchmark, the reference comparison is qualitative because Yoon [56] reports the laminar and turbulent diffuser layouts and post-processing behavior rather than a directly matching J_D table for the diffuser.

To bridge this gap, the primary validation metric is the physical viscous dissipation integrated over the full exported CFD domain, including the non-design inlet, outlet, and fixed regions that are retained in the final geometry,

$$\Phi_D = \int_\Omega \frac{1}{2}\mu_{\text{eff}} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) : \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) \, d\Omega, \tag{6.1}$$

where Ω denotes the complete sharp-geometry flow domain exported to Ansys and $\mathbf{S} = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)$ is the symmetric strain-rate tensor. The integrand in Eq. 6.1 can therefore also be written as $2\mu_{\text{eff}}\mathbf{S} : \mathbf{S}$. In the Ansys Fluent post-processing, the scalar strain-rate magnitude is $\dot{\gamma} = \sqrt{2\mathbf{S} : \mathbf{S}}$, so the custom field function $\mu_{\text{eff}}\dot{\gamma}^2$ is equivalent to the integrand in Eq. 6.1. The effective viscosity is $\mu_{\text{eff}} = \mu + \mu_t$ for turbulent evaluations and $\mu_{\text{eff}} = \mu$ for laminar reference solves.

The second global validation metric is the static pressure drop between the physical inlet and outlet sections,

$$\Delta p = \bar{p}_{\text{in}} - \bar{p}_{\text{out}}, \qquad \bar{p}_\Gamma = \frac{1}{|\Gamma|}\int_\Gamma p_s \, d\Gamma, \tag{6.2}$$

where p_s is the static pressure and $|\Gamma| = \int_\Gamma 1 \, d\Gamma$ is the boundary length in the two-dimensional model. In Ansys Fluent, p_s is taken directly from the area-weighted average static pressure on the inlet and outlet surfaces. For FEniCS results that use kinematic pressure, the corresponding static pressure is obtained as $p_s = \rho p_k$ before evaluating Eq. 6.2. Pressure drop is retained throughout Chapter 6 because it is the most direct engineering interpretation of the same mechanical losses measured by Φ_D. All two-dimensional dissipation quantities are evaluated per unit depth.

6.3 Pipe Bend Benchmark

The pipe-bend benchmark introduces a direct comparison with the turbulent topology optimization results of Bayat et al. [12]. While the reference study employs a standard k-ε RANS model with wall functions, the present framework tackles the same problem using a wall-resolved SA model under the frozen-turbulence assumption. This comparison tests whether the fundamental topological solution, a streamlined and separation-avoiding bend, remains consistent across different turbulence closures.

6.3.1 Geometry and Boundary Conditions

The boundary conditions and geometric constraints replicate the setup shown in Fig. 10 of Bayat et al. [12], specifically targeting the $V_f = 0.25$ volume-fraction case.

The inlet is placed on the left side of the non-design extension, the outlet is placed along the lower boundary, and the outlet pressure is fixed to zero. The optimization minimizes the average pressure on this prescribed inlet boundary required to drive the flow through the evolving bend. The local benchmark geometry used for the present implementation is shown in Figure 6.1.

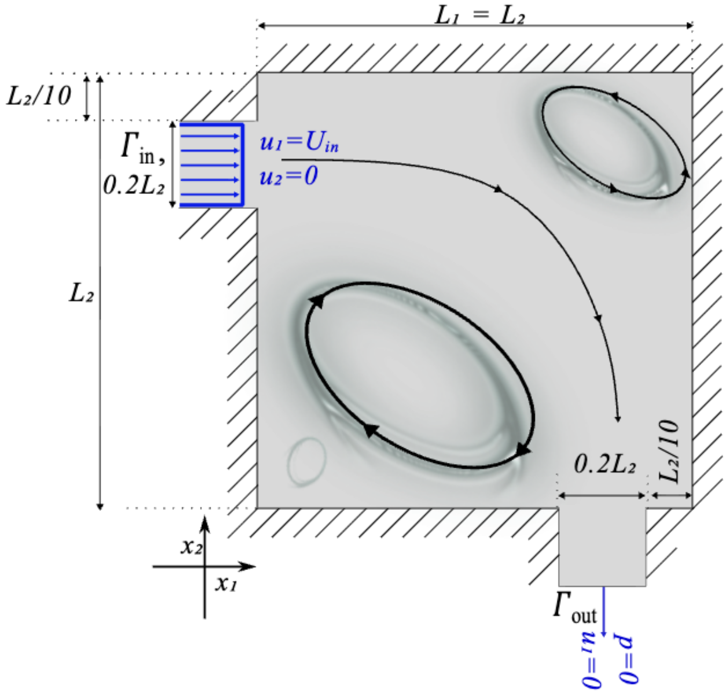


FIGURE 6.1: Geometry and boundary conditions for the pipe-bend benchmark from Bayat et al. [12]. The present evaluation targets the $V_f = 0.25$ case using the SA model.

6.3.2 Sensitivity Verification

Before interpreting the optimized topology, the frozen adjoint is checked directly on the pipe-bend configuration. This is the relevant local test for the final thesis setup because it uses the same topology-dependent SA and wall-distance penalties, inlet-pressure objective, and frozen-turbulence adjoint used during the optimization. No full turbulence-adjoint or wall-distance-adjoint comparison is made; the purpose is to compare the derivative supplied to MMA with objective changes obtained by re-solving the perturbed velocity-pressure state. The check follows the common literature practice of combining coordinate finite differences on selected design variables with a directional Taylor test over multiple step sizes. It therefore verifies the implemented derivative of the objective actually used in the pipe-bend run, J_p. If a later pipe-bend or U-bend campaign switches from J_p to the dissipation objective

J_D, the same diagnostic should be repeated because the adjoint right-hand side and the direct objective derivative change.

The check is performed at the initial pipe-bend design state, with base objective $J_0 = 1.8363$. In every perturbation, the SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0 are held fixed. This matches the frozen-turbulence assumption used by the adjoint: the flow state is re-solved, but the turbulence and wall-distance updates are not differentiated.

Five active DG_0 design cells are selected using a fixed pseudo-random sampling procedure, so that the same control volumes are used whenever the verification script is rerun. These sampled control volumes are labelled CV_1–CV_5 in Table 6.2. For sampled cell i, the raw density vector $\boldsymbol{\gamma}$ is perturbed by $\pm\Delta h\,\mathbf{e}_i$, where $\mathbf{e}_i$ is the unit vector for that active design variable and $\Delta h = 10^{-6}$. The central finite-difference derivative is

$$S_i^{\mathrm{FD}} = \frac{J_p\big(\boldsymbol{\gamma} + \Delta h\,\mathbf{e}_i; \mathbf{u}^+, p^+, \tilde{\nu}^0, G^0\big) - J_p\big(\boldsymbol{\gamma} - \Delta h\,\mathbf{e}_i; \mathbf{u}^-, p^-, \tilde{\nu}^0, G^0\big)}{2\Delta h}. \tag{6.3}$$

The frozen-adjoint value for the same cell is denoted by S_i^{fr}; the superscripts FD and fr denote finite-difference and frozen-adjoint quantities, respectively. The absolute and relative discrepancies are computed as

$$\Delta_i^{\mathrm{abs}} = \left|S_i^{\mathrm{fr}} - S_i^{\mathrm{FD}}\right|, \qquad \varepsilon_i^{\mathrm{fr}} = \frac{\Delta_i^{\mathrm{abs}}}{\max\big(\left|S_i^{\mathrm{fr}}\right|, \left|S_i^{\mathrm{FD}}\right|, 10^{-30}\big)}. \tag{6.4}$$

Table 6.2: Coordinate sensitivity check for the frozen pipe-bend configuration. Values are taken from `SensitivityVerificationTable.tsv`.

Cell	**Finite difference** S_i^{FD}	**Frozen adjoint** S_i^{fr}	**Absolute error** Δ_i^{abs}	**Relative error** $\varepsilon_i^{\mathrm{fr}}$
CV_1	-2.7041×10^{-5}	-2.7042×10^{-5}	4.1371×10^{-10}	1.5299×10^{-5}
CV_2	-2.1494×10^{-6}	-2.1491×10^{-6}	3.1679×10^{-10}	1.4738×10^{-4}
CV_3	-1.6641×10^{-4}	-1.6641×10^{-4}	3.6170×10^{-10}	2.1735×10^{-6}
CV_4	-1.7281×10^{-5}	-1.7281×10^{-5}	1.5118×10^{-12}	8.7488×10^{-8}
CV_5	2.3808×10^{-5}	2.3808×10^{-5}	8.1272×10^{-11}	3.4136×10^{-6}

The coordinate check shows close agreement between the frozen-adjoint derivative and the re-solved central finite differences. The largest absolute discrepancy is 4.14×10^{-10}, and the largest relative discrepancy is 1.47×10^{-4} for CV_2. This relative error remains small and is amplified by the small magnitude of the CV_2 derivative, approximately 2.15×10^{-6}, which places the finite-difference objective change generated by a 10^{-6} perturbation close to roundoff.

The Taylor check provides a complementary directional test. Instead of perturbing one cell at a time, the test perturbs the same five sampled cells together along one fixed direction and checks whether the objective change follows the adjoint's first-order prediction as the step is reduced. The direction signs for CV_1–CV_5 are $(+, +, -, +, -)$, defining a combined perturbation of the coordinate-check cells

Table 6.3: Directional Taylor check for the frozen pipe-bend configuration. Values are taken from `TaylorCheckLog.tsv`. The same five sampled cells as in Table 6.2 are perturbed together.

Step	Adjoint directional derivative	Central finite-difference directional derivative	Relative error	Symmetric Taylor remainder	Observed order
1.0×10^{-3}	9.6135×10^{-5}	9.6135×10^{-5}	1.5795×10^{-8}	2.1515×10^{-11}	–
3.0×10^{-4}	9.6135×10^{-5}	9.6135×10^{-5}	3.8379×10^{-9}	1.9358×10^{-12}	2.0003
1.0×10^{-4}	9.6135×10^{-5}	9.6135×10^{-5}	1.5386×10^{-8}	2.1494×10^{-13}	2.0006
3.0×10^{-5}	9.6135×10^{-5}	9.6135×10^{-5}	3.4657×10^{-8}	1.9096×10^{-14}	2.0107

rather than a separate set of locations. Table 6.3 shows that the central finite-difference directional derivatives agree with the adjoint value to a maximum relative error of 3.47×10^{-8} over the tested steps. The Taylor remainders decrease with approximately second-order rates, which is the expected behavior when the implemented gradient captures the local first-order objective change.

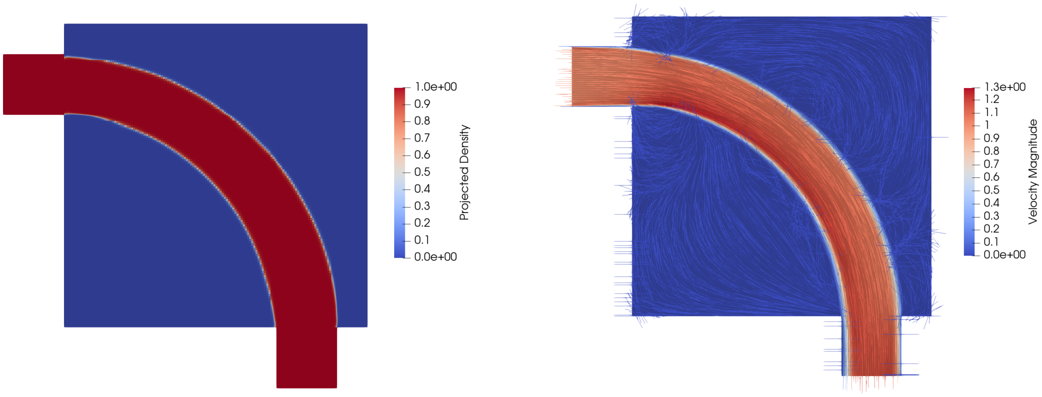
These tests therefore verify the implemented frozen-turbulence derivative for the pipe-bend setup used in the optimization. The conclusion is deliberately limited to that derivative: the SA working variable and reciprocal wall-distance field are held fixed during the checks, matching the optimization adjoint; consequently, the result should not be interpreted as validation of a fully coupled turbulence sensitivity. Repeating the same diagnostic on the U-bend would test the same algebraic implementation path rather than a fundamentally different adjoint formulation. Since there is no independent reference field for this SA frozen derivative, a spatial $dJ_p/d\gamma$ contour is not used as a verification metric; the relevant evidence is the coordinate finite-difference agreement and the second-order Taylor behavior reported above.

6.3.3 Optimization Results

Figure 6.2 illustrates the projected density field $\bar{\gamma}$ and the corresponding velocity magnitude. Despite the differing turbulence models, the FEniCS-SA topology is expected to follow the same broad routing strategy as the k-ε reference: the primary high-speed path should bypass the sharpest corners to minimize momentum loss, while boundary-layer treatment can still affect the exact gray-scale density distribution near the walls.

For the same $V_f = 0.25$, $Re = 5{,}000$ operating point, Bayat et al. [12] report $J_p = 0.4023$ Pa for their conventional k-ε topology optimization. This value is used only as a reference scale, not as a direct validation target, because the turbulence closure and near-wall treatment differ from the present wall-resolved SA formulation.

Figure 6.3 reports the convergence behavior that accompanies the final topology. The pressure objective is normalized by its initial value J_0, while the volume constraint is plotted as a relative residual rather than as the raw volume fraction. Denoting the realized fluid volume fraction at MMA iteration k by $v_f^{(k)}$ and the



(A) Projected density field $\bar{\gamma}$.   (B) Velocity magnitude and vector field.

Figure 6.2: FEniCS-SA optimization result for the pipe-bend benchmark at $V_f = 0.25$. Final objective value $J_p = 0.2032$ Pa.

prescribed target fraction by V_f, the plotted residual is

$$r_V^{(k)} = 100\frac{v_f^{(k)} - V_f}{V_f}. \tag{6.5}$$

With this sign convention, the dashed line at $r_V = 0$ represents the active volume constraint. Values below this line are feasible and correspond to a design using slightly less fluid volume than the prescribed limit.

The objective decreases rapidly during the first part of the optimization, reaching roughly one tenth of its initial value by the end of the run. The subsequent oscillations are expected for the continuation-based MMA solve, since changes in the interpolation and projection parameters alter the effective optimization landscape. The volume residual starts at zero because the initial uniform density is set equal to the prescribed volume fraction, $V_f = 0.25$. It then briefly becomes negative as the optimizer removes fluid while reducing the pressure objective, before returning close to the active constraint. The small negative peaks in the residual occur immediately after new continuation stages are introduced, most clearly near global MMA iterations 110 and 190. At these iterations the continuation update changes the mapping between the design variables and the projected material distribution, so the same design can momentarily sit below the target volume fraction until MMA rebalances the objective and constraint in the following iterations.

6.3.4 CFD Validation

The structural robustness of the extracted pipe-bend topology is verified in Table 6.4. Because the SST $k-\omega$ model is notably more sensitive to adverse pressure gradients than SA, evaluating the extracted geometry under both closures serves as a rigorous stress test for the optimized shape.

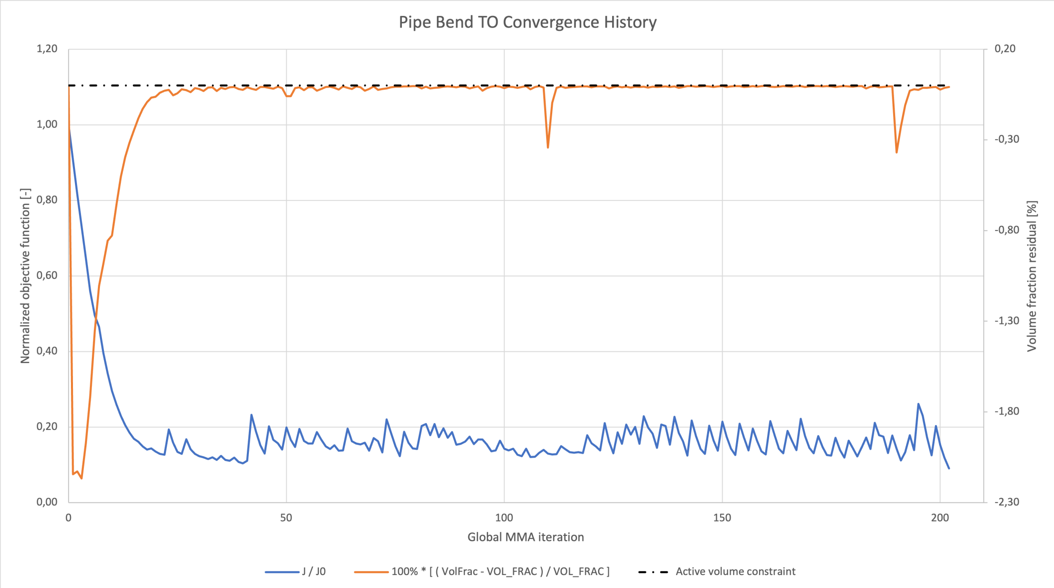


FIGURE 6.3: Optimization convergence history for the pipe-bend benchmark. The blue curve shows the normalized pressure objective J/J_0, while the orange curve shows the relative volume-fraction residual r_V from Eq. (6.5).

TABLE 6.4: CFD validation metrics for the extracted pipe-bend topology.

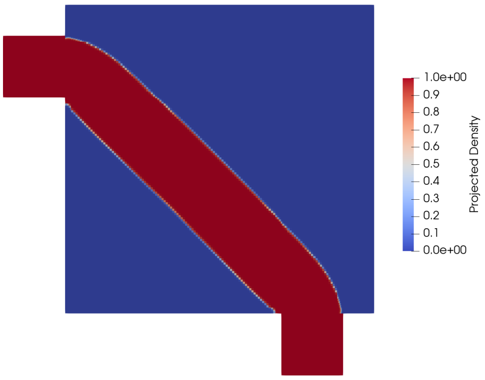
Solver	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Comment**
FEniCS diffuse design	SA	0.0147	0.2032	Optimization state
Ansys sharp geometry	SA	0.0153	0.1457	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	0.0166	0.1522	Turbulence-model sensitivity

Using the Ansys-SA result as the reference, the SST $k-\omega$ evaluation increases the pressure drop by approximately 4.5% and the viscous dissipation by approximately 8.5%. The SST $k-\omega$ model therefore predicts slightly higher losses, as expected for a closure that is more sensitive to adverse pressure gradients, but the changes remain modest. This suggests that the optimized pipe-bend geometry is robust under a change of turbulence closure, rather than being tuned only to the SA model.

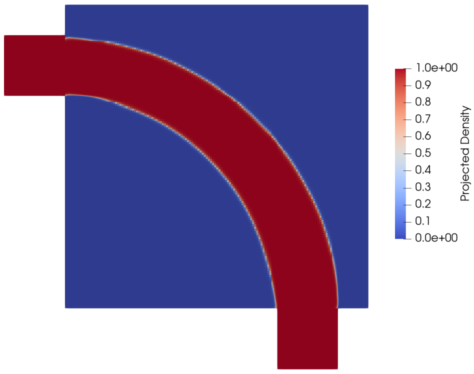
6.3.5 Comparison with Laminar Design

The laminar reference uses the same pipe-bend mesh, design-domain restrictions, inlet and outlet treatment, objective type, and volume fraction, but replaces the SA turbulent state equations by the laminar flow formulation and is optimized at $Re = 1$. The expected distinction is that a laminar bend can minimize wall length and viscous shear in the creeping-flow limit, but may produce a sharper or less separation-aware turn when evaluated at the target turbulent Reynolds number. The comparison is therefore interpreted under identical turbulent operating conditions, rather than by comparing the laminar and turbulent optimization objectives directly.

Figure 6.4 compares the laminar and turbulent optimized topologies before sharp-geometry extraction. The laminar-design mesh-independence study supporting the SST $k-\omega$ result is reported in Table C.2 of Appendix C.



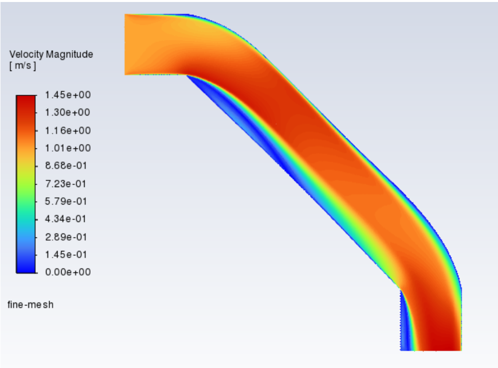
(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$). Final objective value $J_p = 96.751$ Pa.



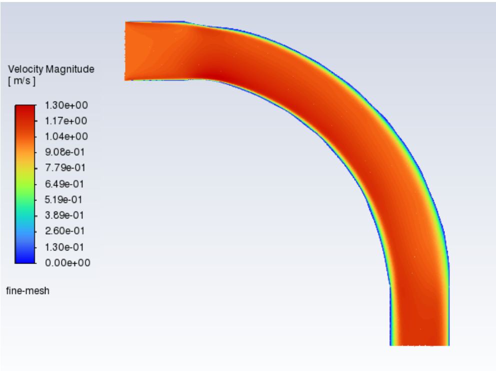
(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 5{,}000$). Final objective value $J_p = 0.2032$ Pa.

Figure 6.4: Laminar and turbulent topology comparison for the pipe-bend benchmark.

Figure 6.5 shows the corresponding SST $k-\omega$ velocity contours through the extracted laminar and turbulent pipe-bend designs. These contours provide the flow-field evidence behind the scalar metrics: sharper turns and corners in the laminar design are expected to promote stronger local separation and momentum loss when the geometry is operated at the turbulent Reynolds number.



(A) Velocity magnitude field u: laminar-optimized design.



(B) Velocity magnitude field u: turbulent-optimized design.

Figure 6.5: Velocity-contour comparison for the extracted pipe-bend designs under identical turbulent operating conditions (SST $k-\omega$ model).

The final SST $k-\omega$ comparison in Table 6.5 evaluates the extracted laminar and turbulent pipe-bend designs under identical turbulent operating conditions. A lower pressure drop and lower viscous dissipation for the turbulent design would support the need for a turbulent topology optimization framework rather than a laminar surrogate.

TABLE 6.5: Performance comparison of the extracted pipe-bend topologies under identical turbulent operating conditions.

Design origin	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Relative performance**
Laminar TO, $Re = 1$	SST $k - \omega$	0.0313	0.5444	Baseline
Turbulent TO, SA	SST $k - \omega$	0.0166	0.1522	47.0% lower Φ_D; 72.0% lower Δp

6.4 U-Bend Benchmark

The U-bend is the companion benchmark to the pipe bend. It uses the same frozen-SA topology optimization framework, but the geometry forces a 180° return path between two ports on the same side of the domain. This makes the case useful for checking whether the optimizer can form a smooth turbulent return duct without relying on the shorter 90° routing available in the pipe-bend case.

6.4.1 Geometry and Boundary Conditions

The geometry and boundary conditions follow Fig. 11 and Table 4 of Bayat et al. [12]. The square design region has side length L, with a non-design lead duct of length $0.2L$ on the left. The inlet and outlet ports both have height $0.2L$; the top port acts as the inlet and the lower port acts as the pressure outlet. A fixed internal separator bar of thickness $0.10L$ extends from the lead duct into the design domain, forcing the flow to turn around its rounded tip. The target volume fraction is $V_f = 0.27$, and the Reynolds number is $Re = 5{,}000$ based on the port height and maximum inlet velocity. Figure 6.6 shows the corresponding local geometry and boundary-condition layout used for this benchmark.

6.4.2 Optimization Results

Figure 6.7 reports the final projected density and velocity-magnitude field for the U-bend. The main feature to assess is whether the design forms a rounded return passage around the separator tip, rather than a short path with tight local curvature. Because the objective is average inlet pressure, a successful layout reduces the pressure required to maintain the imposed inlet flow while respecting the prescribed volume fraction.

For the matching $V_f = 0.27$, $Re = 5{,}000$ U-bend case, the conventional k-ε result reported by Bayat et al. [12] is $J_p = 0.8327$. As for the pipe bend, this value provides a reference scale only; the comparison remains qualitative because the present solver uses a different turbulence model and near-wall treatment.

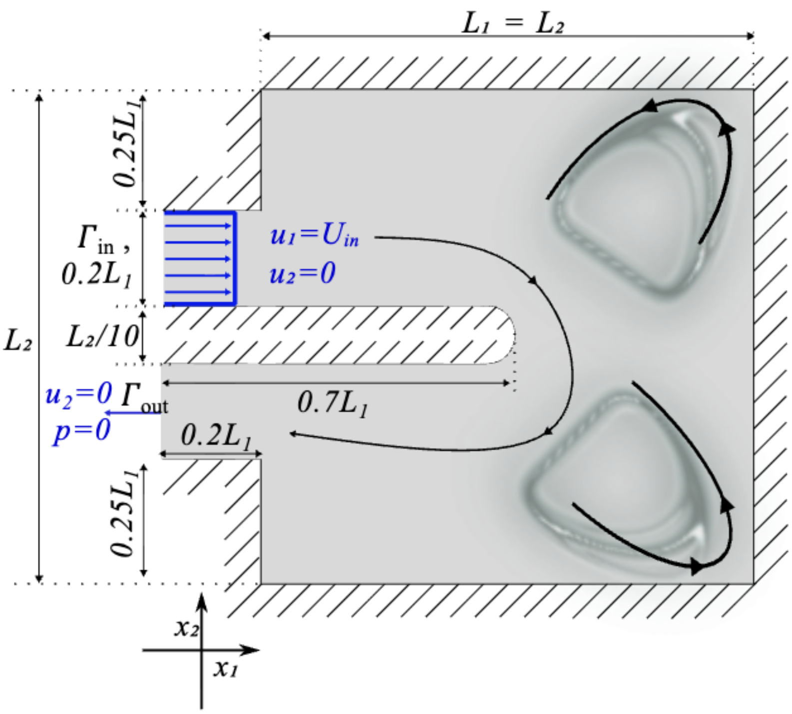


Figure 6.6: Geometry and boundary conditions for the U-bend benchmark from Bayat et al. [12].

Placeholder

Projected density field $\bar{\gamma}$, $V_f = 0.27$

Placeholder

Velocity-magnitude contours

(a) Projected density field $\bar{\gamma}$.

(b) Velocity magnitude and vector field.

Figure 6.7: FEniCS-SA optimization result for the U-bend benchmark at $V_f = 0.27$. Final objective value $J_p = TBD$ Pa.

The U-bend convergence histories are collected in Figure 6.8. As for the pipe bend, the final figure should combine the normalized inlet-pressure objective and the relative volume-fraction residual on shared iteration coordinates so that objective progress and constraint activity can be read together.

Placeholder

Combined convergence history: normalized J_p/J_0 and relative volume residual r_V vs. MMA iteration

FIGURE 6.8: Optimization convergence history for the U-bend benchmark. The final figure should use the same combined-objective and volume-residual format as Figure 6.3.

6.4.3 CFD Validation

The extracted U-bend topology is validated with the same sharp-geometry workflow as the pipe bend. The Ansys SST $k-\omega$ solve tests whether the design remains robust under a two-equation model that is more sensitive to separation and adverse pressure gradients, while the Ansys-SA solve checks cross-code consistency for the same turbulence closure family. The resulting validation metrics are summarized in Table 6.6.

TABLE 6.6: CFD validation metrics for the extracted U-bend topology.

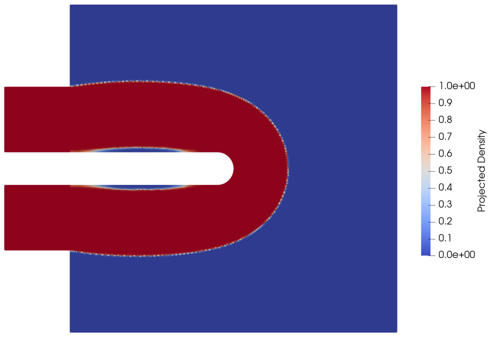
Solver	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Comment**
FEniCS diffuse design	SA	[TBD]	[TBD]	Optimization state
Ansys sharp geometry	SA	[TBD]	[TBD]	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	[TBD]	[TBD]	Turbulence-model sensitivity

6.4.4 Comparison with Laminar Design

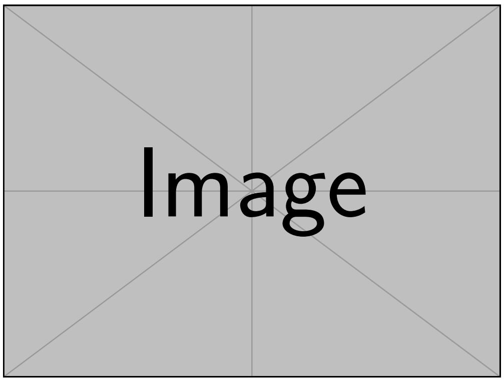
The laminar reference uses the same U-bend mesh, inlet and outlet ports, separator geometry, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent layouts are then evaluated under the same target turbulent operating conditions. This comparison is expected to show whether the laminar optimizer creates an overly tight return path around the separator tip, while the turbulent optimizer allocates fluid volume to reduce separation and momentum loss through the 180° turn.

Figure 6.9 compares the laminar and turbulent optimized U-bend topologies. The laminar-design mesh-independence study supporting the SST $k-\omega$ result is reported in Table C.4 of Appendix C.

Figure 6.10 shows the SST $k-\omega$ velocity contours for the extracted U-bend designs. In this case the comparison is especially useful near the separator tip, where



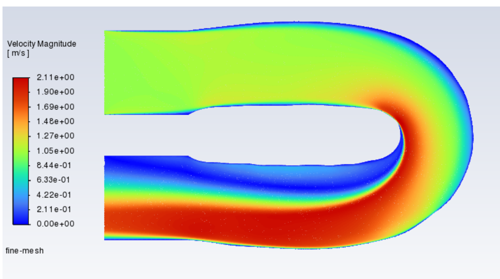
(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$). Final objective value $J_p = 136.181$ Pa.



(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 5{,}000$). Final objective value $J_p = TBD$ Pa.

Figure 6.9: Laminar and turbulent topology comparison for the U-bend benchmark.

a laminar-optimized shortcut can create a tighter return path, stronger adverse-pressure-gradient effects, and larger separated or low-momentum regions than the turbulent design.



(A) Velocity magnitude field u: laminar-optimized design.

Placeholder

SST $k - \omega$ velocity contour through turbulent U-bend design

(B) Velocity magnitude field u: turbulent-optimized design.

Figure 6.10: SST $k - \omega$ velocity-contour comparison for the extracted U-bend designs under identical turbulent operating conditions.

Table 6.7 compares the extracted laminar and turbulent U-bend topologies under identical turbulent operating conditions. If the turbulent design lowers Δp and Φ_D under the SST $k - \omega$ evaluation, the result supports the thesis claim that high-Reynolds topology optimization should include turbulence directly in the design loop.

Table 6.7: Performance comparison of the extracted U-bend topologies under identical turbulent operating conditions.

Design origin	Model	Φ_D [W/m]	Δp [Pa]	Relative performance
Laminar TO, $Re = 1$	SST $k - \omega$	[TBD]	[TBD]	Baseline
Turbulent TO, SA	SST $k - \omega$	[TBD]	[TBD]	[TBD]% vs. laminar

6.5 Diffuser Benchmark

The diffuser benchmark follows the third numerical example of Yoon [56]. It differs from the Bayat benchmarks in two important ways. First, the inlet and outlet are both prescribed by parabolic velocity profiles, with the outlet placed only on the center third of the right boundary. Second, the optimization objective is the dissipation objective J_D, not the average inlet-pressure objective J_p. The benchmark therefore tests whether the frozen-SA implementation can reproduce the expected turbulent diffuser behavior while optimizing the same energy-loss functional used in the reference formulation.

6.5.1 Geometry and Boundary Conditions

The geometry and boundary conditions follow Fig. 20 of Yoon [56]. The design domain is a square of side length $L = 1$ m. A parabolic inflow enters through the full left boundary with maximum speed $U_{\max,\text{in}} = 3$ m/s, while a parabolic outflow is prescribed on the center third of the right boundary with maximum speed $U_{\max,\text{out}} = 9$ m/s. The remaining external boundaries are treated as no-slip walls. The Reynolds number is $Re = 3{,}000$ based on $U_{\max,\text{in}} L\rho/\mu$, and the present density convention converts Yoon's 70% solid mass usage into a maximum fluid volume fraction of $V_f = 0.30$. Figure 6.11 shows the corresponding local geometry and boundary-condition layout used for this benchmark.

6.5.2 Optimization Results

Figure 6.12 reports the final projected density and velocity-magnitude field for the diffuser. The main feature to assess is whether the turbulent design expands the flow near the inlet and the contracted outlet, rather than forming only the simple cone-like diffuser associated with the laminar reference in Yoon's Fig. 21. Because the objective is J_D, the optimizer directly penalizes viscous deformation loss and residual Brinkman drag in the diffuse design domain.

Yoon [56] reports that the turbulent diffuser design develops enlarged inlet and outlet regions, while the laminar solution forms a more cone-shaped diffuser. The present FEniCS-SA result is interpreted against this qualitative behavior rather than against a direct numerical objective target, because the present implementation uses a wall-resolved frozen-SA solve, active-mask filtering, and Heaviside projection.

The diffuser convergence histories are collected in Figure 6.13. Because this benchmark minimizes J_D rather than J_p, the convergence plot reports the normal-

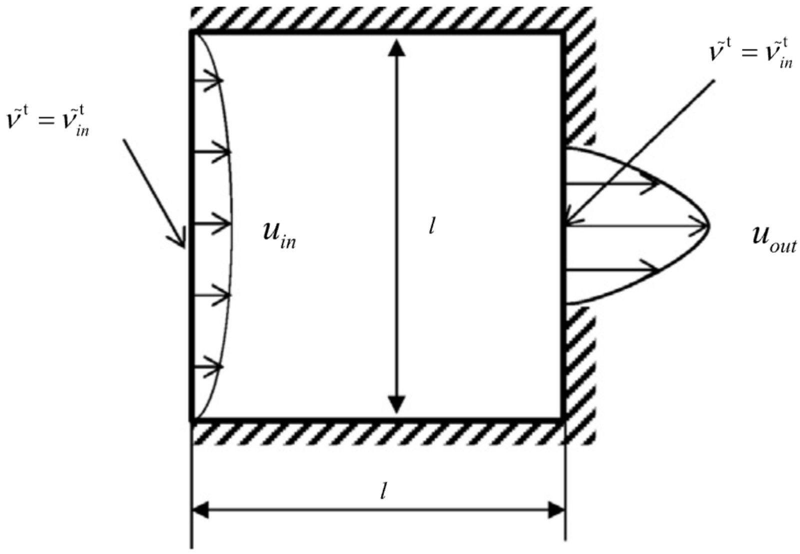
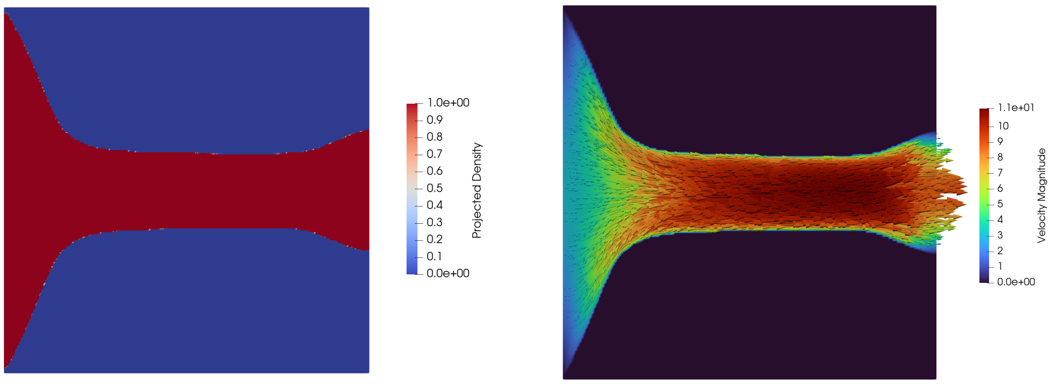


Figure 6.11: Geometry and boundary conditions for the diffuser benchmark from Yoon [56].



(A) Projected density field $\bar{\gamma}$.

(B) Velocity magnitude and vector field.

Figure 6.12: FEniCS-SA optimization result for the diffuser benchmark at $V_f = 0.30$. Final objective value $J_D = 2.8264 \times 10^5$ W/m.

ized dissipation objective and the relative volume-fraction residual on one combined figure.

6.5.3 CFD Validation

The extracted diffuser topology is validated with the same sharp-geometry workflow as the pipe bend and U-bend. The Ansys SST $k-\omega$ solve tests whether the diffuser remains robust under a two-equation model, while the Ansys-SA solve checks cross-code consistency for the same turbulence closure family. Because the optimization

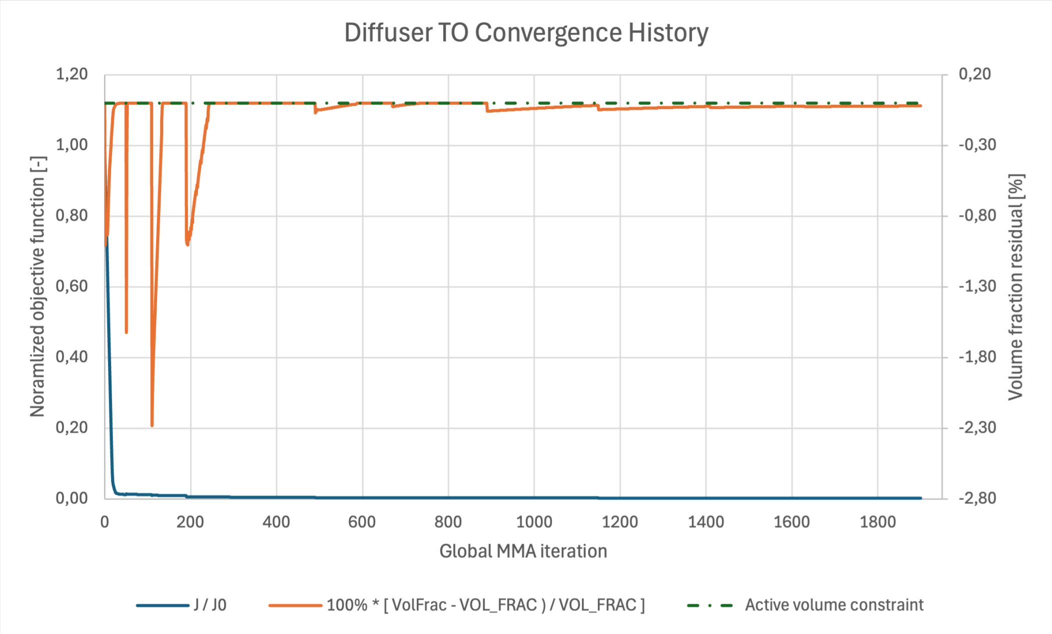


FIGURE 6.13: Optimization convergence history for the turbulent diffuser benchmark. The blue curve shows the normalized dissipation objective $J_D/J_{D,0}$, while the orange curve shows the relative volume-fraction residual.

objective is J_D, the validation emphasizes both physical viscous dissipation and static pressure drop. The resulting validation metrics are summarized in Table 6.8.

TABLE 6.8: CFD validation metrics for the extracted diffuser topology.

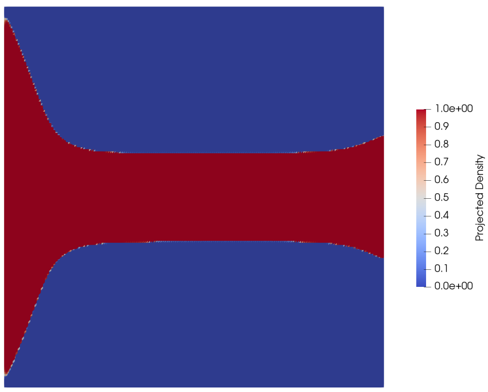
Solver	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Comment**
FEniCS diffuse design	SA	2.7553×10^5	2.6666×10^5	Diffuse diagnostic; $J_D = 2.8264 \times 10^5$ W/m
Ansys sharp geometry	SA	1.2995×10^5	1.2422×10^5	Cross-code comparison
Ansys sharp geometry	SST $k-\omega$	1.5725×10^4	4.1833×10^4	Turbulence-model sensitivity

The difference between the Ansys-SA and Ansys-SST $k-\omega$ results in Table 6.8 is therefore interpreted as a turbulence-closure effect rather than a residual mesh effect. The mesh-independence study in Appendix B gives sub-percent medium–fine changes for both closures, but the fine-mesh SST $k-\omega$ result is lower than the SA result by a factor of 8.3 in Φ_D and a factor of 3.0 in Δp. Since the validation dissipation integrand is $2\mu_{\text{eff}}\mathbf{S}:\mathbf{S}$ through Eq. 6.1, differences in the modeled eddy viscosity directly change the reported loss. The SA model computes μ_t from the transported working variable through $\mu_t = \rho\tilde{\nu}f_{v1}$ [44, 39]; in the present Ansys-SA diffuser runs the effective-viscosity ratio is of order 10^2–10^3. By contrast, the SST model computes eddy viscosity using Menter's shear-stress-transport definition, in which μ_t is bounded by the larger of an ω term and a strain-rate term [32, 39]. Although the raw inlet quantity $\rho k/(\mu\omega)$ was verified to match the specified turbulent-viscosity ratio of 1000, the converged SST effective-viscosity ratio remains of order 10^1 in this geometry. The much lower SST μ_{eff} is consistent with the lower SST dissipation and pressure drop; the diffuser validation should therefore be read as strongly closure-sensitive in absolute loss prediction.

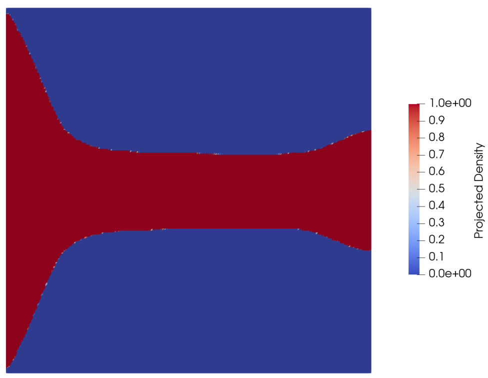
6.5.4 Comparison with Laminar Design

The laminar reference uses the same diffuser mesh, inlet and outlet profile shapes, objective type, and volume fraction, but it is optimized at $Re = 1$ without the SA turbulence model. The extracted laminar and turbulent layouts are then evaluated under the same target turbulent operating conditions. This comparison is expected to show whether the laminar cone-like diffuser remains competitive once turbulent eddy viscosity and the outlet contraction are considered.

Figure 6.14 compares the laminar and turbulent optimized diffuser topologies. The expected distinction follows Yoon [56]: the laminar design tends toward a cone-like diffuser, while the turbulent design should allocate additional fluid area near the inlet and outlet where eddy-viscosity effects increase the local loss. The laminar FEniCS configuration and optimization diagnostics are reported in Appendix C, and the laminar-design mesh-independence study supporting the SST $k-\omega$ result is reported in Table C.6.



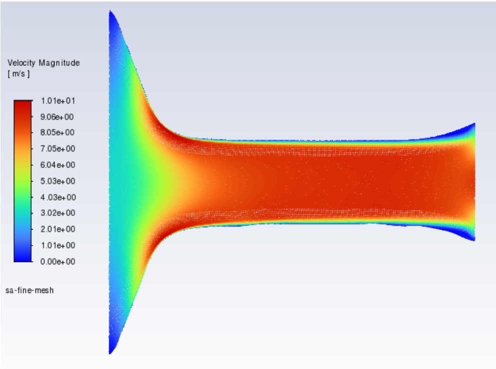
(A) Projected density field $\bar{\gamma}$: laminar-optimized design ($Re = 1$). Final objective value $J_D = 1.1655 \times 10^7$ W/m.



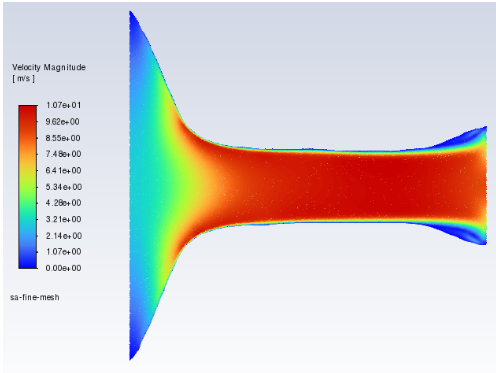
(B) Projected density field $\bar{\gamma}$: turbulent-optimized design ($Re = 3{,}000$). Final objective value $J_D = 2.8264 \times 10^5$ W/m.

Figure 6.14: Laminar and turbulent topology comparison for the diffuser benchmark.

Figure 6.15 shows Ansys SA velocity contours for the extracted diffuser designs. The SA contours are used because both diffuser topologies are assessed with the same closure used by the turbulent optimization loop, and because the SA re-analysis is the model under which the turbulent design outperforms the laminar design. The contour comparison complements the topology plot by indicating whether the cone-like laminar layout accelerates or separates near the contraction, while the turbulent layout distributes the expansion more smoothly near the inlet and outlet.



(A) Velocity magnitude field u: laminar-optimized design.



(B) Velocity magnitude field u: turbulent-optimized design.

FIGURE 6.15: Ansys SA velocity-magnitude comparison for the extracted diffuser designs under identical turbulent operating conditions.

Table 6.9 compares the extracted laminar and turbulent diffuser topologies under identical turbulent operating conditions. Because the diffuser was optimized with J_D, lower Φ_D is the primary performance indicator, while the static pressure drop remains the most direct hydraulic interpretation of the extracted sharp geometry. Both SST k-ω and SA re-analyses are included because the diffuser benchmark is strongly turbulence-model-sensitive.

TABLE 6.9: Performance comparison of the extracted diffuser topologies under identical turbulent operating conditions.

Design origin	**Model**	Φ_D **[W/m]**	Δp **[Pa]**	**Relative performance**
Laminar TO, $Re = 1$	SST $k - \omega$	1.3077×10^4	3.6464×10^4	Baseline (SST)
Turbulent TO, SA	SST $k - \omega$	1.5725×10^4	4.1833×10^4	20.3% higher Φ_D; 14.7% higher Δp
Laminar TO, $Re = 1$	SA	1.4333×10^5	1.4200×10^5	Baseline (SA)
Turbulent TO, SA	SA	1.2995×10^5	1.2422×10^5	9.3% lower Φ_D; 12.5% lower Δp

Under SST k-ω, the laminar-optimized diffuser gives the lower Ansys re-analysis metrics, whereas the SA re-analysis reverses that ranking and favors the turbulent-optimized diffuser. This split is consistent with the model sensitivity observed in Table 6.8: the diffuser conclusion depends on the turbulence closure used for the sharp-geometry validation, not only on mesh refinement.

6.6 Conclusion

This chapter answers the third research question by evaluating whether the turbulent topology optimization framework produces designs that remain credible after independent CFD re-analysis. The sensitivity verification is performed on the same penalized wall-distance, topology-penalized SA, frozen-turbulence formulation used for the pipe-bend optimization; therefore, the coordinate finite-difference and Taylor checks directly test the derivative supplied to MMA for J_p. The pipe-bend result provides the strongest completed evidence: the frozen derivative agrees with finite differences to a maximum relative discrepancy of 1.47×10^{-4} in the coordinate check, the Taylor remainder shows approximately second-order behavior, and the extracted design remains robust when re-solved in Ansys Fluent with both SA and SST $k-\omega$.

For the pipe bend, changing the Ansys closure from SA to SST $k-\omega$ increases the validated pressure drop by only about 4.5% and the viscous dissipation by about 8.5%, indicating that the optimized geometry is not narrowly tuned to the SA model. Under identical SST $k-\omega$ operating conditions, the turbulent design also reduces the pressure drop and viscous dissipation relative to the laminar-optimized design, supporting the claim that high-Reynolds turbulence physics should be included directly in the design loop. The U-bend and diffuser sections extend the same evaluation structure to a stronger flow-reversal case and to a J_D-driven diffuser case; once their remaining placeholder metrics are replaced, they should be judged by the same standard: convergence of the optimization, sharp-geometry validation under at least one independent closure, and comparison with an equivalent laminar-optimized layout.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

The objective of this thesis was to build and assess a flow-only turbulent topology optimization framework for internal configurations that are relevant to energy and thermal-management systems. The word flow-only is essential. The framework developed here does not solve the energy equation and does not yet optimize conjugate heat transfer. Instead, it addresses the hydraulic prerequisite for those applications: generating low-loss internal flow paths under turbulent operating conditions. This narrower scope is deliberate, because turbulent momentum transport, wall-distance treatment, Brinkman penalization, and adjoint sensitivity assembly already form a difficult and only partially mature optimization problem.

The thesis structure follows that scope. Chapter 2 establishes the turbulence-model background and the expected limits of the Spalart-Allmaras (SA) model. Chapter 3 presents the FEniCS implementation of the SA forward solver, and Chapter 4 verifies it against Ansys Fluent before it is used for topology optimization. Chapter 5 then introduces the density-based turbulent optimization framework, including Brinkman penalization, topology-dependent SA and wall-distance penalties, filtering, projection, MMA, and the frozen-turbulence adjoint. Finally, Chapter 6 evaluates the resulting designs through independent sharp-geometry CFD re-analysis.

Research question 1: How accurately does the Spalart-Allmaras turbulence model predict curvature-dominated internal turbulent flows, and where do its physical limitations appear? The answer is conditional. SA is a practical baseline model for the present optimization framework because its one-equation, wall-resolved formulation is cheaper and simpler than two-equation closures while still representing near-wall eddy-viscosity effects. The pipe-bend validation supports this use: the optimized bend produces a smooth attached flow path, and Ansys-SA and Ansys-SST $k-\omega$ predict similar pressure-drop and dissipation levels for the extracted turbulent design. However, the diffuser case exposes the important limitation. There, separated low-velocity regions appear near the diverging outlet part, and the Ansys-SA and SST $k-\omega$ re-analyses give substantially different dissipation and pressure-drop values. This behavior is consistent with the known weakness of eddy-viscosity closures, and especially one-equation closures, in separated, anisotropic, adverse-pressure-gradient flow. SA is therefore suitable as a bounded engineering optimization closure, but not as a model whose diffuser predictions can be accepted without independent validation and flow-field diagnosis.

Research question 2: Can a custom FEniCS implementation of the Spalart-Allmaras turbulence model achieve quantitative agreement with Ansys Fluent for turbulent internal-flow benchmarks? Within the two-dimensional and wall-resolved scope of this thesis, the answer is yes. The implemented solver reproduces the governing ingredients needed by the optimizer: incompressible flow coupling, SA transport, eddy-viscosity reconstruction, wall-distance evaluation, and near-wall resolution. The periodic channel case verifies the near-wall turbulence balance in a canonical geometry, while the U-bend verification tests the same solver under curvature and stronger convection. These comparisons do not prove that SA is universally accurate, but they do show that the FEniCS implementation is consistent enough with Ansys-SA to serve as the forward model for the topology optimization studies.

Research question 3: To what extent do SA-driven topology optimizations produce final designs whose predicted performance remains consistent under independent Ansys re-analysis with SA and SST $k{-}\omega$, and how do they compare with laminar-optimized designs under identical turbulent operating conditions? The strongest completed evidence is the pipe-bend benchmark. The frozen derivative used by MMA was checked directly through coordinate finite differences and a directional Taylor test, giving a maximum coordinate relative discrepancy of 1.47×10^{-4} and approximately second-order Taylor behavior. The extracted pipe-bend topology also remains credible under independent Ansys re-analysis: switching from Ansys-SA to SST $k-\omega$ changes the validated losses only modestly, and the turbulent design substantially outperforms the laminar-optimized reference under identical turbulent operating conditions. The U-bend remains the most important remaining benchmark because it combines high Reynolds number operation with strong flow reversal; any final claim for that case should only be made after the $Re = 5{,}000$ turbulent optimization and sharp-geometry validation placeholders are replaced. The diffuser already provides a useful negative result: under SA validation the turbulent design improves on the laminar design, but under SST $k - \omega$ the ranking reverses. That split does not invalidate the framework, but it shows that separated diffuser flows require model-sensitive interpretation rather than a single scalar performance claim.

Overall, the thesis demonstrates that a frozen-SA density-based topology optimization framework can produce credible low-loss turbulent internal-flow designs when the flow remains reasonably attached and when the final geometry is independently re-analyzed. It also shows where the framework is not yet sufficient: strongly separated diffuser behavior, objective oscillations during some MMA continuation stages, and incomplete high-Reynolds U-bend optimization still require additional computation and diagnosis. The most defensible conclusion is therefore not that the method is a finished turbulent thermal-design tool, but that it is a verified flow-only stepping stone toward one.

7.2 Limitations

An objective evaluation of this methodology reveals several important limitations that must be acknowledged:

1. **The Frozen Adjoint Approximation:** By ignoring the derivative of the eddy viscosity with respect to the design field, the optimizer cannot mathematically anticipate future turbulence production. In flows dominated by massive separation, this can blind the optimizer to the downstream hydrodynamic consequences of upstream geometric changes, potentially leading to sub-optimal local minima.

2. **Geometry Extraction Uncertainties:** The transition from a continuous, Brinkman-penalized density field ($\gamma \in [0, 1]$) to a Boolean CAD geometry introduces inherent ambiguity. The specific iso-value threshold chosen to extract the fluid volume dictates the final wall placement. Consequently, the performance discrepancies observed between FEniCS and Ansys are inextricably linked to this manual geometry extraction process, making it difficult to isolate pure solver physics from thresholding effects.

3. **Two-Dimensional Simplification:** The benchmark geometries evaluated in this thesis were restricted to 2D planar domains. Because turbulent phenomena—particularly curvature-induced secondary flows like Dean vortices and turbulent energy cascades—are inherently three-dimensional, the 2D assumption artificially suppresses complex mixing mechanisms that would inevitably occur in physical prototypes.

4. **Lack of Experimental Validation:** While independent cross-code verification against commercial finite-volume solvers strongly corroborates the numerical implementation, it is not a substitute for physical validation. The true operational performance of these algorithmically generated turbulent manifolds requires future validation against experimental pressure-drop and flow-field measurements.

5. **Sensitivity Verification Scope:** The finite-difference and Taylor checks verify the frozen derivative used by MMA, not the derivative of a fully coupled turbulence, wall-distance, and geometry-extraction pipeline. This distinction is important when interpreting the sensitivity evidence: the optimizer is internally consistent with its chosen approximation, but the approximation itself remains a modeling decision.

6. **Optimization Robustness:** Some continuation histories still show objective oscillations, especially when pressure-based objectives are used as proxies for energy loss. Although pressure drop and dissipation are physically related for

steady internal flows, a domain-integrated dissipation objective can provide a less localized and more stable optimization signal than an average inlet-pressure objective. The final benchmark conclusions should therefore distinguish completed, validated runs from remaining runs that still require stricter continuation, smaller MMA move limits, or longer fixed-parameter polishing stages.

7.3 Future Work

The verified turbulent fluid mechanics framework developed in this thesis establishes a mathematically stable foundation for several advanced avenues of future research:

7.3.1 Immediate Benchmark Completion and Diagnostics

Before extending the framework physically, the remaining two-dimensional benchmark evidence should be completed. The highest-priority computation is the $Re =$ 5,000 turbulent U-bend topology optimization, preferably with the dissipation objective J_D rather than the average inlet-pressure proxy J_p. If the full high-Reynolds run remains unstable, an intermediate Reynolds number should be used only as supporting evidence for the onset of asymmetric high-inertia topology formation, not as a replacement for the target benchmark.

The pipe-bend case should be repeated with the same J_D objective and stricter MMA damping to test whether the current objective oscillations are caused primarily by the localized pressure objective. If oscillations persist, a long fixed-continuation polishing run at the most stable q, β, and move-limit setting should be used, and the remaining oscillatory behavior should be reported as an optimization-robustness limitation rather than hidden.

For the diffuser, the main unresolved question is not simply whether the turbulent design has a lower scalar loss, but where the turbulence-model discrepancy originates. The separated low-velocity region near the diverging outlet should be isolated by re-running the extracted geometry after trimming the diverging part, by testing a pressure-outlet topology-optimization variant, and by plotting cellwise dissipation density, turbulent viscosity, and the turbulent component of dissipation for both SA and SST $k-\omega$ re-analyses. These plots would directly test whether the large SA dissipation originates in the separated outlet region and whether it is driven by over-predicted eddy viscosity.

7.3.2 Conjugate Heat Transfer (CHT) and Thermal Management

The ultimate engineering application for high-Reynolds topology optimization is not merely fluid routing, but active thermal management. Historically, the immense computational cost of resolving Conjugate Heat Transfer (CHT) has forced optimization frameworks to rely on laminar approximations (e.g., natural convection as demonstrated by Alexandersen et al. [3]) or simplified flow proxies such as Darcy models [58]. Even when high-resolution, density-based solvers are successfully deployed for microelectronics cooling, they typically circumvent the complexities of turbulent momentum transport [11].

However, high-performance heat sinks still require models that capture turbulent convection. Recent work has introduced turbulence into thermal topology optimization using extended finite element (XFEM) level-set approaches with the SA model [33], computationally reduced multi-layer thermofluid approximations [57], and standard density-based frameworks coupled to two-equation $k-\epsilon$ closures [47].

With the turbulent momentum and SA transport equations implemented and verified within the current FEniCS architecture, the framework provides a practical starting point for turbulent CHT optimization. Extending the current methodology to CHT will require implementing a turbulent Prandtl number (Pr_t) approach to model the turbulent heat flux vector, alongside deriving the corresponding thermal adjoint sensitivities to minimize peak temperatures or maximize overall heat transfer coefficients. Furthermore, the capacity for optimizing two-fluid heat exchangers—currently dominated by laminar assumptions [27]—could be improved by adapting this turbulent solver.

7.3.3 Three-Dimensional Topologies

As noted in the limitations, turbulent secondary flows are inherently 3D phenomena. To fully evaluate the geometric limits of the turbulence models and design physically realizable manifolds, the current 2D FEniCS implementation must be extended to three dimensions. The mathematical structure of the governing equations and adjoint derivations remains similar, but the computational cost changes by orders of magnitude because wall-resolved turbulent optimization requires fine near-wall resolution over a three-dimensional design domain. Large-scale topology optimization studies have demonstrated that parallel finite-element optimization at extreme design-variable counts is possible when the formulation is paired with efficient domain decomposition and scalable solvers [1]. A 3D turbulent extension of the present work should therefore prioritize parallel mesh generation, robust algebraic multigrid preconditioning, and automated sharp-geometry extraction.

7.3.4 Adjoint Model Hierarchies

This thesis deliberately uses the frozen-turbulence adjoint as the robust baseline. Future frameworks should investigate stabilized ways of reintroducing selected turbulence-production sensitivities, especially in separated flows where the constant-eddy-viscosity assumption is least accurate. Existing turbulent topology optimization studies already span several levels of adjoint complexity, from frozen or constant-eddy-viscosity approximations [19, 60] to more complete RANS sensitivity formulations and implementation studies [31, 55]. A practical next step is therefore not necessarily to jump directly to a fully coupled turbulence adjoint, but to build an adjoint hierarchy: frozen SA sensitivities as the baseline, selected SA transport sensitivities as an intermediate model, and two-equation closures such as SST $k-\omega$ for cases where curvature, separation, and anisotropic shear dominate the design response.

Appendices

Appendix A

Configuration and Grid Independence of the Spalart-Allmaras Verification Cases

To ensure complete scientific reproducibility and to prevent the disruption of the narrative flow in Chapter 3, all comprehensive configuration matrices and spatial grid independence studies for the fluid benchmarks are documented in this appendix.

Unless stated otherwise, all grid-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ, the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

A.1 Verification Benchmark 1: Channel Flow

A.1.1 Physical and Geometric Parameters

The first verification benchmark consists of a pressure-driven periodic 2D channel. To align with fundamental turbulence research conventions, the characteristic length is defined as the full channel height (H), and the Reynolds number (Re_H) is computed strictly based on this geometric parameter rather than the hydraulic diameter. The complete physical setup is detailed in Table A.1.

A.1.2 Numerical Simulation Parameters

The channel flow was resolved using an Incremental Pressure Correction Scheme (IPCS) embedded within a Picard fixed-point iteration loop. The exact numerical solver configurations, convergence tolerances, and relaxation factors utilized in FEniCS are summarized in Table A.2.

TABLE A.1: Geometric, physical, and boundary-condition parameters for the periodic channel flow benchmark.

Parameter	**Channel flow setting**
Benchmark type	Pressure-driven, streamwise-periodic 2D channel
Domain dimensions	Length $L = 2.0$ m; full channel height $H = 2.0$ m
Characteristic length	Full channel height $H = 2.0$ m
Reference velocity	$U_{\text{ref}} = 18.5$ m/s, utilized to define the target Reynolds number
Kinematic viscosity	$\nu = 1.81818 \times 10^{-3}$ m^2/s
Reynolds number	$Re_H \approx 2.03 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s^2
Driving condition	Fixed pressure drop from $p = 2$ Pa to $p = 0$ Pa across the periodic streamwise direction
Velocity boundary conditions	Periodic velocity between inlet/outlet planes; no-slip on top and bottom walls
Pressure boundary conditions	$p = 2$ Pa at the upstream pressure plane; $p = 0$ Pa at the downstream pressure plane
SA boundary conditions	Periodic $\tilde{\nu}$ between inlet/outlet planes; $\tilde{\nu} = 0$ at walls
Initial fields	$\mathbf{u}_0 = (18.5, 0)$ m/s, $p_0 = 2$ Pa, $\tilde{\nu}_0 = 5.0 \times 10^{-3}$ m^2/s
Verification mesh	`Fine_FEniCS`; first layer $\Delta y_1 = 1.709 \times 10^{-3}$ m

TABLE A.2: Numerical simulation parameters for the FEniCS steady Spalart-Allmaras channel benchmark.

Parameter	**Symbol**	**Setting**
Velocity space	$\mathbf{u}$	Vector CG2, periodic in streamwise direction
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1, periodic in streamwise direction
Quadrature degree	–	2
Maximum outer Picard steps	$N_{\text{Picard,max}}$	200
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-6}
Outer pressure tolerance	ϵ_p	1.0×10^{-6}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	5.0×10^{-3} s
Maximum IPCS steps per Picard step	–	500
IPCS velocity tolerance	–	1.0×10^{-6}
IPCS pressure tolerance	–	1.0×10^{-6}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.3
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.3
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: direct MUMPS; SA transport: default solver

A.1.3 Spatial Verification and Grid Independence

To guarantee that the numerical results were independent of the spatial discretization, a formal grid independence study was conducted. Three distinct boundary-layer inflated meshes (Coarse, Medium, and Fine) were evaluated. The geometric properties, node counts, and resolution parameters of this mesh hierarchy are explicitly detailed in Table A.3. Crucially, the first-layer height was kept constant across all grids to strictly enforce the $y^+ \approx 1$ requirement, while the bulk and inflation-layer resolutions were systematically refined.

Table A.3: Wall-resolved channel mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the streamwise resolution, wall-normal resolution, and inflation-layer resolution are refined.

Parameter	Coarse	Medium	Fine
Mesh settings			
Streamwise cells	10	20	40
Wall-normal cells	42	51	68
Δy_1 [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\min}$ [m]	1.709×10^{-3}	1.709×10^{-3}	1.709×10^{-3}
$\Delta y_{\max}$ [m]	7.707×10^{-2}	8.454×10^{-2}	7.589×10^{-2}
Inflation layers per wall	10	14	18
Growth ratio	1.45	1.35	1.25
Vertices	473	1,092	2,829
Elements	840	2,040	5,440
Grid metric			
U_{bulk} [m/s]	16.923	18.489	18.472
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	9.25	0.09

Note. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

Visual confirmation of this spatial convergence is provided by the wall-normal velocity profiles plotted in Figure A.1. In each case, the profile is extracted at the streamwise mid-plane, $x = 0.5L$, with the wall-normal coordinate y varied across the full channel height. The `Fine_FEniCS` grid was consequently selected for the final verification against the Ansys Fluent baseline, while the coarser grids demonstrate that the wall-normal velocity profile is already asymptotically converged.

Furthermore, to ensure a rigorous cross-code validation, an equivalent grid independence study was performed for the commercial Ansys Fluent baseline. The Ansys domain was discretized using a comparable mapped mesh hierarchy, maintaining a strict $y^+ \approx 1$ resolution at the walls. In addition to the residual monitors, the bulk velocity was monitored and required to change by less than 1.0×10^{-3} over 250 iterations before accepting a converged solution. The continuity residual criterion was set to 1.0×10^{-5}, while the momentum and SA working-variable residual criteria were set to 1.0×10^{-6}. Mass conservation was also checked from the inlet and outlet mass-flow reports. The properties of these meshes are documented in

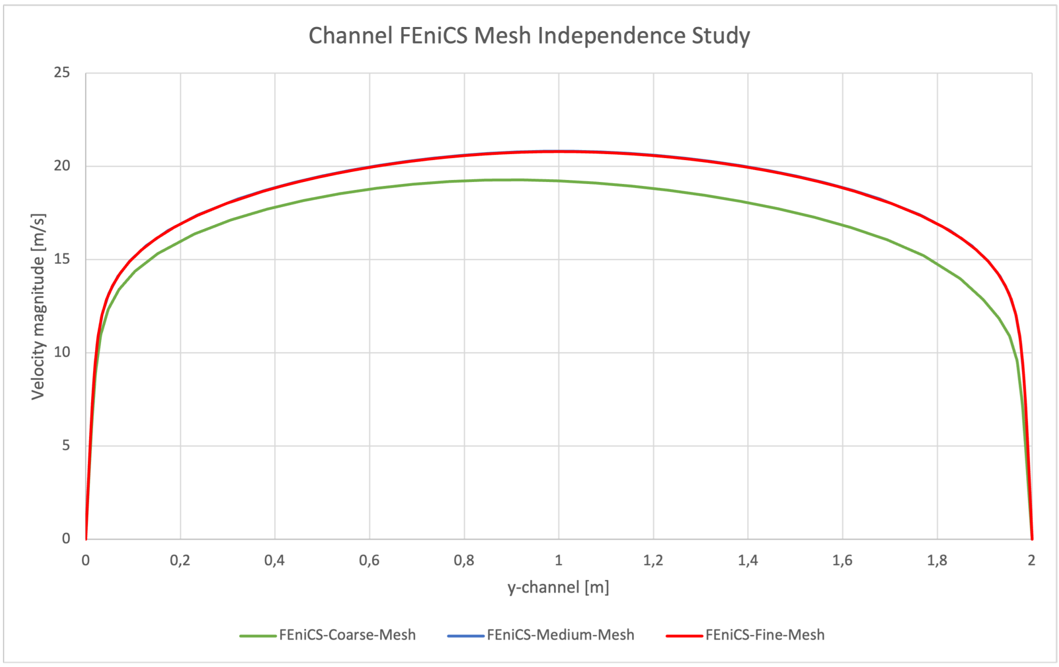


FIGURE A.1: FEniCS wall-normal velocity profiles extracted at $x = 0.5L$ for the Coarse, Medium, and Fine grids in the periodic channel flow benchmark. The overlapping profiles demonstrate asymptotic spatial convergence.

Table A.4. The wall-normal velocity profiles extracted from the Ansys solver at the same mid-channel location, illustrated in Figure A.2, exhibited identical asymptotic convergence, confirming that the baseline reference data is mathematically robust and completely independent of spatial discretization errors.

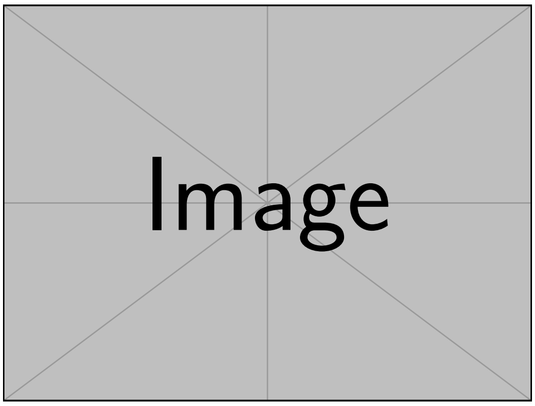


FIGURE A.2: Wall-normal velocity profiles extracted at $x = 0.5L$ for the Coarse, Medium, and Fine grids in the Ansys periodic channel flow benchmark, demonstrating spatial convergence of the baseline data.

TABLE A.4: Wall-resolved channel mesh hierarchy used for the Ansys grid-independence study.

Parameter	Coarse	Medium	Fine
Mesh settings			
Mesh type	MultiZone	MultiZone	MultiZone
Face type	All quad	All quad	All quad
N_x	64	160	220
N_y	80	144	180
Bias type	Two-sided	Two-sided	Two-sided
Bias factor	59	28	21
Approx. Δy_1 [m]	1.700×10^{-3}	1.700×10^{-3}	1.700×10^{-3}
Nodes	7,740	26,448	43,010
Elements	7,565	26,123	42,594
Grid metric			
U_{bulk} [m/s]	*TBD*	18.114	*TBD*
$\epsilon_{\text{rel}}(U_{\text{bulk}})$ [%]	–	*TBD*	*TBD*

Note. N_x and N_y are the streamwise and wall-normal edge divisions. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

A.2 Verification Benchmark 2: U-Bend Geometry

A.2.1 Physical and Geometric Parameters

The second verification benchmark evaluates a highly convective U-bend. Because this FEniCS domain is a 2D planar approximation of the symmetry mid-plane of the 3D Ansys VMFL048 case, the 3D physical pipe diameter (D) is explicitly retained as the characteristic length scale. This ensures the Reynolds (Re_D) and Dean (De) numbers remain non-dimensionally consistent with the commercial baseline. The straight inlet and outlet legs are shorter than in the original VMFL048 Ansys mesh to reduce meshing and computational effort; both the FEniCS and Ansys runs use this same short-leg U-bend geometry. The configuration is detailed in Table A.5.

A.2.2 Numerical Simulation Parameters

Due to the adverse pressure gradients and secondary flow separation induced by the streamline curvature, the U-bend represents a significantly more demanding numerical challenge than the channel. Consequently, the FEniCS solver required a smaller pseudo-time step, tighter under-relaxation, and advanced iterative preconditioners to maintain stability, as detailed in Table A.6.

A.2.3 Spatial Verification and Grid Independence

Managing the highly stretched, non-orthogonal inflation layers around the curved arcs of the U-bend required custom mesh generation via Gmsh. A spatial grid independence study was conducted across three refined unstructured grids. The node counts, element densities, and inflation layer parameters for these generated meshes

TABLE A.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	**U-bend setting**
Benchmark type	Short-leg 2D symmetry-plane U-bend based on Ansys VMFL048
Domain dimensions	Pipe diameter $D = 0.028$ m; centreline bend radius $R_c = 0.125$ m; straight-leg length $30D = 0.840$ m
Characteristic length	Physical pipe diameter $D = 0.028$ m
Reference / inlet velocity	Uniform prescribed inlet velocity $\mathbf{u}_{in} = (0, -1.42)$ m/s
Kinematic viscosity	$\nu = 8.9 \times 10^{-7}$ m^2/s
Reynolds number	$Re_D \approx 4.47 \times 10^4$
Dean number	$De \approx 1.50 \times 10^4$
Body force	$\mathbf{f} = (0, 0)$ m/s^2
Driving condition	Prescribed inlet velocity and zero relative static pressure at the outlet
Velocity boundary conditions	No-slip on all curved and straight pipe walls; natural velocity condition at pressure outlet
Pressure boundary conditions	$p = 0$ Pa at outlet; pressure left free at inlet and walls
SA boundary conditions	$\tilde{\nu}_{in} = 9.34 \times 10^{-5}$ m^2/s at inlet; $\tilde{\nu} = 0$ at walls; natural condition at outlet
Initial fields	$\mathbf{u}_0 = (0, 0)$ m/s, $p_0 = 0$ Pa, $\tilde{\nu}_0 = \tilde{\nu}_{in}$
Verification mesh	`Medium_FEniCS`; first layer $\Delta y_1 = 1.20 \times 10^{-5}$ m

are explicitly documented in Table A.7. Similar to the channel benchmark, the wall-adjacent element height was strictly maintained at 1.20×10^{-5} m to satisfy the low-Reynolds SA wall constraints, while the internal element sizing was systematically refined.

The reported FEniCS static pressure-drop metric is computed from boundary-measure-weighted mean pressures over the inlet and outlet sections, rather than from local point samples. Let Γ_{in} and Γ_{out} denote the inlet and outlet boundaries, and let p_k be the kinematic pressure solved in FEniCS. The physical static pressure drop reported in pascals is

$$\Delta p = \rho \left(\overline{p}_{k,\text{in}} - \overline{p}_{k,\text{out}} \right) = \rho \left[\frac{1}{|\Gamma_{\text{in}}|} \int_{\Gamma_{\text{in}}} p_k \, \mathrm{d}\Gamma - \frac{1}{|\Gamma_{\text{out}}|} \int_{\Gamma_{\text{out}}} p_k \, \mathrm{d}\Gamma \right], \tag{A.1}$$

where $|\Gamma| = \int_\Gamma 1 \, \mathrm{d}\Gamma$ is the FEniCS boundary measure and $\rho = 997$ kg/m^3. In the implementation this corresponds to integrating over the marked inlet and outlet facets with `ds(2)` and `ds(3)`, respectively. The density factor is required because the IPCS formulation stores pressure in kinematic form, while the benchmark table reports the static pressure drop in Pa. Although the outlet pressure is prescribed as zero relative pressure, the outlet term is still evaluated as a boundary average so that the diagnostic follows the same inlet-minus-outlet definition on every grid.

Convergence was visually verified by extracting the velocity profiles directly at the bend exit across the pipe diameter $D = 0.028$ m, a highly sensitive region for curvature-induced separation, as illustrated in Figure A.3. The chosen

Table A.6: Numerical simulation parameters for the highly convective FEniCS U-bend benchmark.

Parameter	**Symbol**	**Setting**
Velocity space	$\mathbf{u}$	Vector CG2
Pressure space	p	Scalar CG1
SA working-variable space	$\tilde{\nu}$	Scalar CG1
Quadrature degree	–	4
Maximum outer Picard steps	$N_{\mathrm{Picard,max}}$	180
SA solves per Picard step	–	1
Outer velocity tolerance	$\epsilon_{\mathbf{u}}$	1.0×10^{-4}
Outer pressure tolerance	ϵ_p	1.0×10^{-4}
Outer SA tolerance	$\epsilon_{\tilde{\nu}}$	1.0×10^{-6}
IPCS pseudo-time step	Δt	1.0×10^{-4} s
Maximum IPCS steps per Picard step	–	180
IPCS velocity tolerance	–	1.0×10^{-4}
IPCS pressure tolerance	–	2.0×10^{-3}
Velocity relaxation factor	α_u	0.3
Pressure relaxation factor	α_p	0.1
SA relaxation factor	$\alpha_{\tilde{\nu}}$	0.01
SA lower bound	$\tilde{\nu}_{\min}$	1.0×10^{-12}
Wall-distance equation	–	Relaxed Eikonal, $\sigma_w = 0.1$, $G_0 = 20$
Wall-distance Newton settings	–	Relative tolerance 1.0×10^{-8}; absolute tolerance 1.0×10^{-10}; maximum 80 iterations; relaxation 0.5
Linear solver strategy	–	IPCS blocks: GMRES with ILU velocity/correction preconditioning and Hypre-AMG pressure preconditioning; SA transport: default solver

Table A.7: Wall-resolved U-bend mesh hierarchy used for the FEniCS grid-independence study. The first-layer height is kept fixed to maintain $y^+ \approx 1$, while the bulk element size and inflation-layer resolution are refined.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
h_{bulk} [m]	2.80×10^{-3}	1.40×10^{-3}	8.00×10^{-4}
Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	14	18
Growth ratio	1.45	1.30	1.25
t_{infl} [m]	1.07×10^{-3}	1.53×10^{-3}	2.62×10^{-3}
Vertices	23,751	73,978	182,450
Elements	45,998	144,952	359,646
Grid metric			
Δp [Pa]	*TBD*	*TBD*	*TBD*
$\epsilon_{\mathrm{rel}}(\Delta p)$ [%]	–	*TBD*	*TBD*

Note. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

`Medium_FEniCS` grid perfectly captured the velocity gradients near the wall and provided sufficient isotropic resolution in the bulk flow to resolve the separation bubbles without numerical instability.

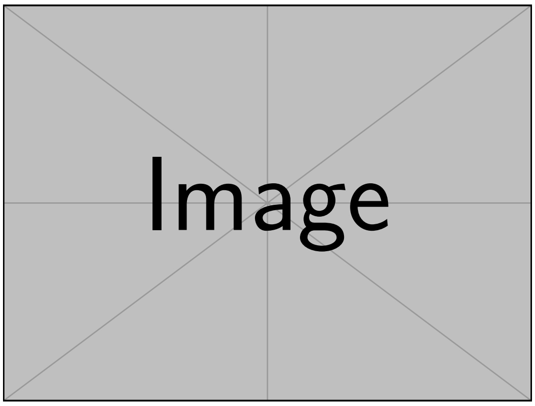


FIGURE A.3: Velocity profiles at the U-bend exit, extracted across the pipe diameter $D = 0.028$ m, for the Coarse, Medium, and Fine grids in FEniCS. The profiles demonstrate spatial convergence, justifying the selection of the Medium grid for the final verification.

Similarly, to guarantee the integrity of the cross-code validation in this highly convective regime, an equivalent mesh independence study was executed for the Ansys Fluent baseline. Using a parallel unstructured meshing strategy with strict $y^+ \approx 1$ boundary layer inflation (documented in Table A.8), the Ansys velocity profiles at the bend exit across the same pipe-diameter line, shown in Figure A.4, were checked for spatial convergence. For each Ansys solve, convergence was accepted only after the scaled continuity, momentum, and Spalart–Allmaras residuals reached 1.0×10^{-6} and the monitored static pressure drop changed by less than a relative tolerance of 5.0×10^{-4} over the final 100 iterations. The inlet and outlet mass-flow rates were also compared for each grid to verify the global mass balance before accepting the pressure-drop values. The `Fine_Ansys` grid is retained as the commercial verification baseline, while the deliberately coarser `Coarse_Ansys` level is selected to bracket the mesh-convergence trend.

Table A.8: Wall-resolved U-bend mesh hierarchy used for the Ansys grid-independence study.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Quad-dom.	Quad-dom.	Quad-dom.
h_{face} [m]	4.00×10^{-3}	2.00×10^{-3}	1.25×10^{-3}
Curvature angle	22°	12°	8°
Inflation option	First layer	First layer	First layer
Δy_1 [m]	1.20×10^{-5}	1.20×10^{-5}	1.20×10^{-5}
Inflation layers	10	16	22
Growth rate	1.35	1.22	1.16
t_{infl} [m]	6.55×10^{-4}	1.59×10^{-3}	2.17×10^{-3}
Nodes	14,525	47,707	106,460
Elements	13,984	46,640	104,763
Grid metric			
Δp [Pa]	814.109	796.804	792.995
$\epsilon_{\text{rel}}(\Delta p)$ [%]	–	2.13	0.48

Note. Quad-dom. = quad-dominant. For the ϵ_{rel} row, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

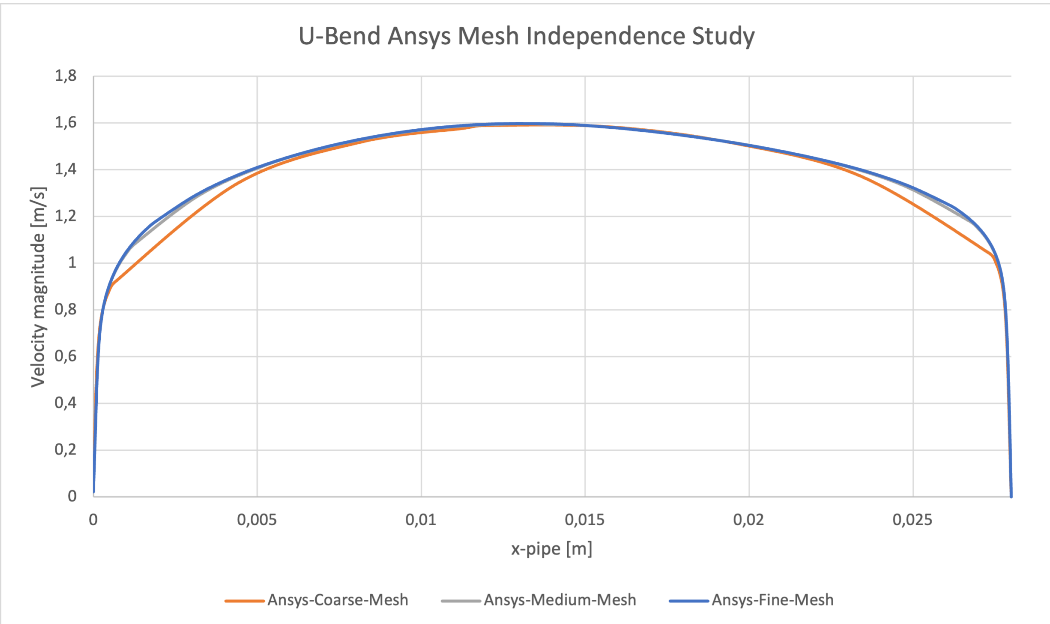


Figure A.4: Velocity profiles at the U-bend exit, extracted across the pipe diameter $D = 0.028$ m, for the Coarse, Medium, and Fine Ansys grids, demonstrating spatial convergence of the commercial baseline.

Appendix B

Configuration and Grid Independence of the Topology Optimization Benchmarks

This appendix collects the detailed setup information for the topology optimization benchmarks used in Chapter 6. The pipe-bend benchmark, U-bend benchmark, and diffuser benchmark are included. For each case, the appendix records the geometry and physical parameters, the FEniCS simulation and optimization parameters, and the Ansys Fluent mesh-independence tables used for sharp-geometry CFD validation.

B.1 Validation and Meshing Strategy

The FEniCS meshes used during topology optimization are treated as fixed optimization discretizations rather than as a separate mesh-convergence hierarchy. This distinction is important because refining the FEniCS mesh would also change the DG_0 design space, the filter stencil, the number of MMA design variables, and the diffuse Brinkman interface location. The pipe-bend optimization uses a rectangular-coordinate triangular mesh with near-wall clustering based on a smooth-wall $y^+ \approx 1$ estimate at the prescribed inlet Reynolds number. The U-bend optimization uses the corresponding wall-resolved triangular Gmsh mesh, with a near-wall size field based on the same $y^+ \approx 1$ estimate and a far-field target size of $h_{\max} = 0.007L$ used to set the density-filter scale. The diffuser benchmark from Yoon [56] uses a wall-resolved rectangular-coordinate triangular mesh over the square diffuser domain, refined around the full-height inlet, the center-third outlet, and the no-slip walls. In all cases, the mesh is chosen to provide sufficient wall-resolved SA and design resolution for the optimization problem; quantitative grid-convergence evidence is instead reported for the extracted sharp geometries in Ansys Fluent.

For each extracted turbulent topology optimization design, the Ansys validation mesh is refined through coarse, medium, and fine resolutions. Both SA and SST $k-\omega$ validation runs use the same extracted geometry, inlet condition, outlet condition, and fluid properties. The mesh-independence comparison is reported for both turbulence models because near-wall behavior and separated-flow predictions may respond differently to grid refinement.

All Fluent mesh-independence runs in this appendix use the same convergence and acceptance criteria. The static pressure drop must change by less than 5×10^{-4} over the final 100 iterations, while the viscous dissipation must change by less than 10^{-3} over the final 100 iterations. The fine mesh is accepted only when the medium–fine relative difference is below 1% for static pressure drop and below 2% for viscous dissipation for both SA and SST $k - \omega$. The reported refinement differences are computed as $|\phi_{\mathrm{coarse}} - \phi_{\mathrm{medium}}|/|\phi_{\mathrm{coarse}}|$ and $|\phi_{\mathrm{medium}} - \phi_{\mathrm{fine}}|/|\phi_{\mathrm{fine}}|$ for each validation metric.

B.2 Shared Turbulence-Topology Coupling Conventions

All three turbulent topology optimization benchmarks use the same basic coupling idea between the density field, the SA working variable, and the reciprocal wall-distance equation. Topology-created solids act as walls for the SA working-variable penalty and the reciprocal wall-distance solve. The wall-density source is therefore `design`, the wall-distance mode is `reciprocal_penalized`, and both the SA and wall-distance penalties use the power-law solid indicator defined in Chapter 5. The same frozen-turbulence adjoint convention is also used: the forward loop updates $\tilde{\nu}$, ν_t, and G, while the derivative freezes $\tilde{\nu}$ and G during gradient assembly.

The pipe-bend and U-bend benchmarks then share a Bayat-style pressure-loss setup, while the diffuser follows the Yoon dissipation benchmark. The common mechanisms and the benchmark-specific parameter values are summarized in Table B.1.

The pipe-bend finite-difference sensitivity verification in Chapter 6 is not assigned a separate configuration table. It uses the same geometry, physical parameters, wall-distance convention, penalty settings, objective, filter, and solver setup as the pipe-bend topology optimization run. The only additional settings are the finite-difference sampling parameters listed in Table B.3.

TABLE B.1: Shared turbulence-topology coupling conventions and benchmark-specific parameter values.

Parameter	Pipe bend and U-bend	Diffuser
SA wall-distance mode	`reciprocal_penalized`	`reciprocal_penalized`
SA wall-density source	`design`	`design`
SA penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance penalty interpolation	Power-law solid indicator	Power-law solid indicator
Wall-distance recovery	Reciprocal recovery from the relaxed Eikonal variable G	Reciprocal recovery from the relaxed Eikonal variable G
Adjoint treatment	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly	Frozen-turbulence adjoint; $\tilde{\nu}$ and G frozen during gradient assembly
Solid threshold for wall source	$\bar{\gamma} = 0.50$	No separate threshold parameter in the diffuser configuration
SA working-variable penalty	Amplitude schedule ending at 10^5; exponent $n = 4$	Amplitude 10^5; exponent $n = 3$
Wall-distance penalty	Amplitude schedule ending at 10^5; exponent $n = 4$	Amplitude 10^5; exponent $n = 3$
Wall-distance parameters	$\sigma_w = 0.01$, $G_0 = 20$	$\sigma_w = 0.1$, $G_0 = 20$
SA inlet treatment	Turbulence-intensity estimate: $I = 7.5\%$, $\ell/H = 0.05$, eddy-viscosity ratio capped at 50	Prescribed $\tilde{\nu} = 1$ at inlet and outlet
Optimization objective	Average inlet pressure with fixed outlet pressure $p = 0$	Dissipation objective J_D with prescribed inlet and outlet velocity profiles

B.3 Pipe Bend Benchmark

The pipe-bend case follows the setup of Bayat et al. [12]. The reference paper uses a standard k-ε model with wall functions; the present thesis uses the wall-resolved SA implementation and evaluates the $V_f = 0.25$ design.

The extracted turbulent pipe-bend geometry is meshed in Ansys Fluent with a patch-conforming triangular bulk mesh and wall-resolved inflation on all no-slip walls. The first-layer height is kept fixed from the smooth-wall $y^+ \approx 1$ estimate, while the mesh hierarchy refines the bulk face size, curvature angle, and inflation distribution.

Table B.2: Geometric, physical, and boundary-condition parameters for the pipe-bend benchmark.

Parameter	**Setting**
Reference case	Pipe-bend benchmark from Bayat et al. [12]
Reference figure for setup	Bayat et al. Fig. 10
Reference velocity figure	Bayat et al. Fig. 15
Reference turbulence model	Standard k-ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [-0.2, 1]$
Non-design inlet duct	Length $0.2L$, height $0.2L$, inlet over $y/L \in [0.7, 0.9]$
Non-design outlet duct	Width $0.2L$, length $0.2L$, outlet over $x/L \in [0.7, 0.9]$ on the lower boundary
Volume fraction	$V_f = 0.25$
Reynolds number	$Re = 5{,}000$, based on $U_{\text{in}} H_{\text{in}} \rho/\mu$ with $H_{\text{in}} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table B.1
Outlet condition	Pressure outlet $p = 0$ on the lower outlet, with the normal/-tangential component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST k−ω

TABLE B.3: Numerical FEniCS solver, topology-optimization, and sensitivity-verification parameters for the pipe-bend benchmark.

Parameter	**Symbol / option**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
hh	–	`mesh_yplus1.xdmf`; 39,128 vertices, 77,334 triangles, $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.25$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 100$
RAMP continuation	q	Code schedule $q = 1/q_\alpha = [1/150,\ 1/75,\ 1/30,\ 1/15]$
Projection continuation	β	$[4,\ 6,\ 9,\ 13]$ with threshold $\eta = 0.50$
MMA move limits	–	$[0.05,\ 0.04,\ 0.03,\ 0.02]$
Maximum optimization iterations	–	Four continuation stages with 25 MMA iterations each; stop if objective change is below 10^{-5} for five iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.20
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly
Sensitivity verification samples	–	Five active DG0 cells, seed 13
Sensitivity verification step	Δh	10^{-6}, central finite difference
Sensitivity verification fields held fixed	–	SA working variable $\tilde{\nu}^0$ and reciprocal wall-distance field G^0

Table B.4: Ansys Fluent mesh-independence table for the extracted turbulent pipe-bend benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	1.10×10^{-2}	5.00×10^{-3}	3.25×10^{-3}
Curvature angle	15°	7°	4.5°
Near-wall	WR infl.	WR infl.	WR infl.
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	8	14	18
Growth rate	1.15	1.10	1.08
t_{infl} [m]	8.24×10^{-3}	1.68×10^{-2}	2.25×10^{-2}
Nodes	5,147	21,343	44,042
Elements	7,551	32,780	68,743
Validation metrics			
Φ_D^{SA} [W/m]	0.0143	0.0151	0.0153
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SA}})$ [%]	–	5.38	1.32
Φ_D^{SST} [W/m]	0.0150	0.0164	0.0166
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SST}})$ [%]	–	9.04	1.22
Δp^{SA} [Pa]	0.1423	0.1449	0.1457
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SA}})$ [%]	–	1.84	0.55
Δp^{SST} [Pa]	0.1481	0.1516	0.1522
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SST}})$ [%]	–	2.36	0.40
Selected validation mesh		Fine_Ansys	

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

B.4 U-Bend Benchmark

The U-bend case follows Fig. 11 and Table 4 of Bayat et al. [12]. It uses the same turbulent topology optimization framework as the pipe bend, but forces a 180° return path around a fixed internal separator.

TABLE B.5: Geometric, physical, and boundary-condition parameters for the U-bend benchmark.

Parameter	**Setting**
Reference case	U-bend benchmark from Bayat et al. [12]
Reference figure/table for setup	Bayat et al. Fig. 11 and Table 4
Reference turbulence model	Standard k-ε with wall functions
Turbulence model in this thesis	Spalart-Allmaras
Design-domain dimensions	Square design region $L \times L$ with $L = 1$
Full computational domain	$x/L \in [-0.2, 1]$, $y/L \in [0, 1]$
Port geometry	Two left-side ports of height $0.2L$; inlet over $y/L \in [0.55, 0.75]$, outlet over $y/L \in [0.25, 0.45]$
Fixed separator	Bar thickness $0.10L$, rounded-tip radius $0.05L$, extending from $x/L = -0.2$ to tip at $x/L = 0.5$
Volume fraction	$V_f = 0.27$
Reynolds number	$Re = 5{,}000$, based on $U_{\mathrm{in}} H_{\mathrm{port}} \rho/\mu$ with $H_{\mathrm{port}} = 0.2L$
Density and viscosity	$\rho = 1.0$, $\mu = 4.0 \times 10^{-5}$
Inlet condition	Uniform inlet velocity $\mathbf{u} = (1, 0)$; SA inlet estimated from the shared turbulence-intensity settings in Table B.1
Outlet condition	Pressure outlet $p = 0$ on the lower left port, with the component constraint used by the FEniCS configuration
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on external walls and the fixed separator; topology-created solids enter through the design-density penalties
Validation metrics	Average inlet pressure, pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST k–ω

Table B.6: Numerical FEniCS solver and topology-optimization parameters for the U-bend benchmark.

Parameter	Symbol / option	Setting
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	`mesh_yplus1.xdmf`; wall-resolved triangular Gmsh mesh with 45,280 vertices, 85,780 triangles, near-wall refinement based on $y^+ \approx 1$, and far-field $h_{\max} = 0.007L$
Objective	J_p	Average inlet pressure
Initial design density	γ_0	$V_f = 0.27$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 100$
RAMP continuation	q	Code schedule $q = 1/q_\alpha = [1/150,\ 1/75,\ 1/35,\ 1/12]$
Projection continuation	β	$[8,\ 10,\ 18,\ 18]$ with threshold $\eta = 0.50$
MMA move limits	–	$[0.05,\ 0.04,\ 0.03,\ 0.02]$
Maximum optimization iterations	–	Four continuation stages with 25 MMA iterations each; stop if objective change is below 10^{-5} for five iterations
Filter radius	r_f	$4h_{\max} = 0.028L$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.20
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly

TABLE B.7: Ansys Fluent mesh-independence table for the extracted turbulent U-bend benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	[TBD]	[TBD]	[TBD]
Near-wall	[TBD]	[TBD]	[TBD]
Δy_1 [m]	[TBD]	[TBD]	[TBD]
Inflation layers	[TBD]	[TBD]	[TBD]
Growth rate	[TBD]	[TBD]	[TBD]
Elements / cells	[TBD]	[TBD]	[TBD]
$y^+_{\max}$, SA	[TBD]	[TBD]	[TBD]
$y^+_{\max}$, SST	[TBD]	[TBD]	[TBD]
Validation metrics			
Φ_D^{SA} [W/m]	[TBD]	[TBD]	[TBD]
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SA}})$ [%]	–	[TBD]	[TBD]
Φ_D^{SST} [W/m]	[TBD]	[TBD]	[TBD]
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SST}})$ [%]	–	[TBD]	[TBD]
Δp^{SA} [Pa]	[TBD]	[TBD]	[TBD]
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SA}})$ [%]	–	[TBD]	[TBD]
Δp^{SST} [Pa]	[TBD]	[TBD]	[TBD]
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SST}})$ [%]	–	[TBD]	[TBD]
Selected validation mesh		[TBD]	

Note. Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

B.5 Diffuser Benchmark

The diffuser case follows Fig. 20–22 of Yoon [56]. Unlike the pipe-bend and U-bend benchmarks, the diffuser prescribes parabolic velocity profiles at both the inlet and the outlet and uses the dissipation objective J_D. Its turbulence-density coupling still uses the shared `design`, `reciprocal_penalized`, power-law, and frozen-adjoint conventions summarized in Table B.1.

For the body-fitted Fluent evaluations, the parabolic inlet and outlet velocity profiles are shifted to the extracted inlet and outlet segments. The listed heights are used as the profile heights, while the lower vertices define the local profile origins. The inlet maximum velocity is kept at 3 m/s so that the nominal Reynolds number remains Re = 3,000; the outlet maximum velocity is adjusted by incompressible flow-rate conservation.

The extracted turbulent diffuser geometry is meshed with the same wall-resolved triangular strategy as the laminar diffuser comparison in Table C.6, but with a distinct coarse–medium–fine hierarchy. The first-layer height is kept fixed at $\Delta y_1 = 1.50 \times 10^{-3}$ m to match the laminar-design diffuser meshes, while the bulk size, curvature angle, and inflation-layer distribution are refined independently for the turbulent topology.

Table B.8: Geometric, physical, and boundary-condition parameters for the diffuser benchmark.

Parameter	**Setting**
Reference case	Diffuser benchmark from Yoon [56]
Reference figures for setup and topology	Yoon Fig. 20–22
Reference turbulence model	Spalart-Allmaras
Turbulence model in this thesis	Spalart-Allmaras with frozen-turbulence adjoint
Design-domain dimensions	Square design region $L \times L$ with $L = 1$ m
Full computational domain	$x/L \in [0, 1]$, $y/L \in [0, 1]$
Inlet geometry	Exported Fluent inlet segment from $(1.16 \times 10^{-3}, 1.38 \times 10^{-2})$ to $(1.16 \times 10^{-3}, 0.979)$, with $H_{\mathrm{in}} = 0.96558$ m
Outlet geometry	Exported Fluent outlet segment from $(0.999, 0.336)$ to $(0.999, 0.670)$, with $H_{\mathrm{out}} = 0.33333$ m
Volume fraction	$V_f = 0.30$ fluid, corresponding to Yoon's 70% solid mass usage
Reynolds number	$Re = 3{,}000$, based on $U_{\mathrm{max,in}} L\rho/\mu$
Density and viscosity	$\rho = 1000$ kg/m^3, $\mu = 1.0$ Pa s
Inlet condition	Shifted parabolic velocity profile with $U_{\mathrm{max,in}} = 3$ m/s; $\tilde{\nu} = 1$ for the FEniCS-SA solve
Outlet condition	Shifted parabolic outflow profile with the maximum velocity set by incompressible flow-rate conservation; $\tilde{\nu} = 1$ for the FEniCS-SA solve; pressure pinned at $(0, 0)$ for gauge fixing
Wall conditions	No-slip velocity and $\tilde{\nu} = 0$ on the top and bottom walls and on the right boundary outside the outlet
Brinkman solid resistance	$\alpha_{\mathrm{solid}} = 10^9$, $\alpha_{\mathrm{fluid}} = 0$
Optimization objective	Dissipation objective J_D, including physical viscous dissipation and Brinkman drag over the design domain
Validation metrics	J_D, pressure drop, Φ_D, and relative changes between FEniCS-SA, Ansys-SA, and Ansys-SST $k - \omega$

Table B.9: Numerical FEniCS solver and topology-optimization parameters for the diffuser benchmark.

Parameter	**Symbol / option**	**Setting**
Flow finite elements	$(\mathbf{u}, p)$	Taylor–Hood P_2/P_1 continuous Galerkin elements
SA working-variable space	$\tilde{\nu}$	P_1 continuous Galerkin
Design-density space	γ	DG_0 cellwise constant density
Reference FEniCS mesh	–	`mesh_yoon_yplus1.xdmf`; wall-resolved triangular mesh with 38,745 vertices, 76,704 triangles, far-field $h_{\max} = L/180 = 5.56 \times 10^{-3}L$, and first-cell width $1.82 \times 10^{-3}L$
Objective	J_D	Dissipation objective over the design domain
Initial design density	γ_0	$V_f = 0.30$, matched to the filtered volume
Brinkman penalty	$\alpha(\bar{\gamma})$	RAMP form with $\alpha_{\text{fluid}} = 0$, $\alpha_{\text{solid}} = 10^9$
RAMP continuation	q	12-stage schedule: 0.01, 0.02, 0.04, 0.08, 0.08, 0.12, 0.20, 0.35, 0.50, 0.75, 1.00, 1.00
Projection continuation	β	12-stage schedule: 1, 2, 4, 8, 8, 12, 16, 32, 64, 96, 128, 128, with threshold $\eta = 0.50$
MMA move limits	–	12-stage schedule: 0.05, 0.04, 0.03, 0.02, 0.012, 0.010, 0.0075, 0.005, 0.0035, 0.0025, 0.0015, 0.0010
Maximum optimization iterations	–	12-stage schedule: 50, 60, 80, 100, 200, 180, 220, 260, 260, 220, 260, 360 MMA iterations; stop if objective change is below 10^{-6} for ten iterations
Filter radius	r_f	$4h_{\max} = 2.22 \times 10^{-2}L$
SA working-variable penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance penalty	–	Power-law solid indicator with amplitude 10^5 and exponent $n = 3$
Wall-distance equation	–	Reciprocal penalized relaxed Eikonal equation with $\sigma_w = 0.1$ and $G_0 = 20$
Forward flow solver	–	IPCS flow solves for frozen-SA Picard updates, followed by a warm-started SNES Newton line-search final flow solve
Frozen NS-SA coupling	–	Three Picard updates per design iteration with SA relaxation factor 0.15
Linear solver	–	MUMPS for SNES/state and adjoint solves; BiCGSTAB with ILU preconditioning for IPCS warm-start solves
Adjoint treatment	–	Frozen-turbulence adjoint in $(\mathbf{u}^*, p^*)$; $\tilde{\nu}$ and G are frozen during gradient assembly

Table B.10: Ansys Fluent mesh-independence table for the extracted turbulent diffuser benchmark geometry.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR ST infl.
h_{bulk} [m]	7.50×10^{-3}	5.00×10^{-3}	4.00×10^{-3}
Curvature angle	9°	6°	5°
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	10	14	16
Growth rate	1.11	1.09	1.08
t_{infl} [m]	2.51×10^{-2}	3.90×10^{-2}	4.55×10^{-2}
Nodes	8,104	16,166	21,238
Elements	12,330	24,570	40,268
Validation metrics			
Φ_D^{SA} [W/m]	1.2921×10^5	1.2966×10^5	1.2995×10^5
$\epsilon_{\text{rel}}(\Phi_D^{\text{SA}})$ [%]	–	0.35	0.22
Φ_D^{SST} [W/m]	1.5296×10^4	1.5570×10^4	1.5725×10^4
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	1.79	0.99
Δp^{SA} [Pa]	1.2460×10^5	1.2449×10^5	1.2422×10^5
$\epsilon_{\text{rel}}(\Delta p^{\text{SA}})$ [%]	–	0.09	0.22
Δp^{SST} [Pa]	4.2412×10^4	4.2034×10^4	4.1833×10^4
$\epsilon_{\text{rel}}(\Delta p^{\text{SST}})$ [%]	–	0.89	0.48
Selected validation mesh		Fine_Ansys	

Note. WR = wall-resolved; ST = smooth-transition; Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

Appendix C

Laminar-Design Diagnostics and Mesh Independence for Benchmark Topologies

This appendix supports the laminar-design comparisons in Chapter 6. It defines the common evaluation pipeline and reports the Ansys mesh-independence studies for the extracted laminar geometries. The topology figures and final laminar-versus-turbulent SST $k-\omega$ metric tables are placed in the corresponding Chapter 6 benchmark sections so that the main comparison can be read together with each optimization result. This appendix also records the FEniCS convergence histories and configuration differences needed to reproduce the laminar $Re = 1$ reference designs.

Unless stated otherwise, all mesh-independence relative differences in this appendix are reported as positive percentage magnitudes. For a mesh-dependent metric ϕ, the coarse–medium difference is computed as $|\phi_{\text{coarse}} - \phi_{\text{medium}}|/|\phi_{\text{coarse}}| \times 100\%$, while the medium–fine difference is computed as $|\phi_{\text{medium}} - \phi_{\text{fine}}|/|\phi_{\text{fine}}| \times 100\%$.

C.1 Methodology and Evaluation Pipeline

Each benchmark is solved twice. The laminar reference design is optimized at $Re = 1$, where the optimizer primarily responds to creeping-flow viscous losses. The turbulent design is optimized at the target high-Reynolds operating condition using the SA topology optimization framework described in Chapter 5. Apart from replacing the SA turbulent state equations by the laminar flow formulation and setting $Re = 1$, the laminar topology-optimization runs retain the benchmark geometry, FEniCS mesh, design-domain restrictions, non-design regions, volume constraint, prescribed inlet and outlet treatment, objective type, Brinkman penalty limits, filter radius, and projection threshold from the corresponding turbulent runs. Thus, the pipe-bend and U-bend comparisons retain the inlet-pressure objective J_p, while the diffuser comparison retains the dissipation objective J_D from Yoon [56]. The laminar-specific viscosity, solver, and optimizer-control differences are listed in the benchmark subsections below.

Because most FEniCS physical and numerical settings are inherited directly from the turbulent benchmark definitions in Appendix B, they are not repeated in full. Instead, each benchmark lists only the laminar-specific differences from the corresponding turbulent setup.

After optimization, both projected density fields are thresholded and extracted as sharp geometries. The extracted laminar and turbulent designs are then evaluated in Ansys Fluent under the same turbulent operating condition. Because Chapter 6 already validates the extracted turbulent topologies with both Ansys-SA and Ansys-SST $k-\omega$, the laminar-design comparison uses the higher-fidelity SST $k-\omega$ closure. This keeps the comparison focused on topology performance under an independent two-equation turbulence model, rather than repeating the SA cross-code validation for the laminar reference geometries. The key performance metrics are viscous dissipation and pressure drop.

All Fluent mesh-independence studies in this appendix use the same convergence and acceptance criteria as Appendix B, but only for the SST $k-\omega$ model. The static pressure drop must change by less than 5×10^{-4} over the final 100 iterations, while the viscous dissipation must change by less than 10^{-3} over the final 100 iterations. The fine mesh is accepted only when the medium–fine relative difference is below 1% for static pressure drop and below 2% for viscous dissipation. Mass conservation is additionally checked from the inlet and outlet mass-flow reports.

For fairness, the extracted laminar geometries receive their own Ansys mesh-independence checks. The turbulent-design grid studies are reported in Table B.4 of Section B.3, Table B.7 of Section B.4, and Table B.10 of Section B.5 in Appendix B; this appendix adds the corresponding laminar-design mesh tables.

C.2 Pipe Bend Benchmark

The pipe bend tests whether a topology optimized under turbulent conditions produces a more separation-aware 90° turning path than a laminar topology optimized at $Re = 1$. The turbulent case in Chapter 6 uses the $V_f = 0.25$ design. The topology-level comparison and final SST $k - \omega$ performance table are reported in Section 6.3.5.

C.2.1 Laminar FEniCS Configuration Differences

The laminar pipe-bend reference is a completed FEniCS topology-optimization run from `Config_PipeBendAlexandersen_Laminar.py`. It reuses the turbulent pipe-bend setup in Tables B.2 and B.3 wherever not stated otherwise: the geometry, wall-resolved mesh, non-design inlet and outlet ducts, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.25$ volume constraint, average inlet-pressure objective J_p, Brinkman limits, filter radius, and projection threshold are unchanged. Table C.1 lists the laminar-specific differences.

Table C.1: Laminar-specific differences from the turbulent pipe-bend FEniCS setup in Appendix B.

Parameter	**Laminar setting**
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}} H_{\text{in}} \rho/\mu$ instead of the turbulent $Re = 5{,}000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{in}} = 0.2L$ as the turbulent case
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton solves with relative tolerance 10^{-6}; the forward absolute tolerance is 10^{-8}, the adjoint absolute tolerance is 10^{-9}, and the maximum forward iteration count is 220
Continuation and optimizer controls	The q and β schedules match the turbulent run, but the laminar run uses move limits $\{0.04, 0.03, 0.02, 0.012, 0.008, 0.005, 0.003\}$ and inner-iteration budgets $\{80, 100, 140, 200, 200, 240, 260\}$

C.2.2 Laminar FEniCS Optimization Diagnostics

Figure C.1 records the completed laminar pipe-bend optimization history. The normalized inlet-pressure objective drops rapidly in the first continuation stage and then remains nearly flat, while the volume-fraction residual repeatedly returns to the active constraint after continuation updates.

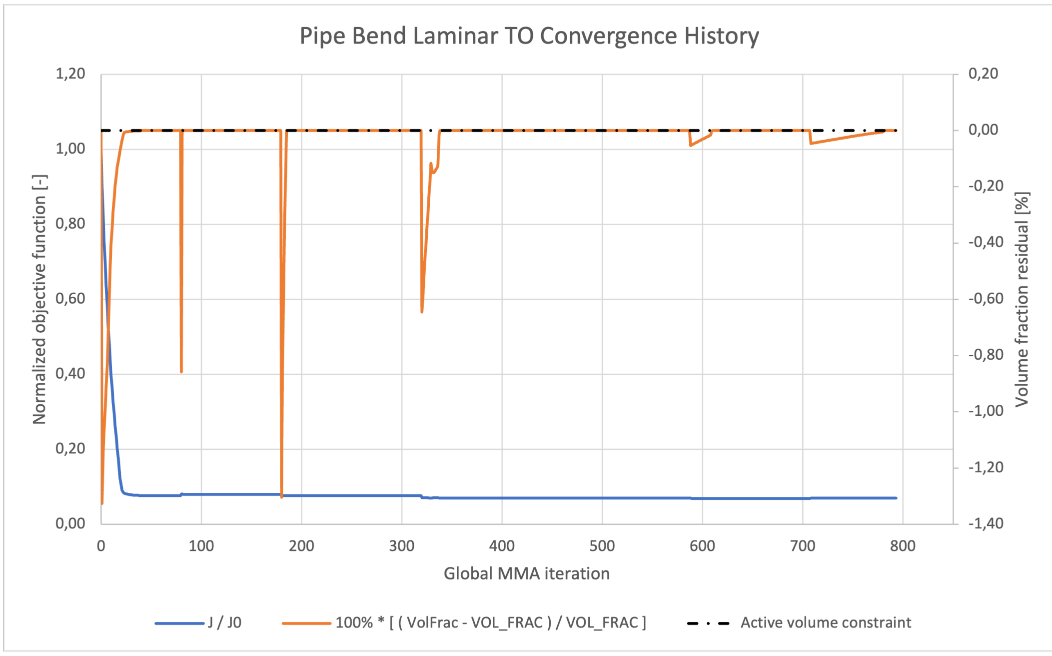


Figure C.1: FEniCS convergence history for the laminar pipe-bend topology-optimization run. The blue curve is the normalized inlet-pressure objective $J_p/J_{p,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.2.3 Laminar-Design Mesh Independence

Table C.2 reports the Ansys mesh-independence study for the extracted laminar pipe-bend design. This table is separate from the turbulent-design mesh study in Table B.4 of Section B.3 because the laminar and turbulent topologies generally produce different wall shapes and near-wall mesh requirements.

TABLE C.2: Ansys mesh-independence table for the extracted laminar pipe-bend design.

Parameter	Coarse	Medium	Fine
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	4.50×10^{-3}	2.60×10^{-3}	2.00×10^{-3}
Curvature angle	8°	4°	3°
Δy_1 [m]	1.20×10^{-3}	1.20×10^{-3}	1.20×10^{-3}
Inflation layers	12	15	16
Growth rate	1.06	1.045	1.04
t_{infl} [m]	2.02×10^{-2}	2.49×10^{-2}	2.62×10^{-2}
Nodes	22,494	56,468	89,804
Elements	35,948	93,828	153,163
Validation metrics			
Φ_D^{SST} [W/m]	0.0293	0.0310	0.0313
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	5.82	0.95
Δp^{SST} [Pa]	0.5229	0.5420	0.5444
$\epsilon_{\text{rel}}(\Delta p^{\text{SST}})$ [%]	–	3.65	0.45
Selected comparison mesh		Fine_Ansys	

Note. Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.3 U-Bend Benchmark

The U-bend tests whether the turbulent optimizer remains advantageous when the flow must complete a 180° return path around a fixed separator. The turbulent case in Chapter 6 uses the $V_f = 0.27$ design. The topology-level comparison and final SST $k - \omega$ performance table are reported in Section 6.4.4.

C.3.1 Laminar FEniCS Configuration Differences

The laminar U-bend reference is a completed FEniCS topology-optimization run from `Config_UBendAlexandersen_Laminar.py`. It reuses the turbulent U-bend setup in Tables B.5 and B.6 wherever not stated otherwise: the geometry, fixed separator, wall-resolved mesh, non-design ports, uniform inlet velocity, pressure-outlet treatment, $V_f = 0.27$ volume constraint, average inlet-pressure objective J_p, Brinkman limits, filter radius, and projection threshold are unchanged. Table C.3 lists the laminar-specific differences.

C.3.2 Laminar FEniCS Optimization Diagnostics

Figure C.2 records the completed laminar U-bend optimization history. As in the pipe-bend case, the objective reduction occurs mainly at the start of the run, after which the optimizer preserves a nearly active volume constraint while only small continuation-related corrections remain.

Table C.3: Laminar-specific differences from the turbulent U-bend FEniCS setup in Appendix B.

Parameter	**Laminar setting**
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, reciprocal wall-distance equation, topology-dependent SA penalties, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\text{in}} H_{\text{port}} \rho/\mu$ instead of the turbulent $Re = 5{,}000$ setup
Density and viscosity	$\rho = 1.0$ and $\mu = 0.2$, with the same $U_{\text{in}} = 1$ and $H_{\text{port}} = 0.2L$ as the turbulent case
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton solves with relative tolerance 10^{-6}; the forward absolute tolerance is 10^{-8}, the adjoint absolute tolerance is 10^{-9}, and the maximum forward iteration count is 260
Continuation and optimizer controls	The q and β schedules match the turbulent run, but the laminar run uses move limits $\{0.02, 0.015, 0.010, 0.006, 0.004, 0.003, 0.002\}$ and inner-iteration budgets $\{80, 100, 140, 200, 200, 240, 260\}$

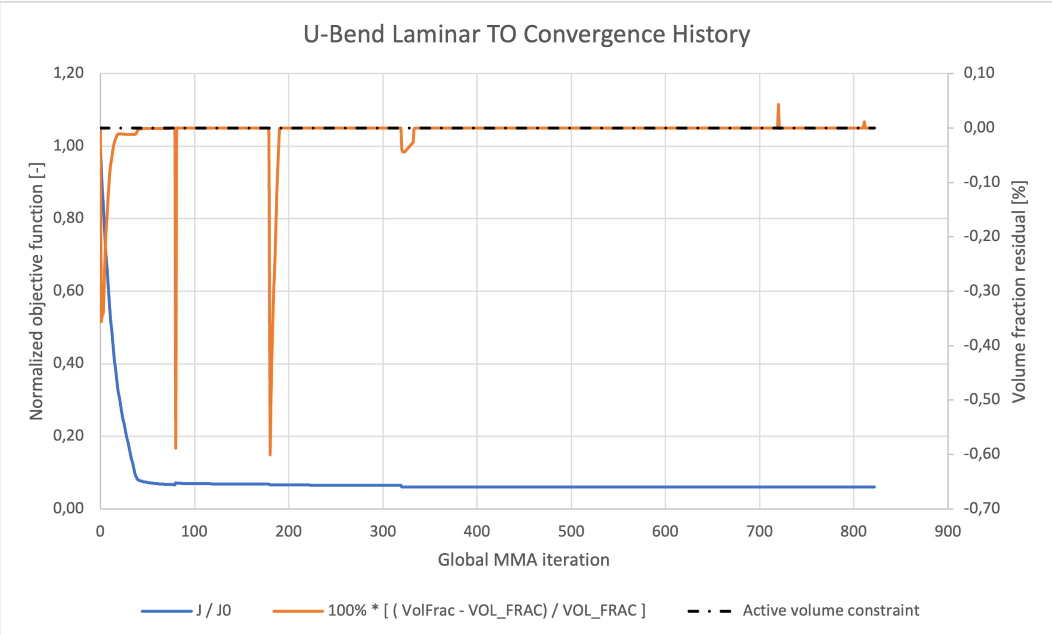


Figure C.2: FEniCS convergence history for the laminar U-bend topology-optimization run. The blue curve is the normalized inlet-pressure objective $J_p/J_{p,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.3.3 Laminar-Design Mesh Independence

Table C.4 reports the Ansys mesh-independence study for the extracted laminar U-bend design. The turbulent U-bend mesh study is reported separately in Table B.7 of Section B.4. For the $Re = 5{,}000$ U-bend operating point, the inlet-port height is $0.2L$ and $U_{\text{in}} = 1\ \text{m}\,\text{s}^{-1}$, giving $\nu = 4.0 \times 10^{-5}\ \text{m}^2\,\text{s}^{-1}$. Using the same smooth-wall estimate as the FEniCS mesh generator gives $u_\tau \approx 6.59 \times 10^{-2}\ \text{m}\,\text{s}^{-1}$ and a wall distance of $y_+ = 1$ at 6.07×10^{-4} m. The Ansys mesh hierarchy therefore keeps the first layer fixed at 6.00×10^{-4} m and refines the bulk triangular mesh, curvature resolution, and inflation-layer distribution around the separator tip and 180° return wall.

TABLE C.4: Ansys mesh-independence table for the extracted laminar U-bend design.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
h_{bulk} [m]	6.00×10^{-3}	4.00×10^{-3}	2.75×10^{-3}
Curvature angle	8°	6°	4°
Δy_1 [m]	6.00×10^{-4}	6.00×10^{-4}	6.00×10^{-4}
Inflation layers	12	16	20
Growth rate	1.10	1.08	1.06
t_{infl} [m]	1.28×10^{-2}	1.82×10^{-2}	2.21×10^{-2}
Nodes	16,106	32,580	62,622
Elements	24,460	50,067	98,084
Validation metrics			
Φ_D^{SST} [W/m]	0.101	0.105	0.107
$\epsilon_{\text{rel}}(\Phi_D^{\text{SST}})$ [%]	–	3.96	1.90
Δp^{SST} [Pa]	1.125	1.141	1.150
$\epsilon_{\text{rel}}(\Delta p^{\text{SST}})$ [%]	–	1.42	0.79
Selected comparison mesh		Fine_Ansys	

Note. Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change.

C.4 Diffuser Benchmark

The diffuser benchmark tests whether the turbulent optimizer remains advantageous when the design must diffuse a full-height inflow into a center-third outflow while minimizing the dissipation objective J_D. The turbulent case in Chapter 6 uses the $V_f = 0.30$, $Re = 3{,}000$ design from Yoon [56]. The topology-level comparison and final SST $k - \omega$ performance table are reported in Section 6.5.4.

C.4.1 Laminar FEniCS Configuration Differences

The laminar diffuser reference is a completed FEniCS topology-optimization run from `Config_DiffuserYoon_Laminar.py`. It reuses the turbulent diffuser setup in Tables B.8 and B.9 wherever not stated otherwise: the square $L = 1$ m domain, full-height inlet, center-third outlet, parabolic velocity profiles, $V_f = 0.30$ volume constraint, dissipation objective J_D, Brinkman limits, mesh, filter radius, projection threshold, continuation schedules, MMA move limits, and maximum MMA iterations are unchanged. Table C.5 therefore lists only the laminar-specific differences.

Table C.5: Laminar-specific differences from the turbulent diffuser FEniCS setup in Appendix B.

Parameter	**Laminar setting**
Flow model	Laminar incompressible Navier–Stokes; the SA working variable, SA wall-distance equation, and frozen-turbulence coupling are omitted
Reynolds number	$Re = 1$, based on $U_{\max,\mathrm{in}} L\rho/\mu$ instead of the turbulent $Re = 3{,}000$ setup
Density and viscosity	$\rho = 1000\ \mathrm{kg/m^3}$ and $\mu = 3000\ \mathrm{Pa\,s}$, with the same $U_{\max,\mathrm{in}} = 3$ m/s and $U_{\max,\mathrm{out}} = 9$ m/s as the turbulent case
Flow finite elements	Taylor–Hood P_2/P_1 for $(\mathbf{u}, p)$; no separate P_1 SA working-variable space is used
Forward and adjoint solves	SNES/MUMPS laminar Newton solves with relative tolerance 10^{-6}; the forward absolute tolerance is 10^{-9}, the adjoint absolute tolerance is 10^{-9}, and the maximum forward iteration count is 200

C.4.2 Laminar FEniCS Optimization Diagnostics

Figure C.3 records the completed laminar diffuser optimization history. The normalized objective drops rapidly during the early continuation stages and then remains nearly flat, while the volume-fraction residual stays close to the active constraint. This figure is kept in the appendix because it documents the baseline optimization run rather than the turbulent performance comparison itself.

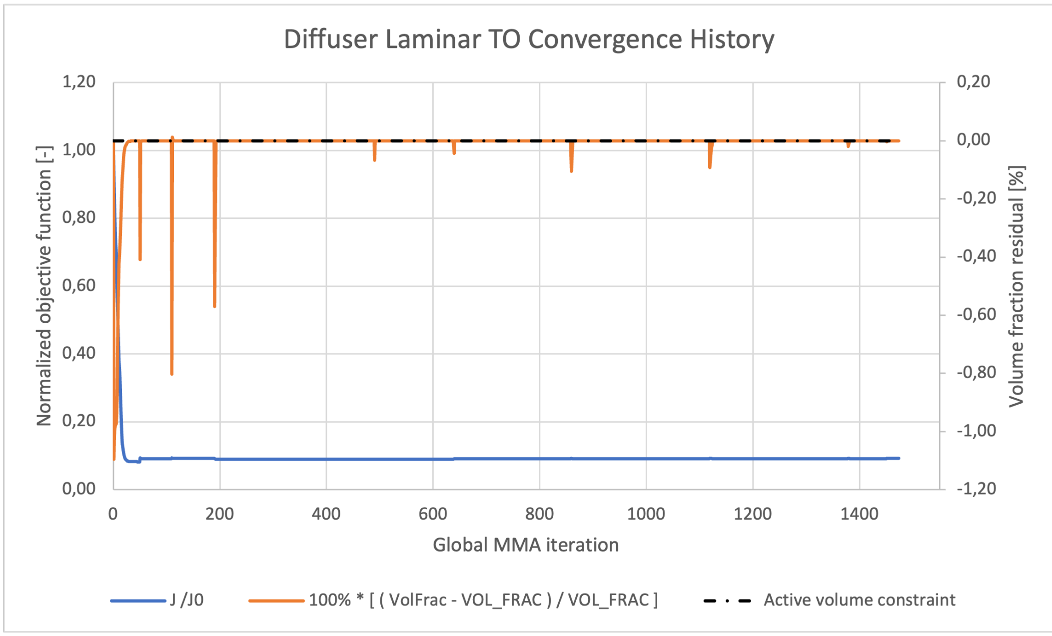


FIGURE C.3: FEniCS convergence history for the laminar diffuser topology-optimization run. The blue curve is the normalized dissipation objective $J_D/J_{D,0}$, the orange curve is the relative volume-fraction residual, and the dashed black line marks the active volume constraint.

C.4.3 Laminar-Design Mesh Independence

Table C.6 reports the Ansys mesh-independence study for the extracted laminar diffuser design. The turbulent diffuser mesh study is reported separately in Table B.10 of Section B.5. The Fluent solves use the same shifted inlet and outlet parabolic velocity-profile convention as the turbulent diffuser validation setup.

Table C.6: Ansys mesh-independence table for the extracted laminar diffuser design.

Parameter	**Coarse**	**Medium**	**Fine**
Mesh settings			
Mesh type	Tri.	Tri.	Tri.
Near-wall	WR infl.	WR infl.	WR infl.
h_{bulk} [m]	1.20×10^{-2}	5.50×10^{-3}	3.75×10^{-3}
Curvature angle	15°	7°	5°
Δy_1 [m]	1.50×10^{-3}	1.50×10^{-3}	1.50×10^{-3}
Inflation layers	6	12	16
Growth rate	1.15	1.10	1.08
t_{infl} [m]	1.31×10^{-2}	3.19×10^{-2}	4.55×10^{-2}
Nodes	3,317	13,444	25,485
Elements	5,085	20,888	39,582
Validation metrics			
Φ_D^{SST} [W/m]	1.2643×10^4	1.2983×10^4	1.3077×10^4
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SST}})$ [%]	–	2.69	0.72
Δp^{SST} [Pa]	3.6628×10^4	3.6693×10^4	3.6464×10^4
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SST}})$ [%]	–	0.18	0.62
Φ_D^{SA} [W/m]	1.4296×10^5	1.4315×10^5	1.4333×10^5
$\epsilon_{\mathrm{rel}}(\Phi_D^{\mathrm{SA}})$ [%]	–	0.13	0.13
Δp^{SA} [Pa]	1.4191×10^5	1.4276×10^5	1.4200×10^5
$\epsilon_{\mathrm{rel}}(\Delta p^{\mathrm{SA}})$ [%]	–	0.60	0.53
Selected comparison mesh		Fine_Ansys	

Note. WR = wall-resolved; Φ_D = viscous dissipation; Δp = static pressure drop; SST denotes SST k–ω. For ϵ_{rel} rows, the Medium column gives the Coarse–Medium change and the Fine column gives the Medium–Fine change. The SA rows are re-evaluations for turbulence-model sensitivity.

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Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025–2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL · E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools.** More info about what sensitive and personal data are, is described in the three confidentiality levels of the KU Leuven data classification model (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter. Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use M**365 Copilot Chat with your KU Leuven**

account. If you use another GenAI tool, you must be **absolutely certain** it does n**ot store or reuse the data you enter.**

- GenAI assistance may not be used for data or topics covered by a Non-Disclosure Agreement (NDA). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.

- If your master's thesis is under embargo, you should first discuss with your supervisor whether the use of GenAI is permitted.

- Before using a GenAI tool, always consider whether its use is responsible, including from a sustainability perspective (e.g. when using GenAI as a search engine, language assistant,. . .).

- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.

- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools. **Exam Regulations Article 84:** *"Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity."*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students (Dutch/English).

- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** (Dutch/English)

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is

allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: .... **Student number:** ....

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

O This form is related to a **course assignment.**
Course name: .... **Course number:** ....

This form is related to my O **Bachelor's** or O **Master's thesis.**
Title Bachelor's or Master's thesis: .... **Supervisor:** ....

Daily supervisor: ....

Please indicate with "X":

O I **did not use** any GenAI assistance tool.

O I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot, ...): ....

GenAI assistance used as/for:	**Name of the GenAI tool(s) used** If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	**Code of conduct:** **For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,...).**
As a language assistant for reviewing or improving texts I wrote myself	*add text here*	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • When using GenAI tools on texts you did not write yourself to improve the text, **you have to refer to the original source or author**, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. **However, be careful**: • If it entails text by an author other than yourself, **you are not allowed to include that paraphrased text without reference to the original source or author**. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others	*add text here*	This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. **However, be careful:** • **You are not allowed to include that translated text without reference to the original source or author.** Without such reference, you would be committing plagiarism and thus an irregularity. • Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	*add text here*	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. **You only have to refer to the existing research and references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, **verify the information and do not just copy-paste it** as you should understand and critically process everything you are writing.

For literature search	*add text here*	This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; **you only have to refer to the literature references you have checked and used** (without such references you would be committing plagiarism and thus committing an irregularity). **However, be careful:** • Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. **Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it** as you should understand and critically process everything you are writing. • Be aware that some GenAI tools may **output no or wrong references.** As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	*add text here*	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.

For generating new (research) ideas	*add text here*	This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. **If so, that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus committing an irregularity).
For generating synthetic data	*add text here*	Use of GenAI for generating synthetic data is allowed, **provided that it is methodologically and ethically justifiable**, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparent about the use of GenAI. **Be careful:** • Always **carefully evaluate** the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.

For generating blocks of text (other than the allowed use without referencing mentioned above)		According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. **Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed.** **Be careful:** • If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparent about the use of GenAI. • However, in general, such GenAI use should be kept to an **absolute minimum**; you should always check the original sources.
For generating visuals, video or audio	*add text here*	Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. **Be careful:** • If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. • If you work with existing visuals, audio or video, **that existing work should be correctly referenced in the text** (without such reference you would be committing plagiarism and thus an irregularity). • Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).

Other use (specify here; this may also include a combination of types of use mentioned above):	*add text here*	To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis **beforehand** and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparent about the use of GenAI.